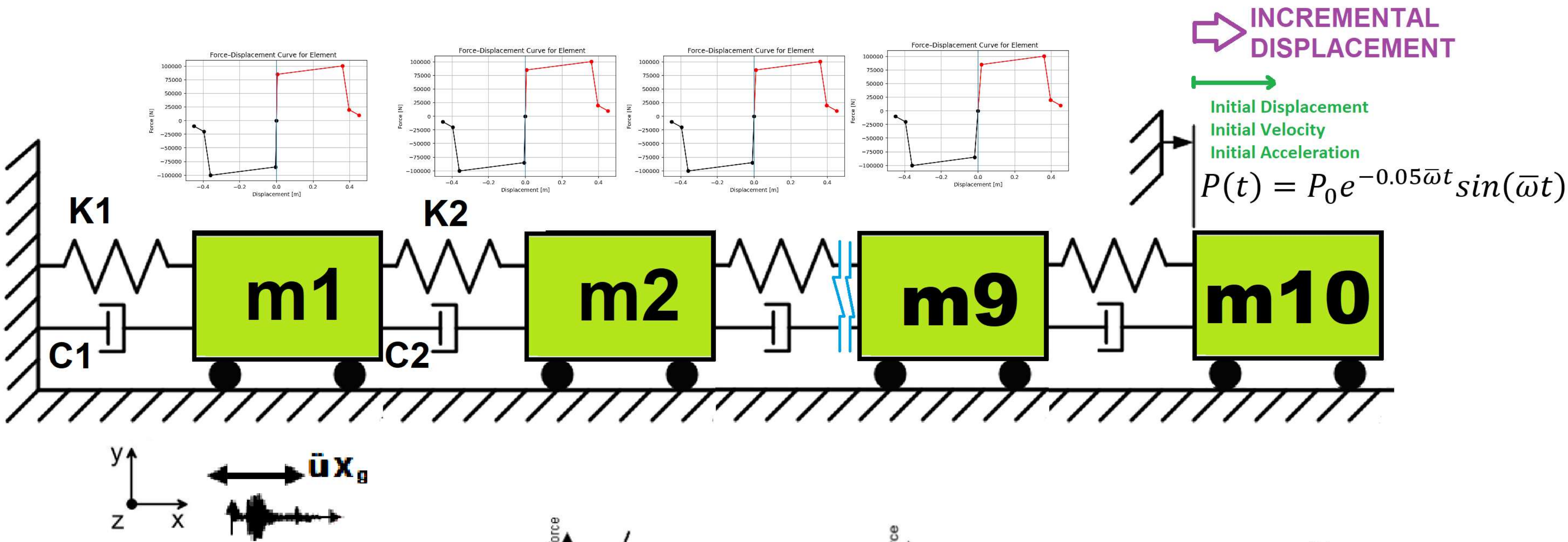
>> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<

COMPREHENSIVE NONLINEAR SEISMIC ASSESSMENT OF A MULTI-DEGREE-FREEDOM (MDOF) STRUCTURE: AN OPENSEES FRAMEWORK FOR STATIC PUSHOVER, CYCLIC DEGRADATION, STATIC TIME-HISTORY AND DYNAMIC TIME-HISTORY ANALYSIS

THIS PROGRAM IS WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)



$$Structural\ Ductility\ Damage\ Index = \frac{\Delta_d - \Delta_y}{\Delta_u - \Delta_y}$$

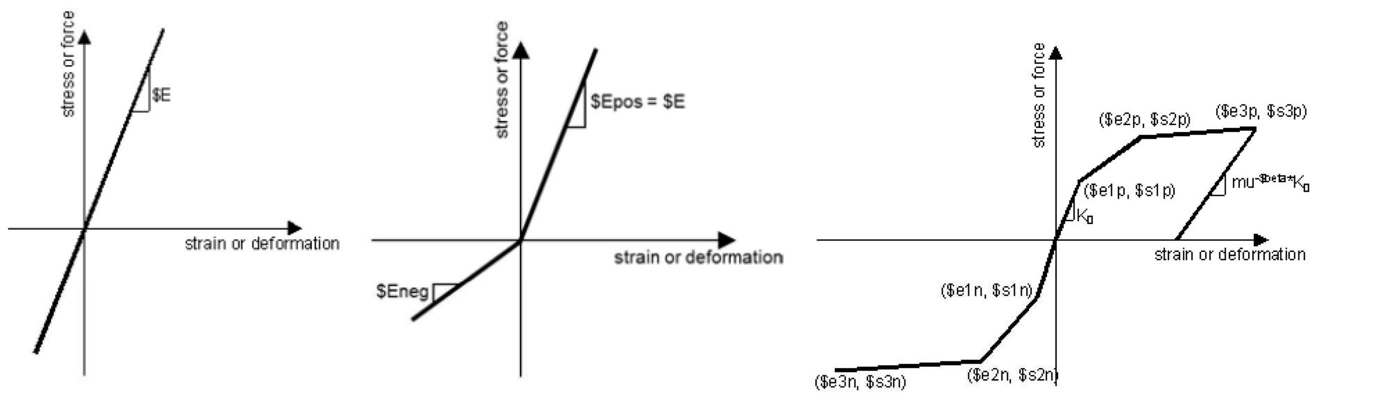
$$\Delta_d = Lateral\ Displacement\ from\ Dynamic\ Analysis$$

$$\Delta_y = Lateral\ Yield\ Displaement\ from\ Pushover\ Analysis$$

$$\Delta_u = Lateral\ Ultimate\ Displaement\ from\ Pushover\ Analysis$$

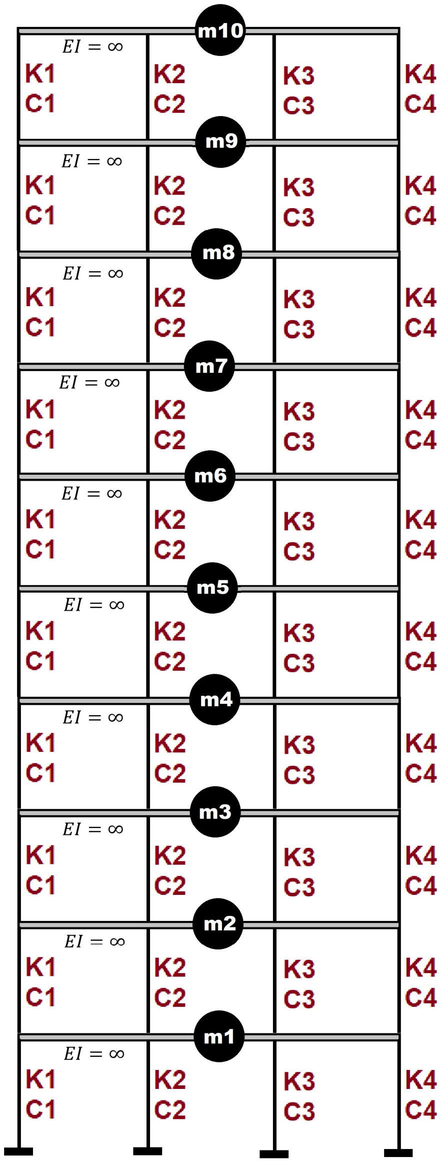
$$P(t) = P_0 e^{-0.05\bar{\omega}t} sin(\bar{\omega}t)$$

$$m\ddot{u} + c\dot{u} + ku = P(t)$$



ELASTIC MATERIAL

INELASTIC MATERIAL



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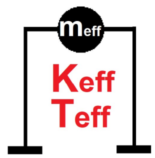
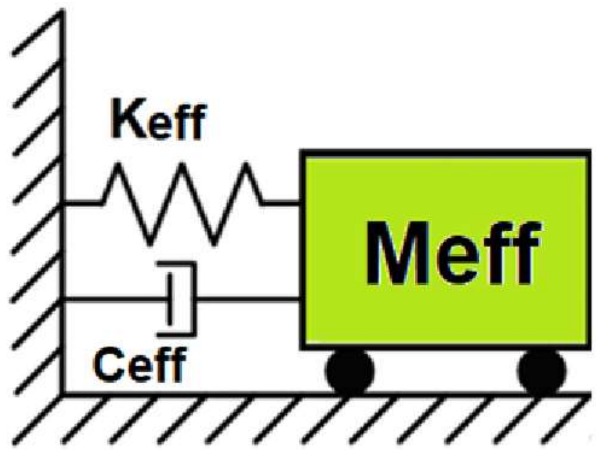
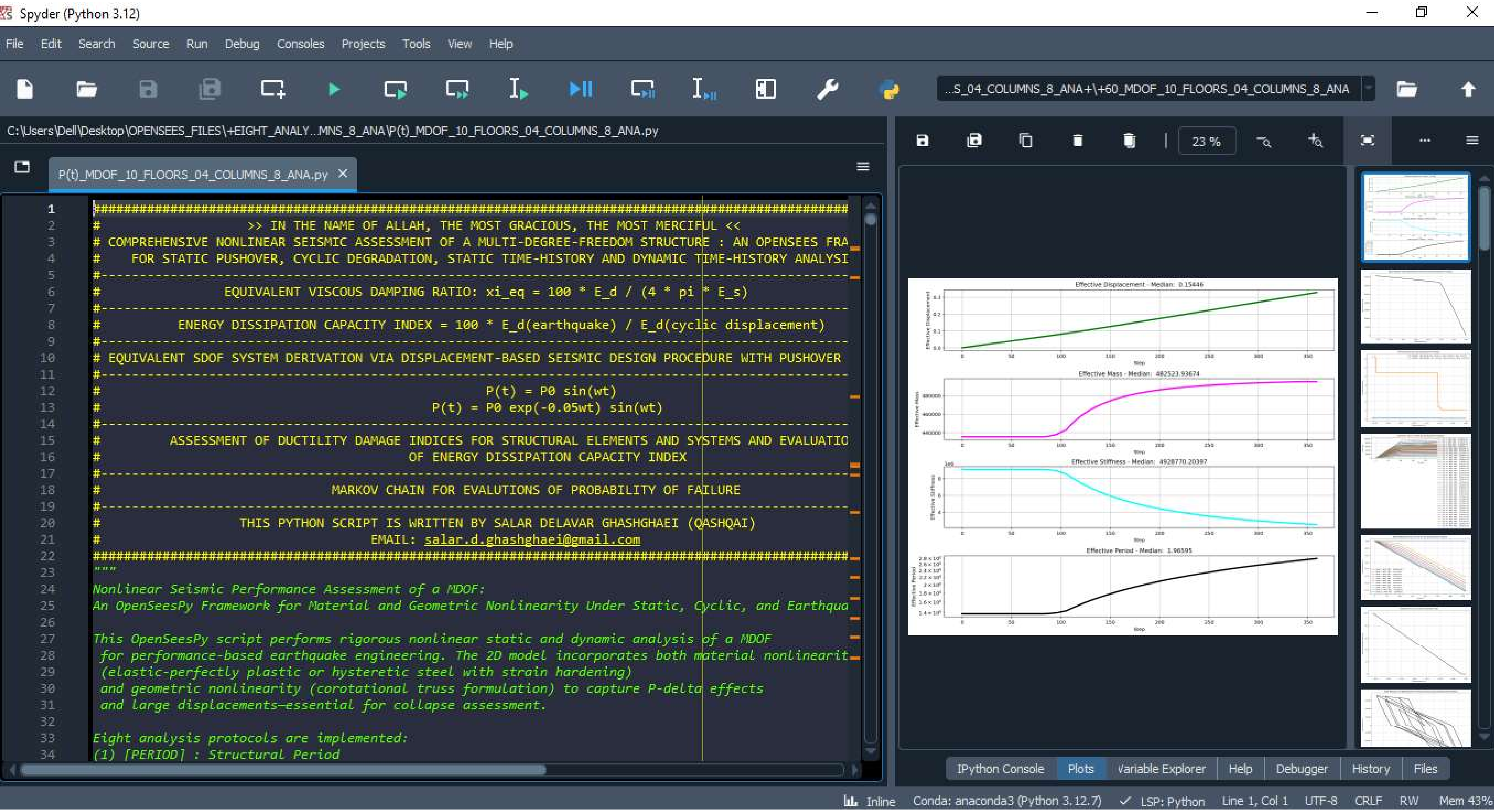


Displacement-based seismic analysis transformation, converting a multi-degree-of-freedom (MDOF) system into an equivalent single-degree-of-freedom (SDOF) system for seismic assessment

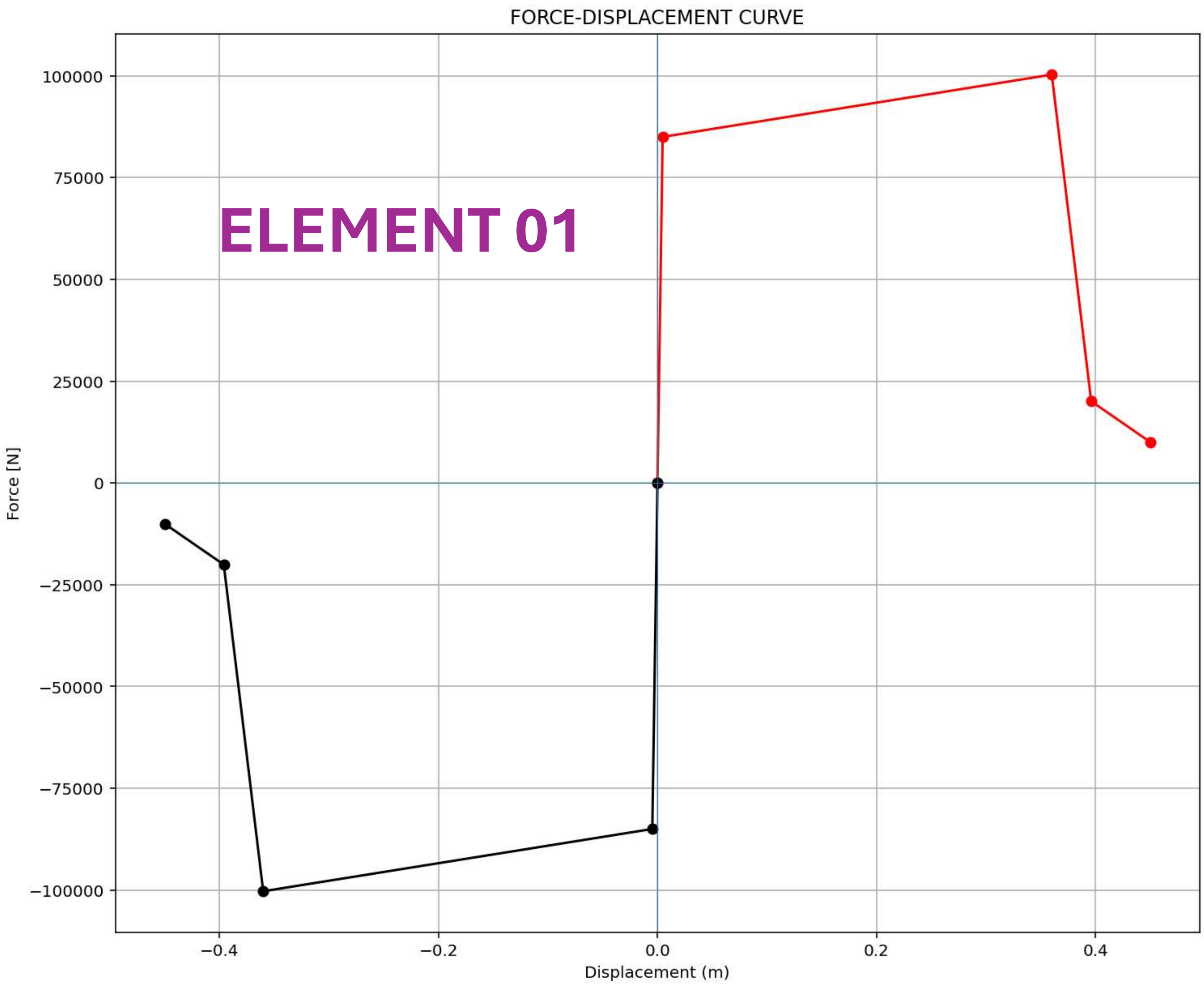
$$\Delta_d = \frac{\sum_{i=1}^{n} m_i \Delta_i^2}{\sum_{i=1}^{n} m_i \Delta_i} \qquad m_{eff} = \frac{\sum_{i=1}^{n} m_i \Delta_i}{\Delta_d}$$

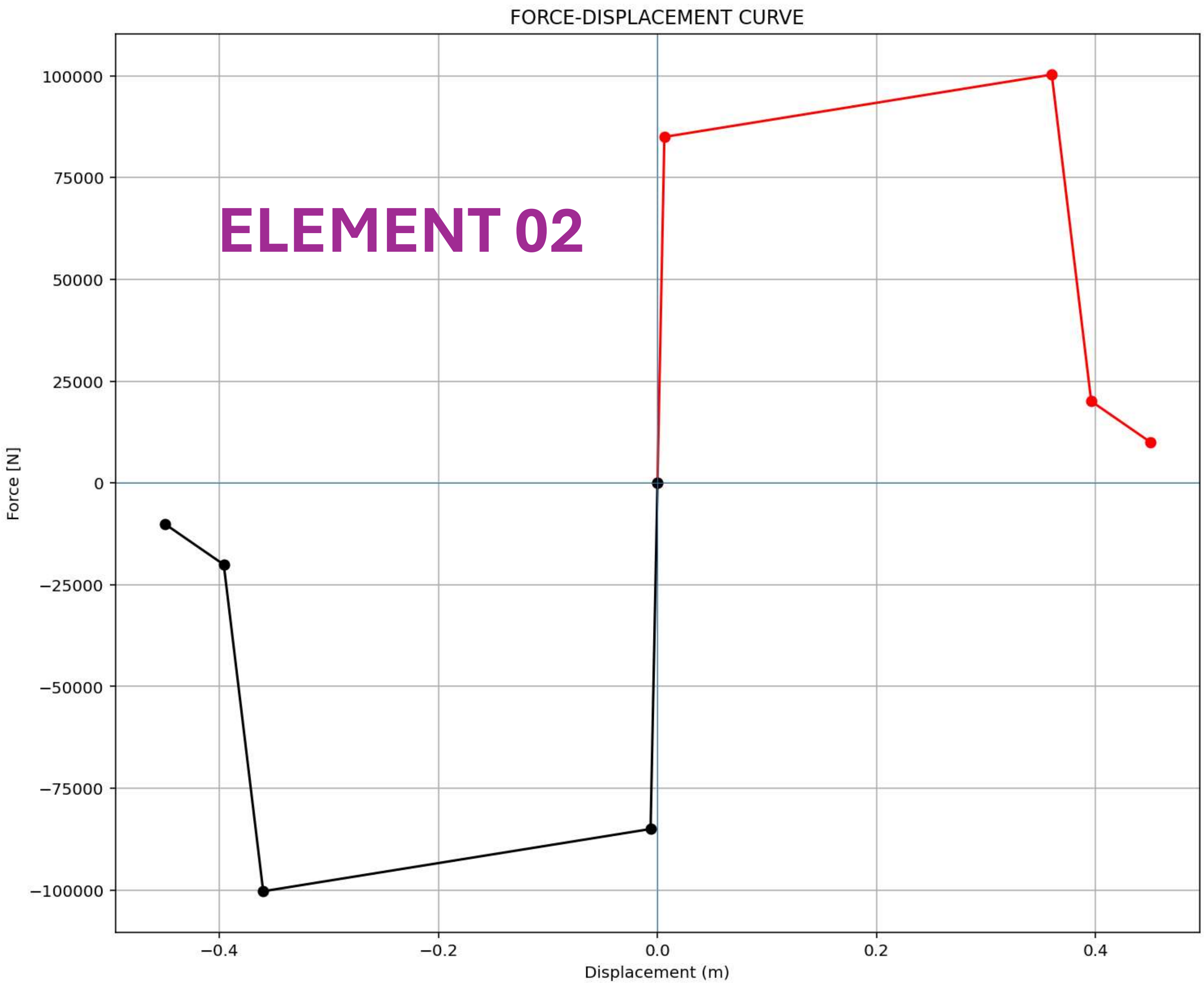
$$k_{eff} = \frac{\sum_{i=1}^{n} k_i \Delta_i}{\Delta_d} \qquad T_{eff} = \frac{2\pi}{\sqrt{\frac{k_{eff}}{m_{eff}}}}$$

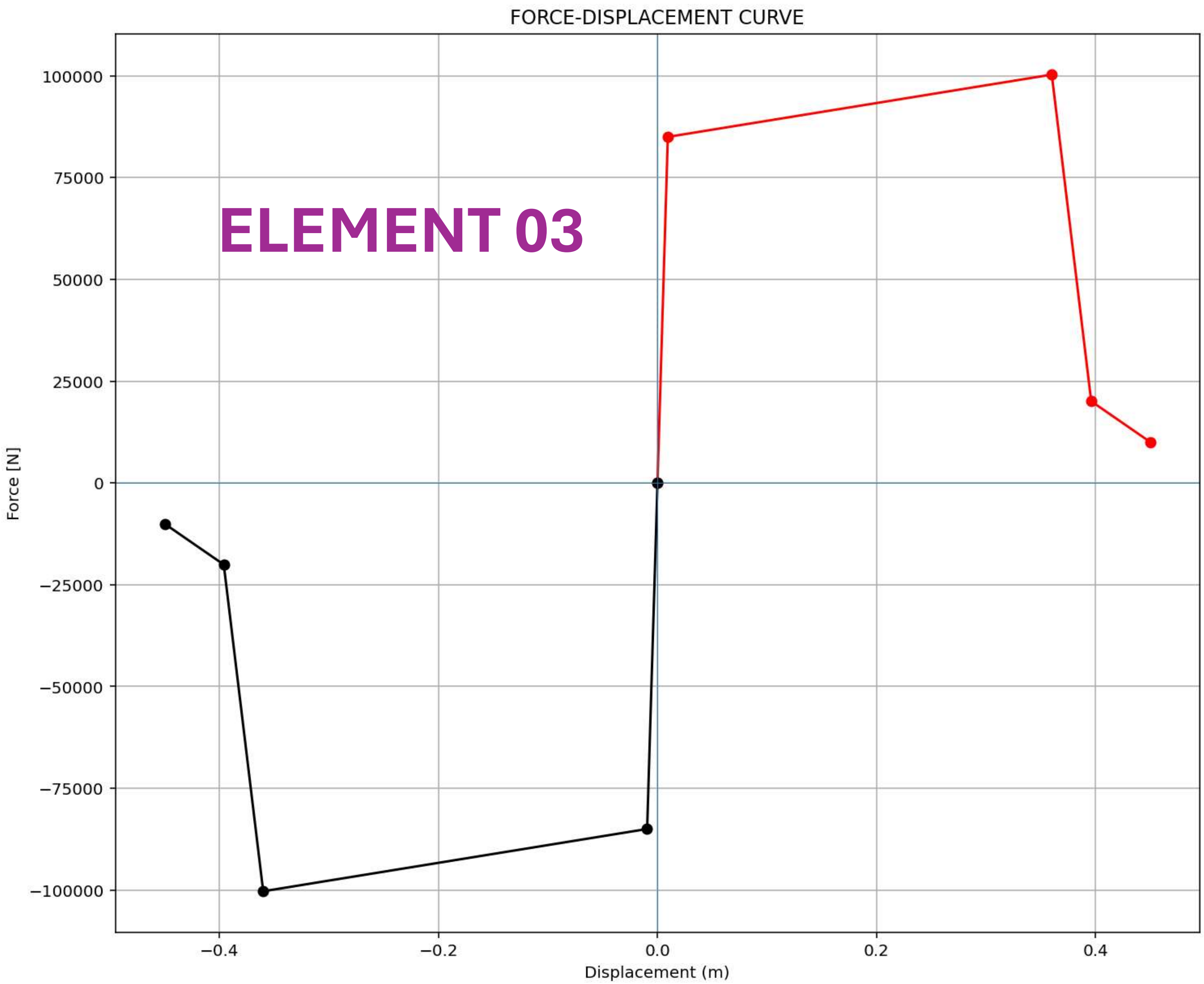


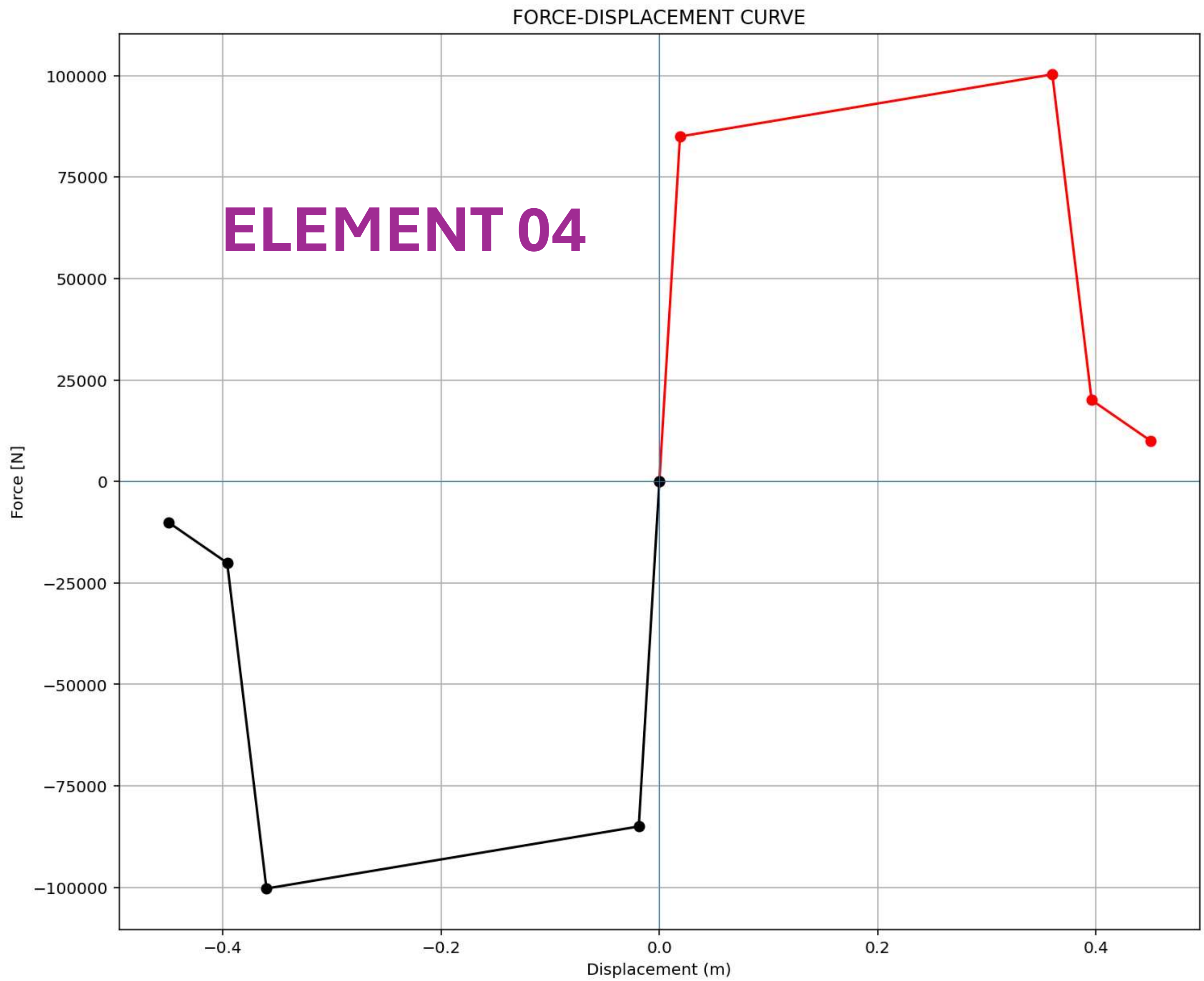


Spyder (Python 3.12)
File Edit Search Source Run Debug Consoles Projects Tools View Help
..S_04_COLUMNS_8_ANA+\+60_MDOF_10_FLOORS_04_COLUMNS_8_ANA
C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY..MNS_8_ANA\P(t)_MDOF_10_FLOORS_04_COLUMNS_8_ANA.py
P(t)_MDOF_10_FLOORS_04_COLUMNS_8_ANA.py
>> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<
COMPREHENSIVE NONLINEAR SEISMIC ASSESSMENT OF A MULTI-DEGREE-FREEDOM STRUCTURE : AN OPENSEES FRA
FOR STATIC PUSHOVER, CYCLIC DEGRADATION, STATIC TIME-HISTORY AND DYNAMIC TIME-HISTORY ANALYSI
EQUIVALENT VISCOUS DAMPING RATIO: xi_eq = 100 * E_d / (4 * pi * E_s)
ENERGY DISSIPATION CAPACITY INDEX = 100 * E_d(earthquake) / E_d(cyclic displacement)
EQUIVALENT SDOF SYSTEM DERIVATION VIA DISPLACEMENT-BASED SEISMIC DESIGN PROCEDURE WITH PUSHOVER
P(t) = P0 sin(wt)
P(t) = P0 exp(-0.05wt) sin(wt)
ASSESSMENT OF DUCTILITY DAMAGE INDICES FOR STRUCTURAL ELEMENTS AND SYSTEMS AND EVALUATIO
OF ENERGY DISSIPATION CAPACITY INDEX
MARKOV CHAIN FOR EVALUTIONS OF PROBABILITY OF FAILURE
THIS PYTHON SCRIPT IS WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)
EMAIL: salar.d.ghashghaei@gmail.com
"""
Nonlinear Seismic Performance Assessment of a MDOF:
An OpenSeesPy Framework for Material and Geometric Nonlinearity Under Static, Cyclic, and Earthqua
This OpenSeesPy script performs rigorous nonlinear static and dynamic analysis of a MDOF
for performance-based earthquake engineering. The 2D model incorporates both material nonlinearit
(elastic-perfectly plastic or hysteretic steel with strain hardening)
and geometric nonlinearity (corotational truss formulation) to capture P-delta effects
and large displacements—essential for collapse assessment.
Eight analysis protocols are implemented:
(1) [PERIOD] : Structural Period
23 %
Effective Displacement - Median: 0.15446
Effective Mass - Median: 482523.93674
Effective Stiffness - Median: 4928770.20397
Effective Period - Median: 1.96595
IPython Console Plots Variable Explorer Help Debugger History Files
Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 1, Col 1 UTF-8 CRLF RW Mem 43%

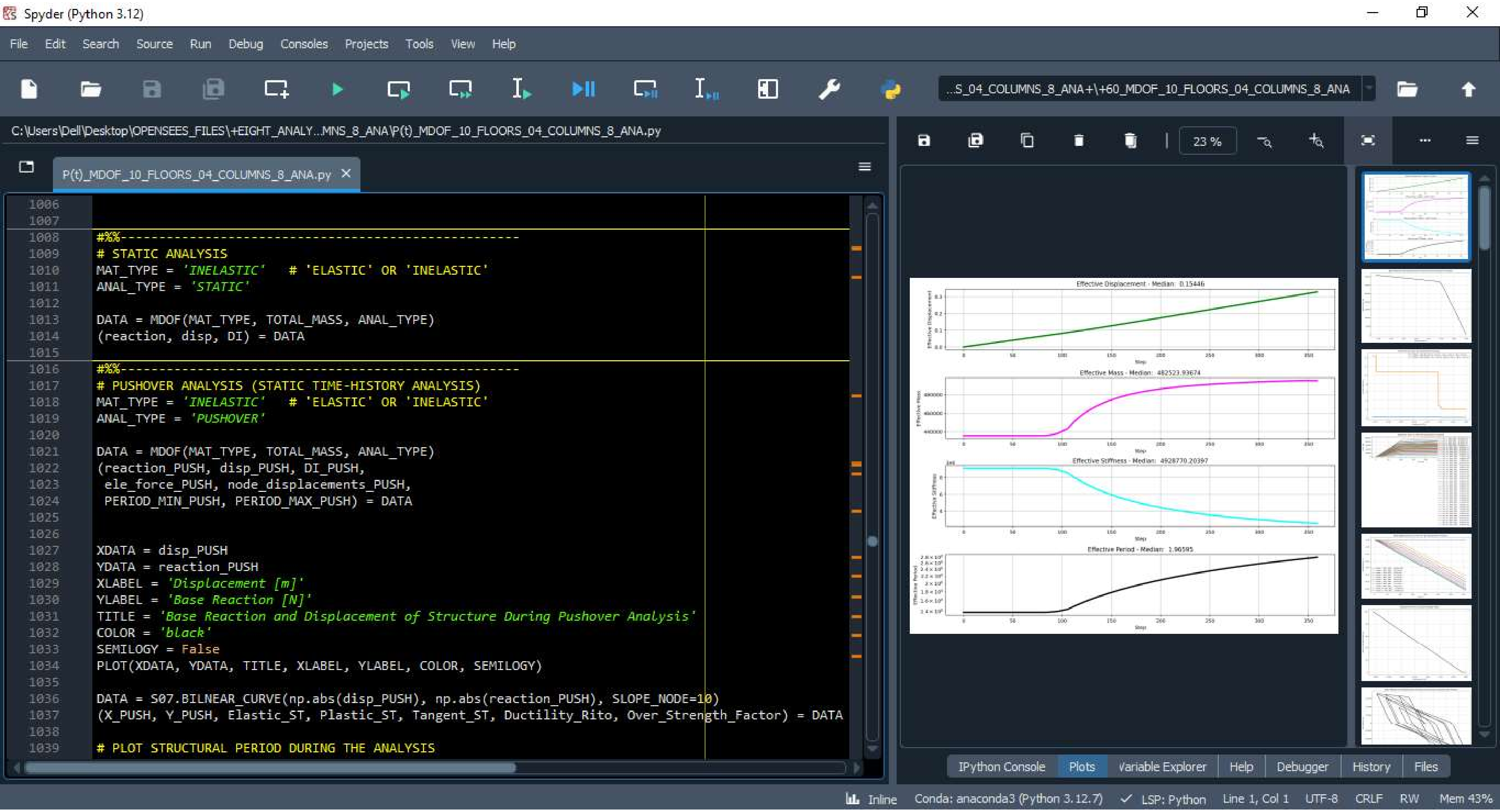

FORCE-DISPLACEMENT CURVE
ELEMENT 01
Force [N]
Displacement (m)
100000
75000
50000
25000
0
−25000
−50000
−75000
−100000
−0.4
−0.2
0.0
0.2
0.4


FORCE-DISPLACEMENT CURVE
ELEMENT 02
100000
75000
50000
25000
0
−25000
−50000
−75000
−100000
Force [N]
−0.4
−0.2
0.0
0.2
0.4
Displacement (m)


FORCE-DISPLACEMENT CURVE
ELEMENT 03
Force [N]
100000
75000
50000
25000
0
−25000
−50000
−75000
−100000
−0.4
−0.2
0.0
0.2
0.4
Displacement (m)


FORCE-DISPLACEMENT CURVE
ELEMENT 04
Force [N]
Displacement (m)
100000
75000
50000
25000
0
−25000
−50000
−75000
−100000
−0.4
−0.2
0.0
0.2
0.4

STATIC ANALYSIS



Spyder (Python 3.12)
File Edit Search Source Run Debug Consoles Projects Tools View Help
...S_04_COLUMNS_8_ANA+\+60_MDOF_10_FLOORS_04_COLUMNS_8_ANA
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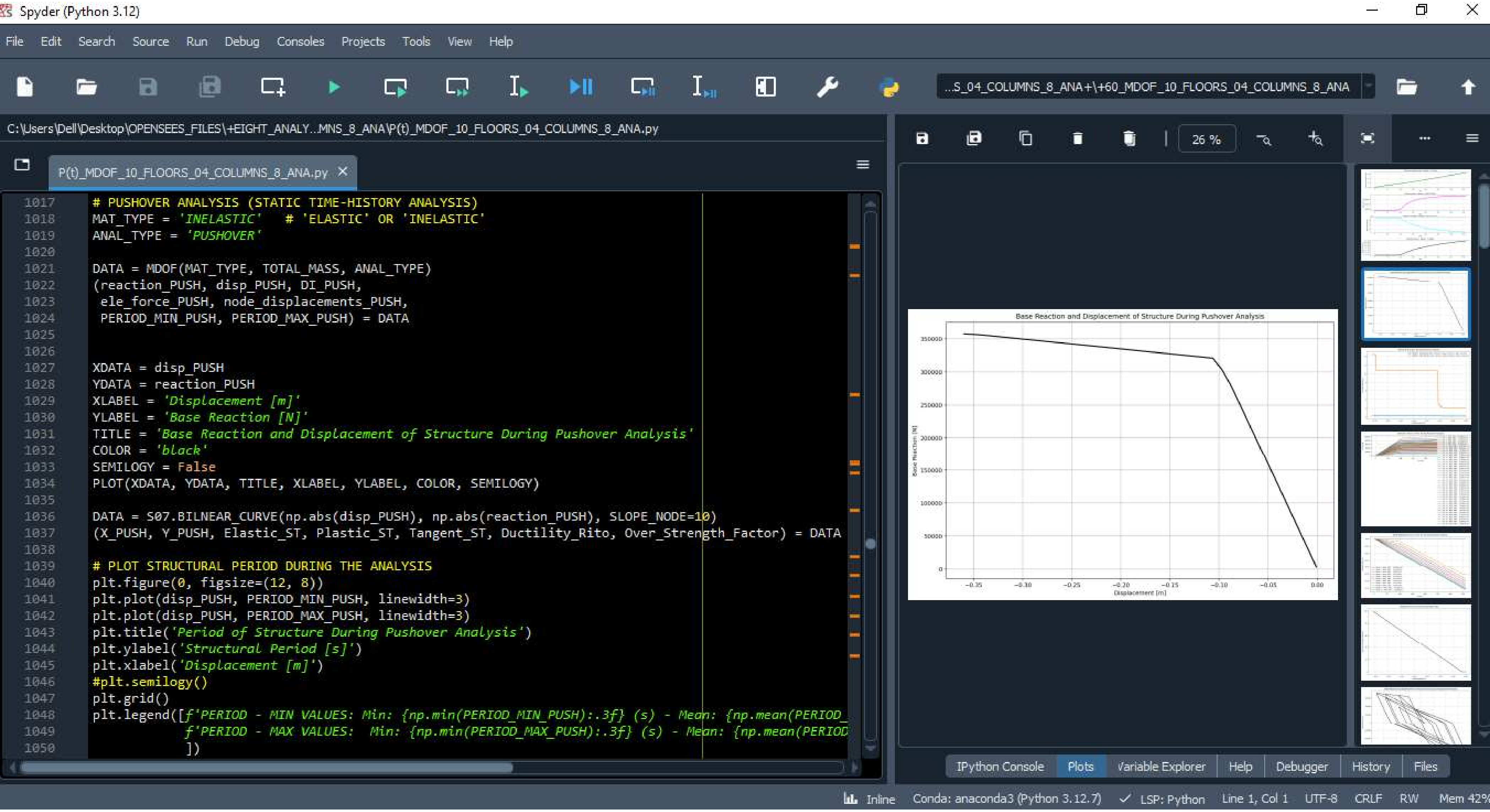
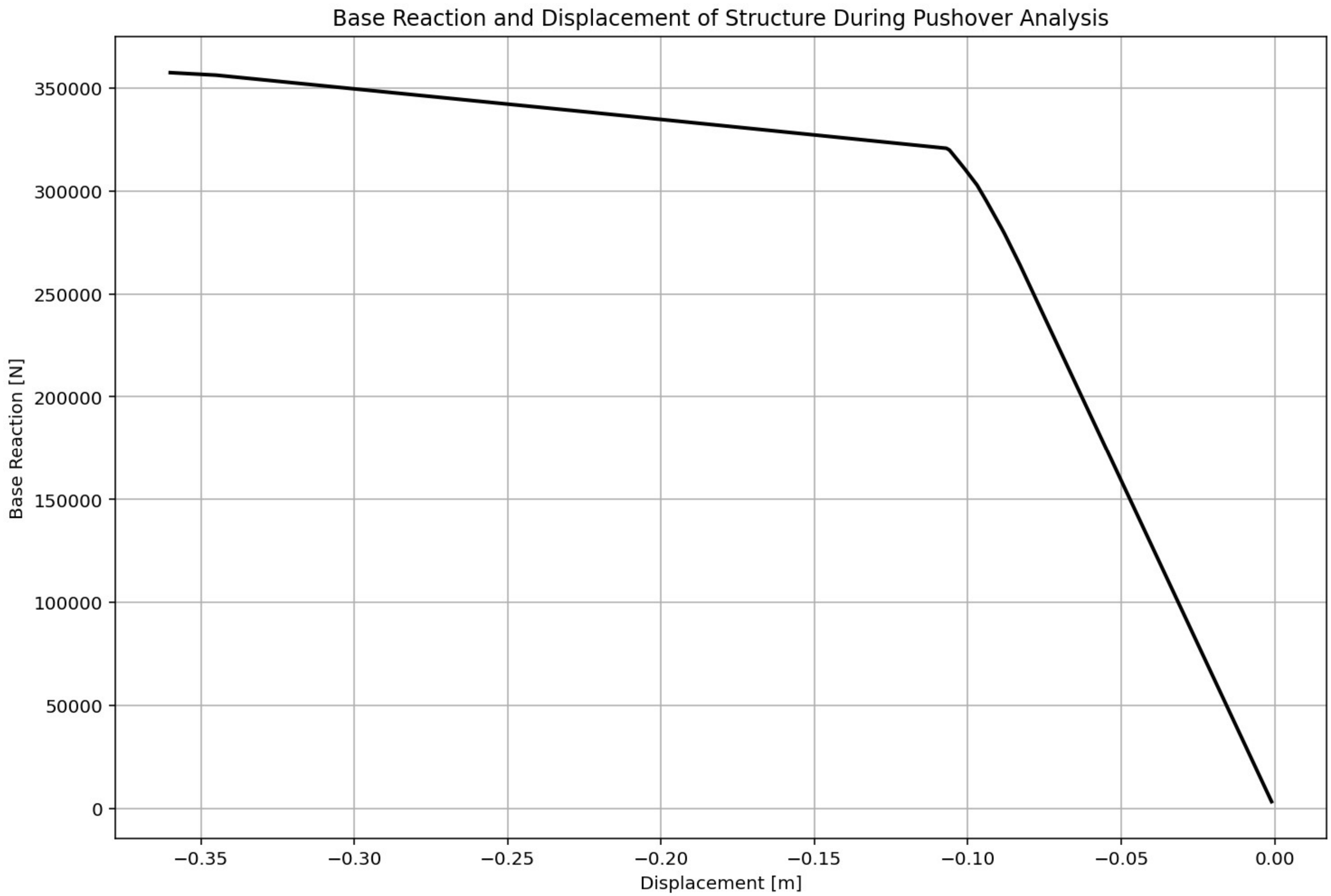
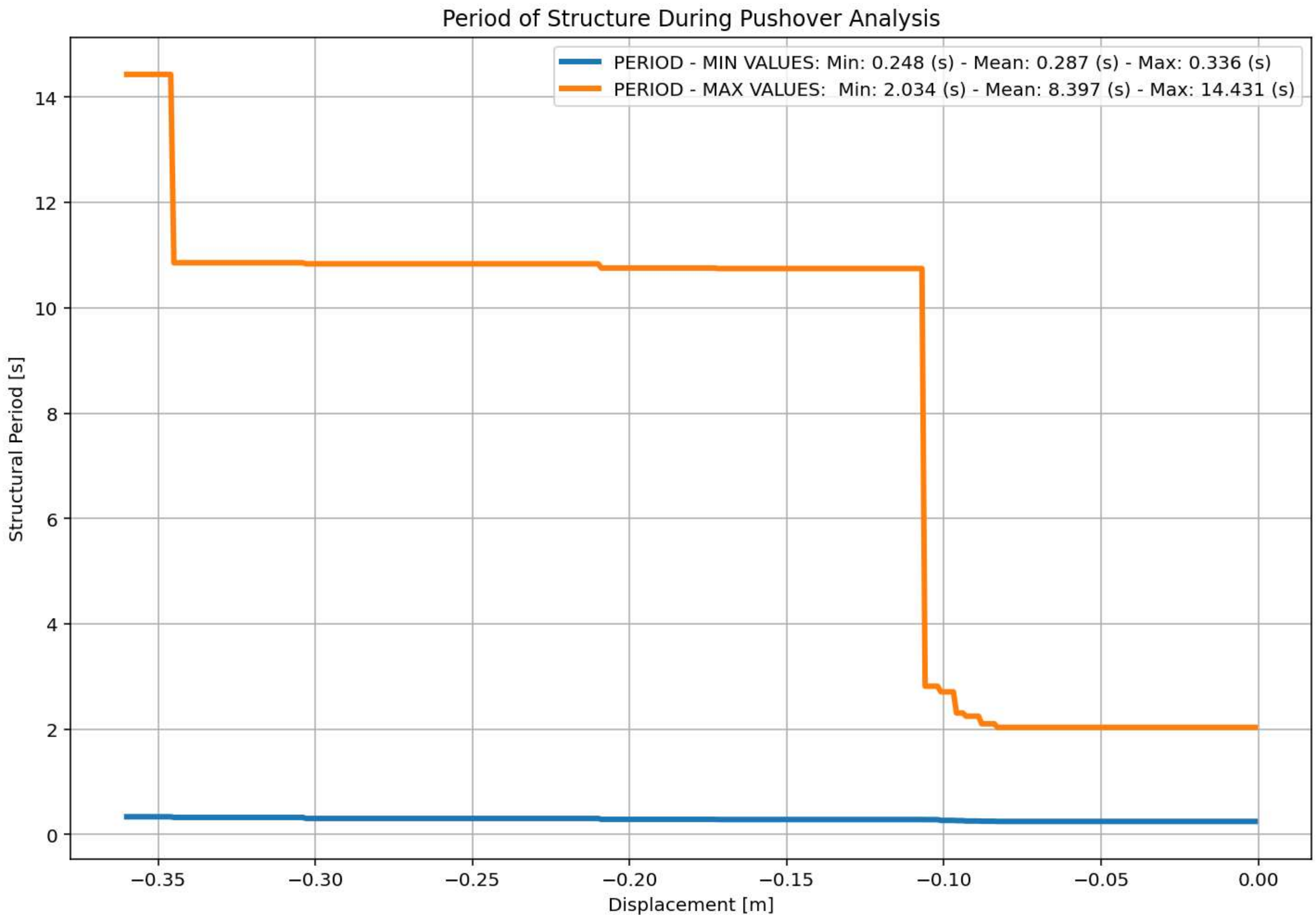
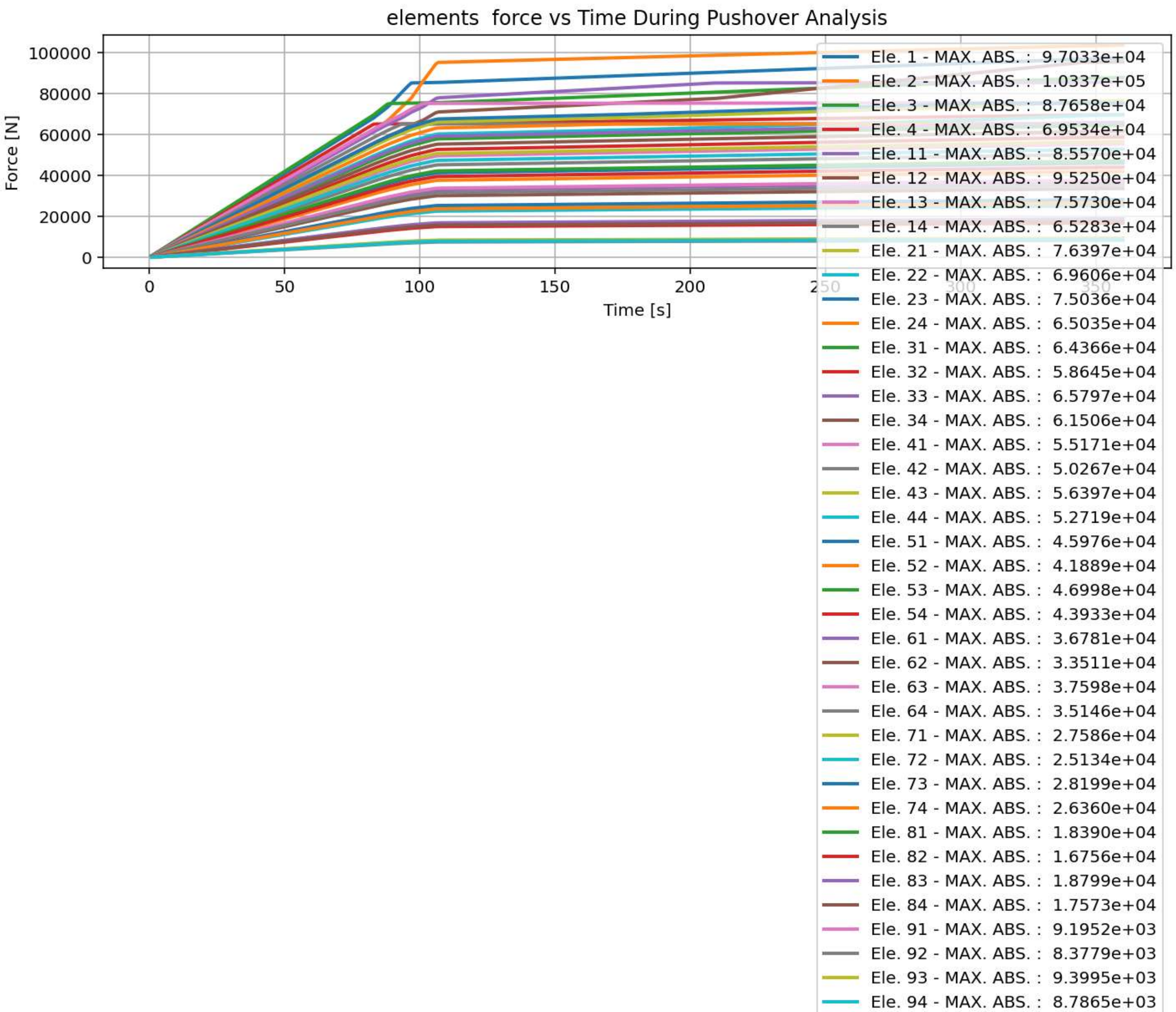
P(t)_MDOF_10_FLOORS_04_COLUMNS_8_ANA.py
#%%--
# STATIC ANALYSIS
MAT_TYPE = 'INELASTIC'   # 'ELASTIC' OR 'INELASTIC'
ANAL_TYPE = 'STATIC'
DATA = MDOF(MAT_TYPE, TOTAL_MASS, ANAL_TYPE)
(reaction, disp, DI) = DATA
#%%--
# PUSHOVER ANALYSIS (STATIC TIME-HISTORY ANALYSIS)
MAT_TYPE = 'INELASTIC'   # 'ELASTIC' OR 'INELASTIC'
ANAL_TYPE = 'PUSHOVER'
DATA = MDOF(MAT_TYPE, TOTAL_MASS, ANAL_TYPE)
(reaction_PUSH, disp_PUSH, DI_PUSH,
 ele_force_PUSH, node_displacements_PUSH,
 PERIOD_MIN_PUSH, PERIOD_MAX_PUSH) = DATA
XDATA = disp_PUSH
YDATA = reaction_PUSH
XLABEL = 'Displacement [m]'
YLABEL = 'Base Reaction [N]'
TITLE = 'Base Reaction and Displacement of Structure During Pushover Analysis'
COLOR = 'black'
SEMILOGY = False
PLOT(XDATA, YDATA, TITLE, XLABEL, YLABEL, COLOR, SEMILOGY)
DATA = S07.BILNEAR_CURVE(np.abs(disp_PUSH), np.abs(reaction_PUSH), SLOPE_NODE=10)
(X_PUSH, Y_PUSH, Elastic_ST, Plastic_ST, Tangent_ST, Ductility_Rito, Over_Strength_Factor) = DATA
PLOT STRUCTURAL PERIOD DURING THE ANALYSIS
23 %
Effective Displacement - Median: 0.15446
Effective Mass - Median: 482523.93674
Effective Stiffness - Median: 4928770.20397
Effective Period - Median: 1.96595
IPython Console Plots Variable Explorer Help Debugger History Files
Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 1, Col 1 UTF-8 CRLF RW Mem 43%

PUSHOVER ANALYSIS

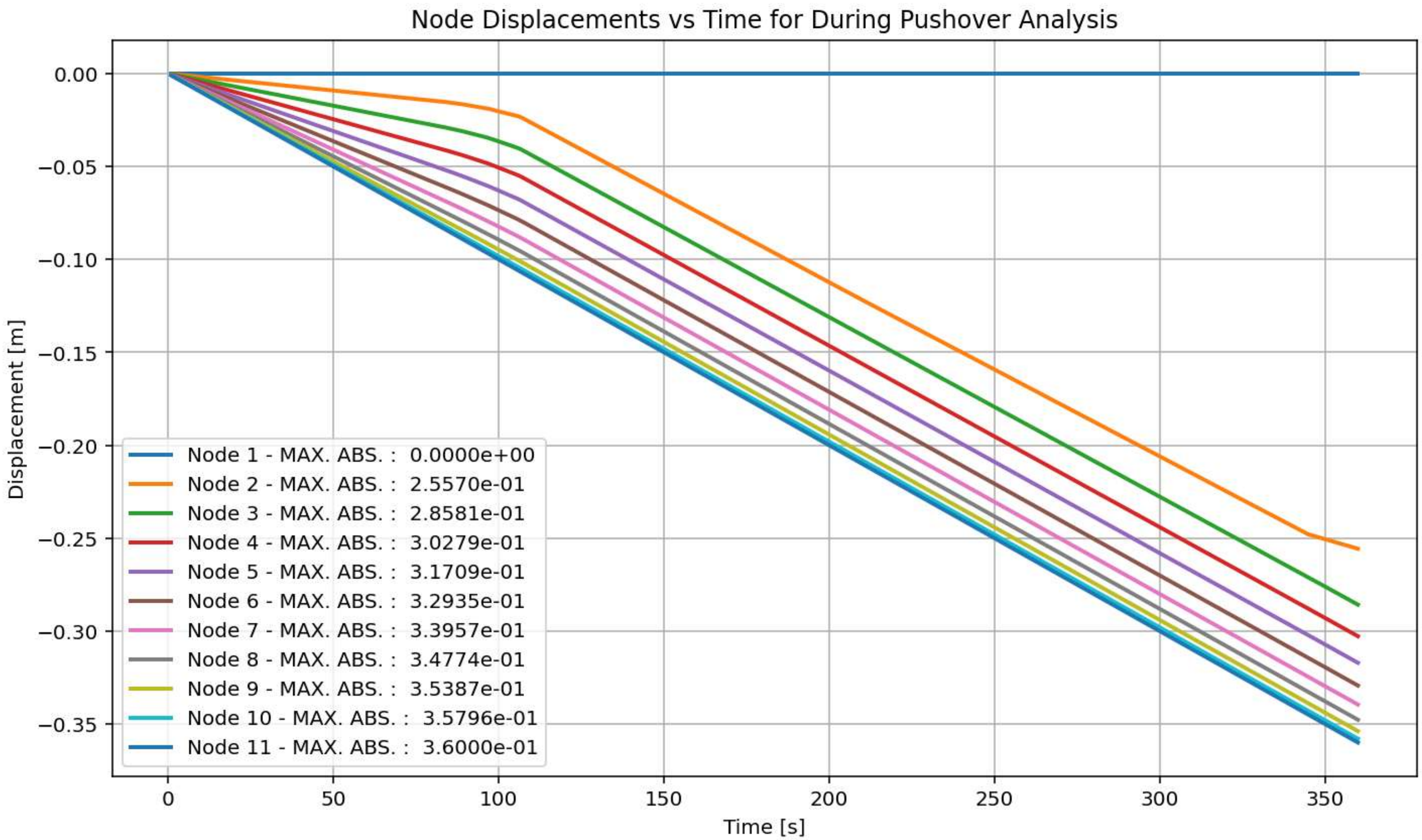

Spyder (Python 3.12)
File Edit Search Source Run Debug Consoles Projects Tools View Help
...S_04_COLUMNS_8_ANA+\+60_MDOF_10_FLOORS_04_COLUMNS_8_ANA
C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...MNS_8_ANA\P(t)_MDOF_10_FLOORS_04_COLUMNS_8_ANA.py
P(t)_MDOF_10_FLOORS_04_COLUMNS_8_ANA.py
# PUSHOVER ANALYSIS (STATIC TIME-HISTORY ANALYSIS)
MAT_TYPE = 'INELASTIC'   # 'ELASTIC' OR 'INELASTIC'
ANAL_TYPE = 'PUSHOVER'
DATA = MDOF(MAT_TYPE, TOTAL_MASS, ANAL_TYPE)
(reaction_PUSH, disp_PUSH, DI_PUSH,
 ele_force_PUSH, node_displacements_PUSH,
 PERIOD_MIN_PUSH, PERIOD_MAX_PUSH) = DATA
XDATA = disp_PUSH
YDATA = reaction_PUSH
XLABEL = 'Displacement [m]'
YLABEL = 'Base Reaction [N]'
TITLE = 'Base Reaction and Displacement of Structure During Pushover Analysis'
COLOR = 'black'
SEMILOGY = False
PLOT(XDATA, YDATA, TITLE, XLABEL, YLABEL, COLOR, SEMILOGY)
DATA = S07.BILNEAR_CURVE(np.abs(disp_PUSH), np.abs(reaction_PUSH), SLOPE_NODE=10)
(X_PUSH, Y_PUSH, Elastic_ST, Plastic_ST, Tangent_ST, Ductility_Rito, Over_Strength_Factor) = DATA
# PLOT STRUCTURAL PERIOD DURING THE ANALYSIS
plt.figure(0, figsize=(12, 8))
plt.plot(disp_PUSH, PERIOD_MIN_PUSH, linewidth=3)
plt.plot(disp_PUSH, PERIOD_MAX_PUSH, linewidth=3)
plt.title('Period of Structure During Pushover Analysis')
plt.ylabel('Structural Period [s]')
plt.xlabel('Displacement [m]')
#plt.semilogy()
plt.grid()
plt.legend([f'PERIOD - MIN VALUES: Min: {np.min(PERIOD_MIN_PUSH):.3f} (s) - Mean: {np.mean(PERIOD_
            f'PERIOD - MAX VALUES: Min: {np.min(PERIOD_MAX_PUSH):.3f} (s) - Mean: {np.mean(PERIOD
            ])
26 %
Base Reaction and Displacement of Structure During Pushover Analysis
350000
300000
250000
200000
150000
100000
50000
0
−0.35
−0.30
−0.25
−0.20
−0.15
−0.10
−0.05
0.00
Displacement [m]
IPython Console
Plots
Variable Explorer
Help
Debugger
History
Files
Inline
Conda: anaconda3 (Python 3.12.7)
LSP: Python
Line 1, Col 1
UTF-8
CRLF
RW
Mem 42%


Base Reaction and Displacement of Structure During Pushover Analysis
Base Reaction [N]
350000
300000
250000
200000
150000
100000
50000
0
−0.35
−0.30
−0.25
−0.20
−0.15
−0.10
−0.05
0.00
Displacement [m]

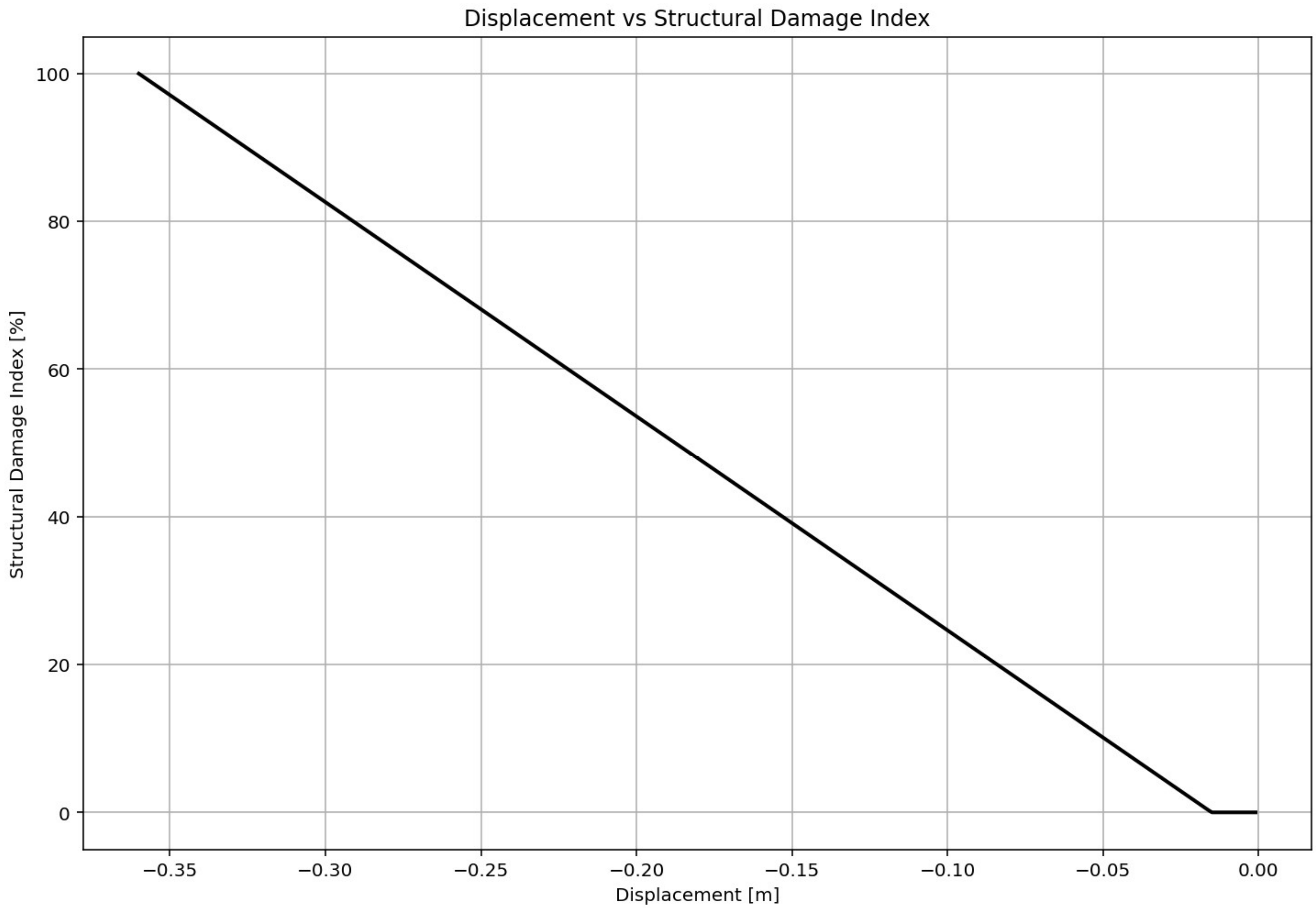

Period of Structure During Pushover Analysis
PERIOD - MIN VALUES: Min: 0.248 (s) - Mean: 0.287 (s) - Max: 0.336 (s)
PERIOD - MAX VALUES: Min: 2.034 (s) - Mean: 8.397 (s) - Max: 14.431 (s)
Structural Period [s]
Displacement [m]
14
12
10
8
6
4
2
0
−0.35
−0.30
−0.25
−0.20
−0.15
−0.10
−0.05
0.00



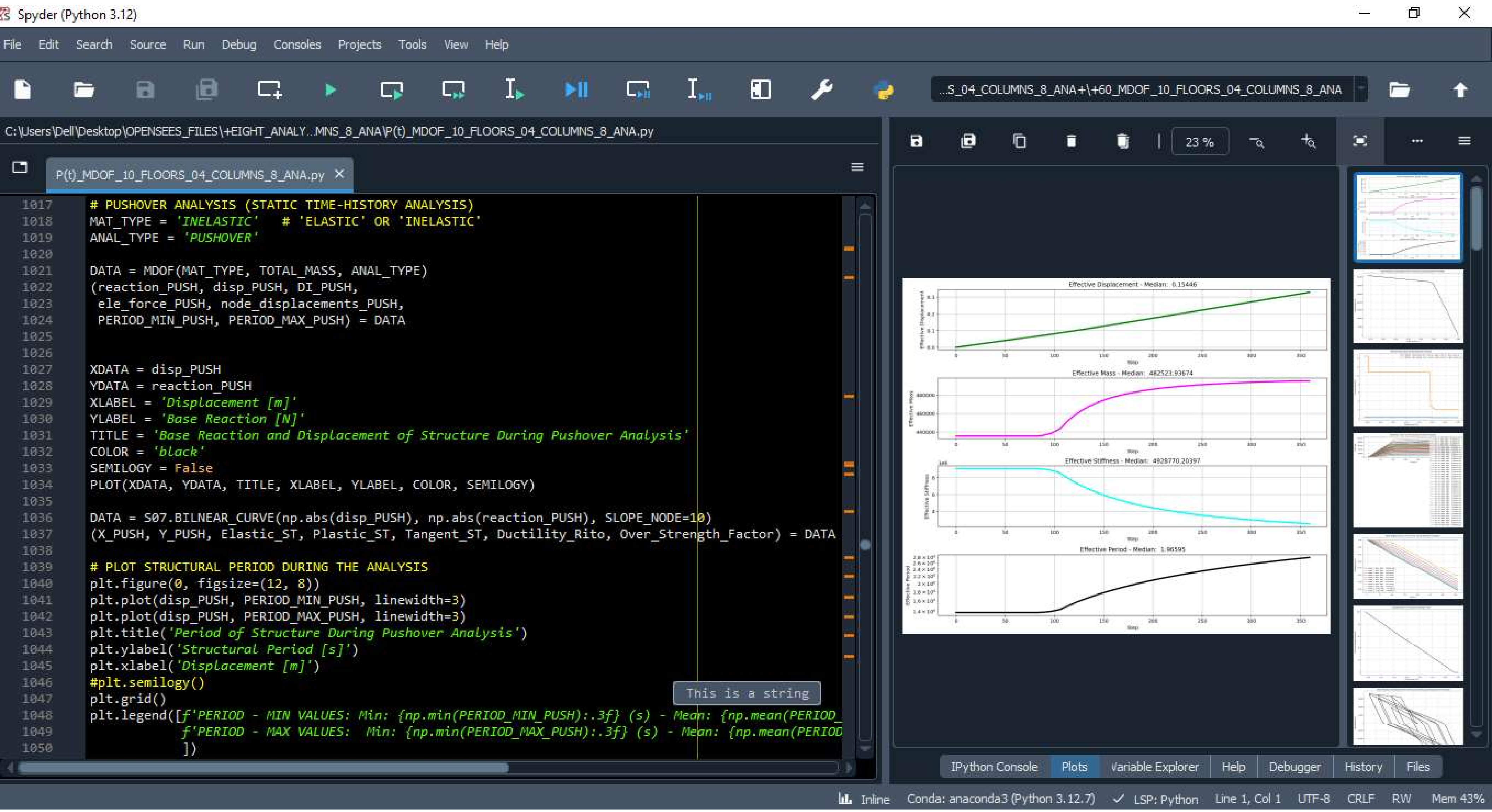
elements force vs Time During Pushover Analysis
Force [N]
Time [s]
0
20000
40000
60000
80000
100000
0
50
100
150
200
250
300
350
Ele. 1 - MAX. ABS. : 9.7033e+04
Ele. 2 - MAX. ABS. : 1.0337e+05
Ele. 3 - MAX. ABS. : 8.7658e+04
Ele. 4 - MAX. ABS. : 6.9534e+04
Ele. 11 - MAX. ABS. : 8.5570e+04
Ele. 12 - MAX. ABS. : 9.5250e+04
Ele. 13 - MAX. ABS. : 7.5730e+04
Ele. 14 - MAX. ABS. : 6.5283e+04
Ele. 21 - MAX. ABS. : 7.6397e+04
Ele. 22 - MAX. ABS. : 6.9606e+04
Ele. 23 - MAX. ABS. : 7.5036e+04
Ele. 24 - MAX. ABS. : 6.5035e+04
Ele. 31 - MAX. ABS. : 6.4366e+04
Ele. 32 - MAX. ABS. : 5.8645e+04
Ele. 33 - MAX. ABS. : 6.5797e+04
Ele. 34 - MAX. ABS. : 6.1506e+04
Ele. 41 - MAX. ABS. : 5.5171e+04
Ele. 42 - MAX. ABS. : 5.0267e+04
Ele. 43 - MAX. ABS. : 5.6397e+04
Ele. 44 - MAX. ABS. : 5.2719e+04
Ele. 51 - MAX. ABS. : 4.5976e+04
Ele. 52 - MAX. ABS. : 4.1889e+04
Ele. 53 - MAX. ABS. : 4.6998e+04
Ele. 54 - MAX. ABS. : 4.3933e+04
Ele. 61 - MAX. ABS. : 3.6781e+04
Ele. 62 - MAX. ABS. : 3.3511e+04
Ele. 63 - MAX. ABS. : 3.7598e+04
Ele. 64 - MAX. ABS. : 3.5146e+04
Ele. 71 - MAX. ABS. : 2.7586e+04
Ele. 72 - MAX. ABS. : 2.5134e+04
Ele. 73 - MAX. ABS. : 2.8199e+04
Ele. 74 - MAX. ABS. : 2.6360e+04
Ele. 81 - MAX. ABS. : 1.8390e+04
Ele. 82 - MAX. ABS. : 1.6756e+04
Ele. 83 - MAX. ABS. : 1.8799e+04
Ele. 84 - MAX. ABS. : 1.7573e+04
Ele. 91 - MAX. ABS. : 9.1952e+03
Ele. 92 - MAX. ABS. : 8.3779e+03
Ele. 93 - MAX. ABS. : 9.3995e+03
Ele. 94 - MAX. ABS. : 8.7865e+03



Node Displacements vs Time for During Pushover Analysis
Displacement [m]
Time [s]
0.00
−0.05
−0.10
−0.15
−0.20
−0.25
−0.30
−0.35
0
50
100
150
200
250
300
350
Node 1 - MAX. ABS. : 0.0000e+00
Node 2 - MAX. ABS. : 2.5570e-01
Node 3 - MAX. ABS. : 2.8581e-01
Node 4 - MAX. ABS. : 3.0279e-01
Node 5 - MAX. ABS. : 3.1709e-01
Node 6 - MAX. ABS. : 3.2935e-01
Node 7 - MAX. ABS. : 3.3957e-01
Node 8 - MAX. ABS. : 3.4774e-01
Node 9 - MAX. ABS. : 3.5387e-01
Node 10 - MAX. ABS. : 3.5796e-01
Node 11 - MAX. ABS. : 3.6000e-01



Displacement vs Structural Damage Index
Structural Damage Index [%]
100
80
60
40
20
0
−0.35
−0.30
−0.25
−0.20
−0.15
−0.10
−0.05
0.00
Displacement [m]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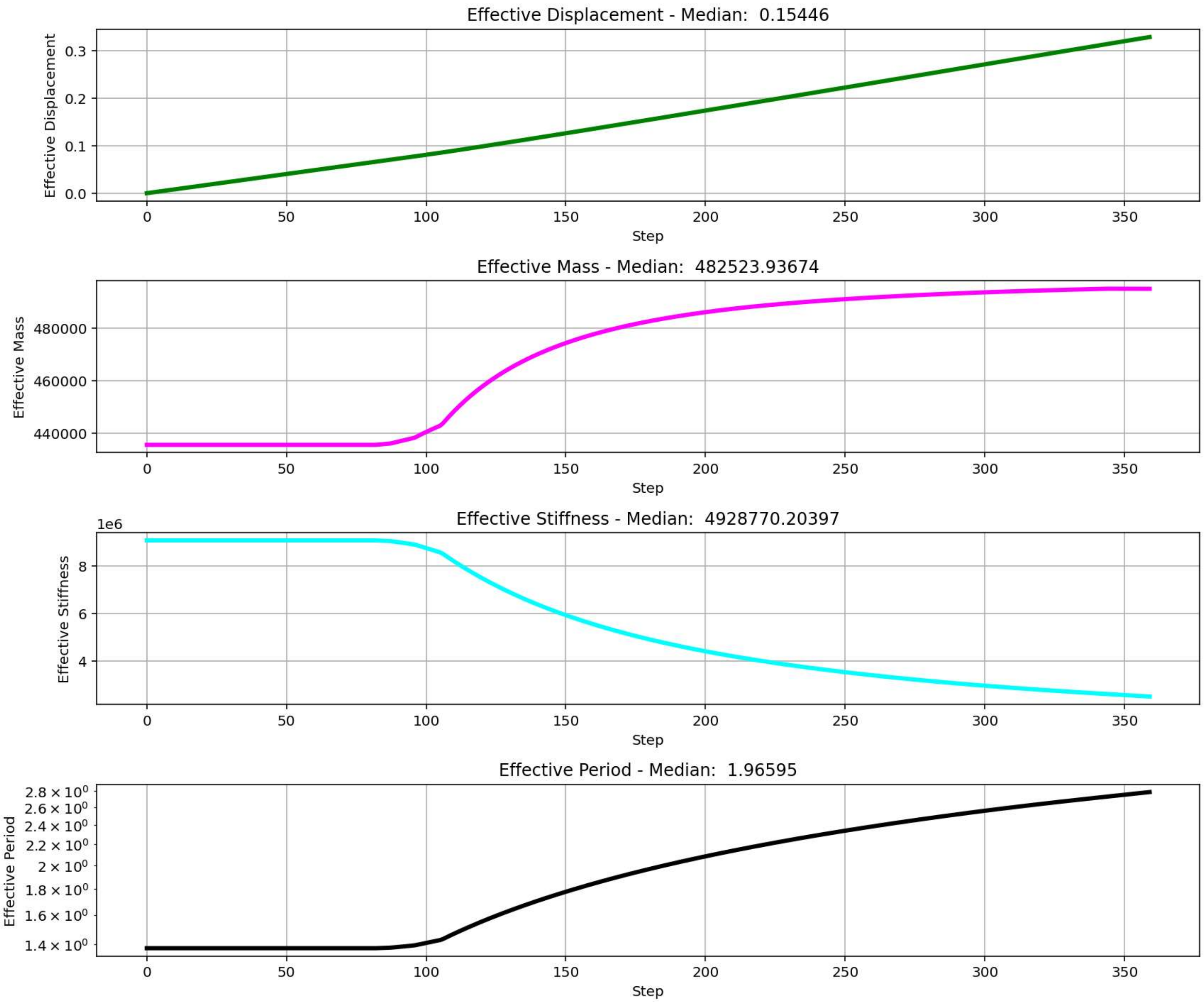
C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...MNS_8_ANA\P(t)_MDOF_10_FLOORS_04_COLUMNS_8_ANA.py
P(t)_MDOF_10_FLOORS_04_COLUMNS_8_ANA.py
# PUSHOVER ANALYSIS (STATIC TIME-HISTORY ANALYSIS)
MAT_TYPE = 'INELASTIC'   # 'ELASTIC' OR 'INELASTIC'
ANAL_TYPE = 'PUSHOVER'
DATA = MDOF(MAT_TYPE, TOTAL_MASS, ANAL_TYPE)
(reaction_PUSH, disp_PUSH, DI_PUSH,
 ele_force_PUSH, node_displacements_PUSH,
 PERIOD_MIN_PUSH, PERIOD_MAX_PUSH) = DATA
XDATA = disp_PUSH
YDATA = reaction_PUSH
XLABEL = 'Displacement [m]'
YLABEL = 'Base Reaction [N]'
TITLE = 'Base Reaction and Displacement of Structure During Pushover Analysis'
COLOR = 'black'
SEMILOGY = False
PLOT(XDATA, YDATA, TITLE, XLABEL, YLABEL, COLOR, SEMILOGY)
DATA = S07.BILNEAR_CURVE(np.abs(disp_PUSH), np.abs(reaction_PUSH), SLOPE_NODE=10)
(X_PUSH, Y_PUSH, Elastic_ST, Plastic_ST, Tangent_ST, Ductility_Rito, Over_Strength_Factor) = DATA
# PLOT STRUCTURAL PERIOD DURING THE ANALYSIS
plt.figure(0, figsize=(12, 8))
plt.plot(disp_PUSH, PERIOD_MIN_PUSH, linewidth=3)
plt.plot(disp_PUSH, PERIOD_MAX_PUSH, linewidth=3)
plt.title('Period of Structure During Pushover Analysis')
plt.ylabel('Structural Period [s]')
plt.xlabel('Displacement [m]')
#plt.semilogy()
plt.grid()
plt.legend([f'PERIOD - MIN VALUES: Min: {np.min(PERIOD_MIN_PUSH):.3f} (s) - Mean: {np.mean(PERIOD_
            f'PERIOD - MAX VALUES: Min: {np.min(PERIOD_MAX_PUSH):.3f} (s) - Mean: {np.mean(PERIOD
            ])
This is a string
23 %
Effective Displacement - Median: 0.15446
Effective Mass - Median: 482523.93674
Effective Stiffness - Median: 4928770.20397
Effective Period - Median: 1.90595
IPython Console Plots Variable Explorer Help Debugger History Files
Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 1, Col 1 UTF-8 CRLF RW Mem 43%



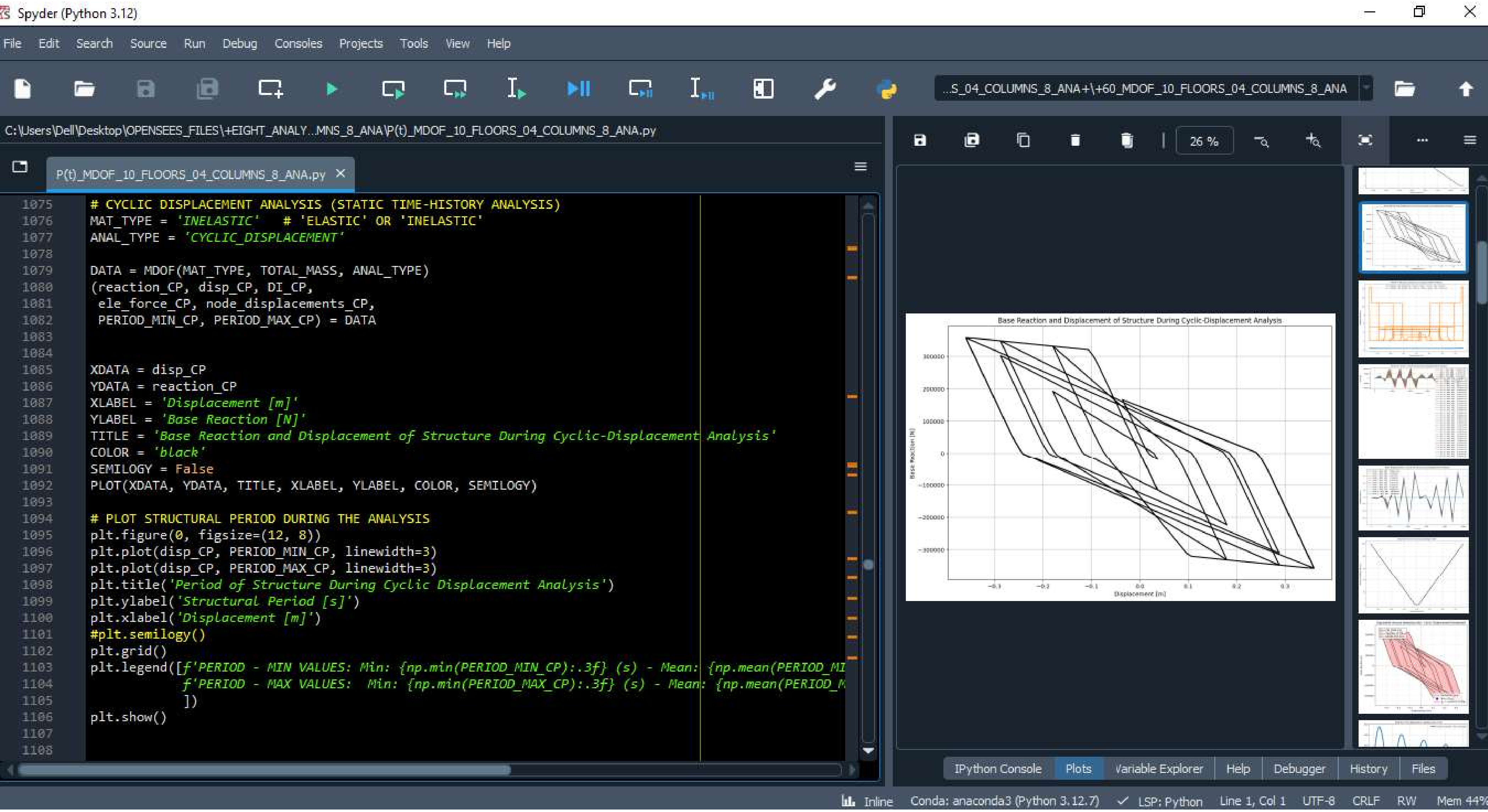
$$\Delta_d = \frac{\sum_{i=1}^{n} m_i \Delta_i^2}{\sum_{i=1}^{n} m_i \Delta_i}$$

$$m_{eff} = \frac{\sum_{i=1}^{n} m_i \Delta_i}{\Delta_d}$$

$$k_{eff} = \frac{\sum_{i=1}^{n} k_i \Delta_i}{\Delta_d}$$

$$T_{eff} = \frac{2\pi}{\sqrt{\frac{k_{eff}}{m_{eff}}}}$$

CYCLIC DISPLACEMENT ANALYSIS


Spyder (Python 3.12)
File Edit Search Source Run Debug Consoles Projects Tools View Help
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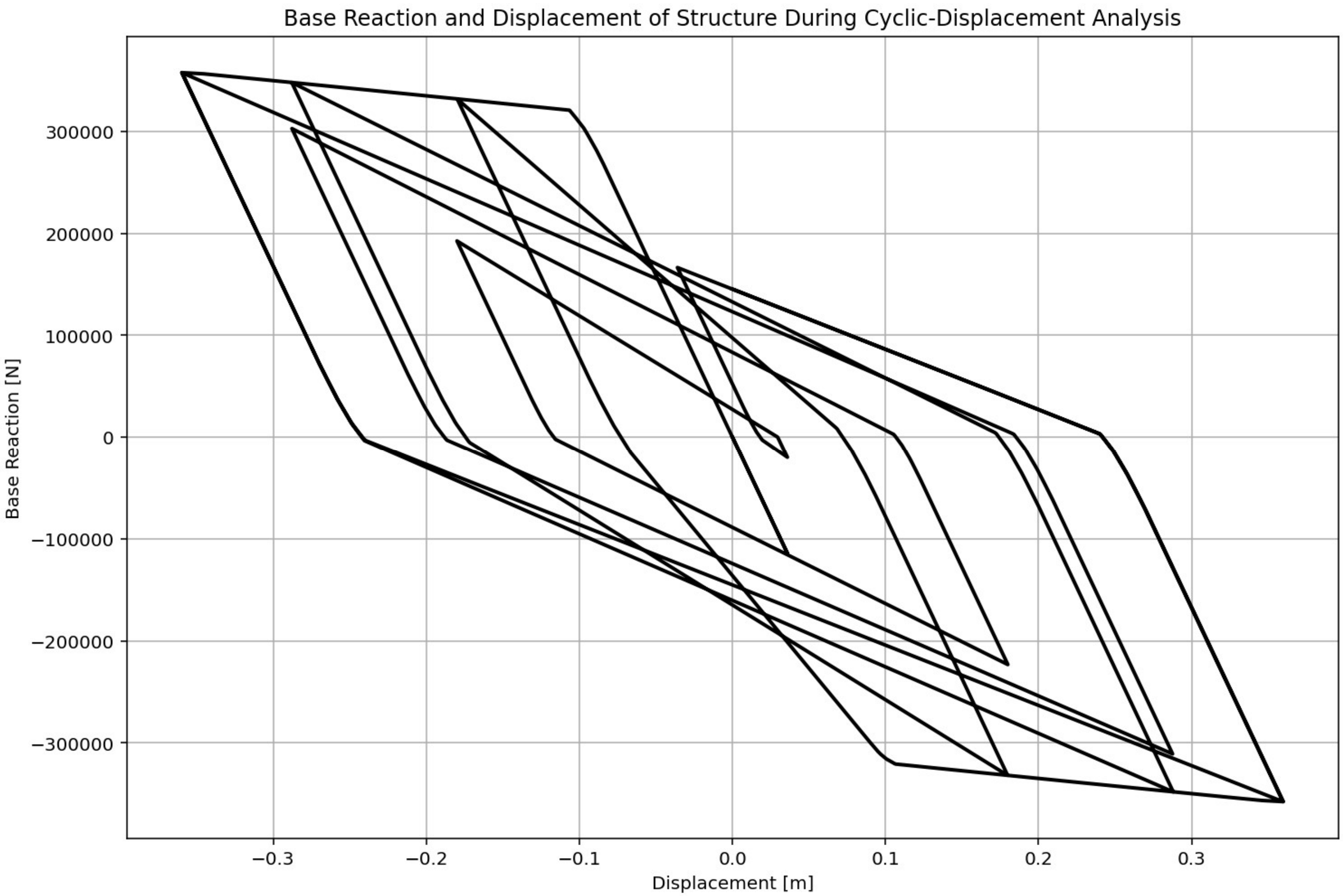
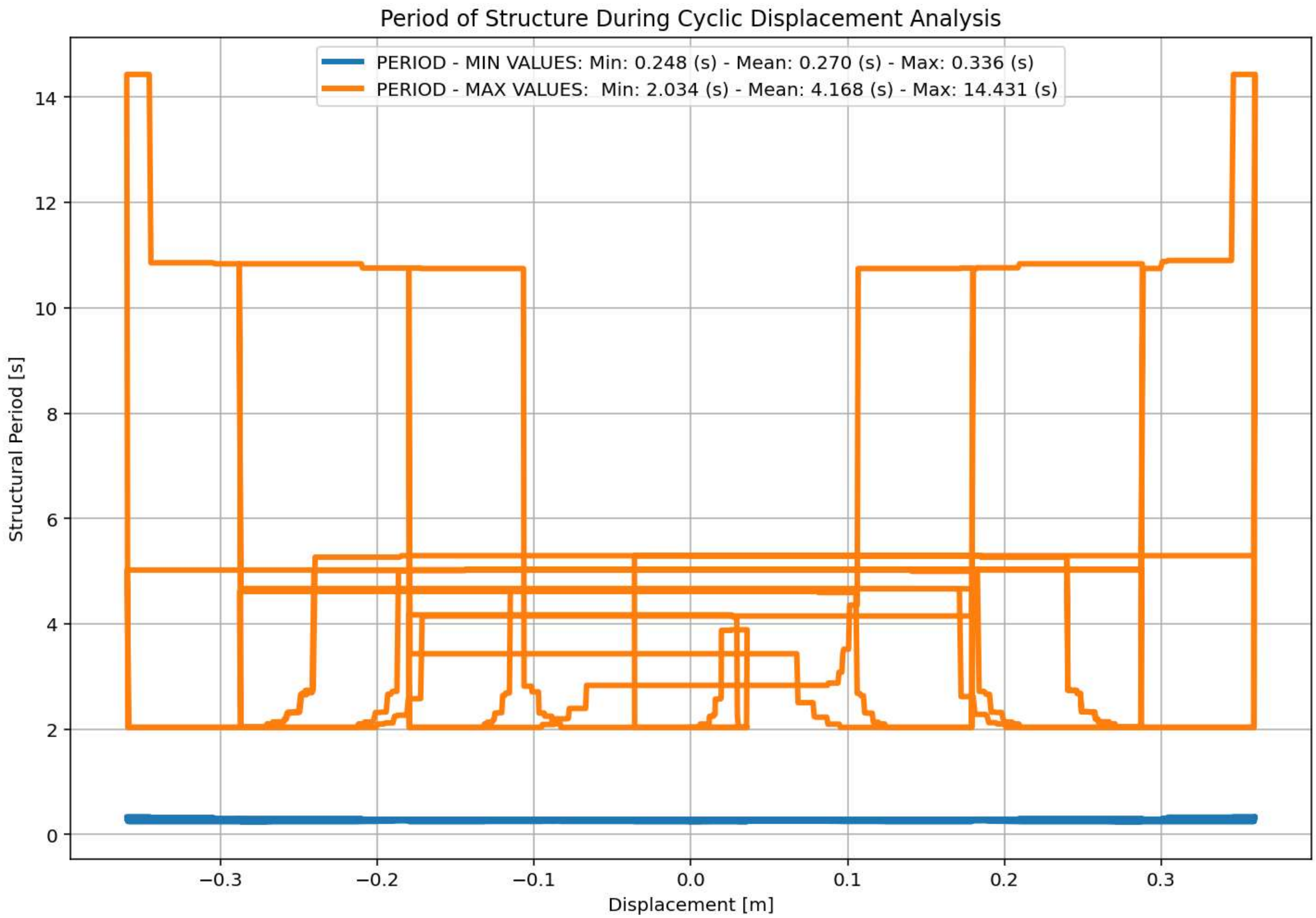
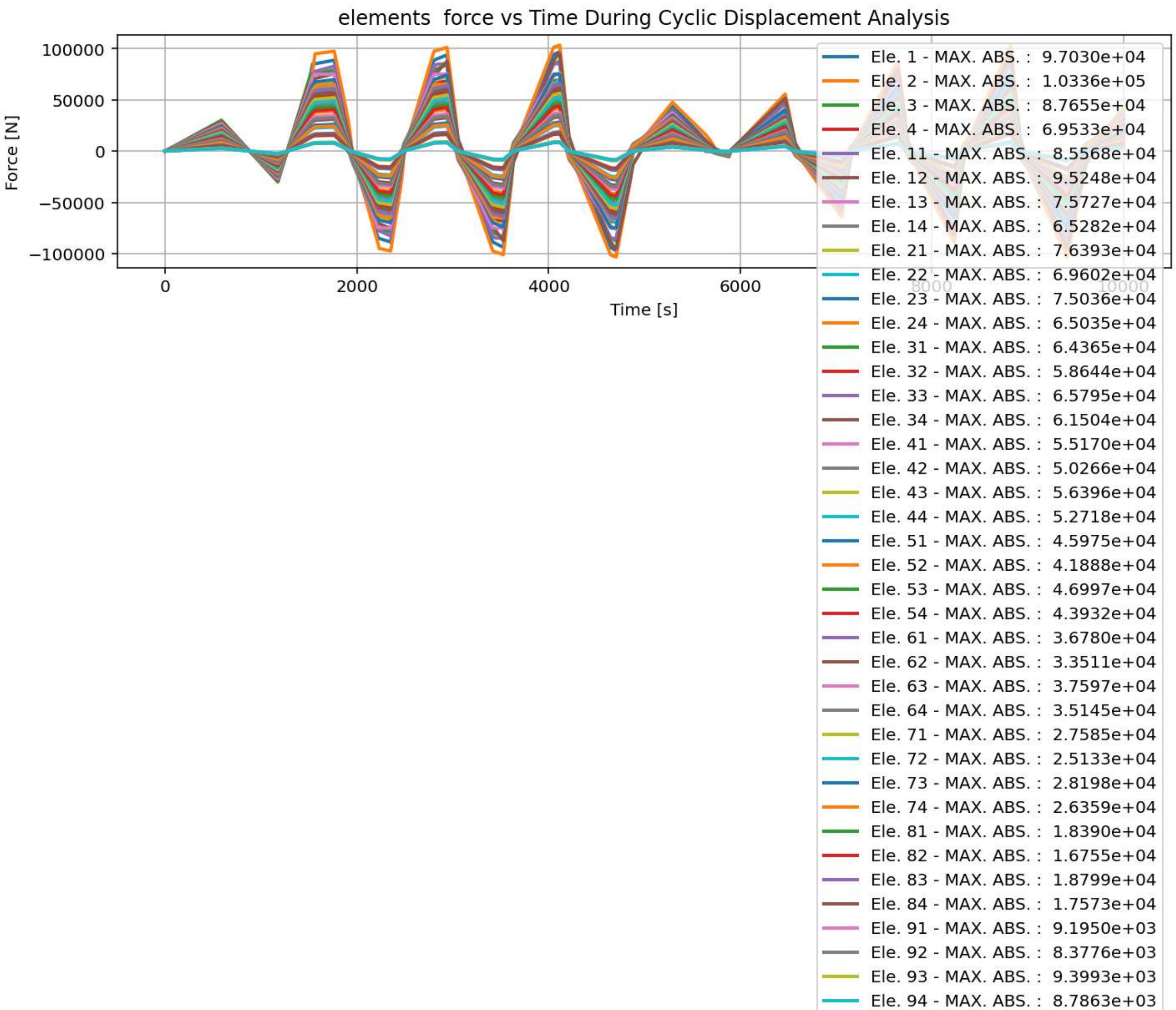
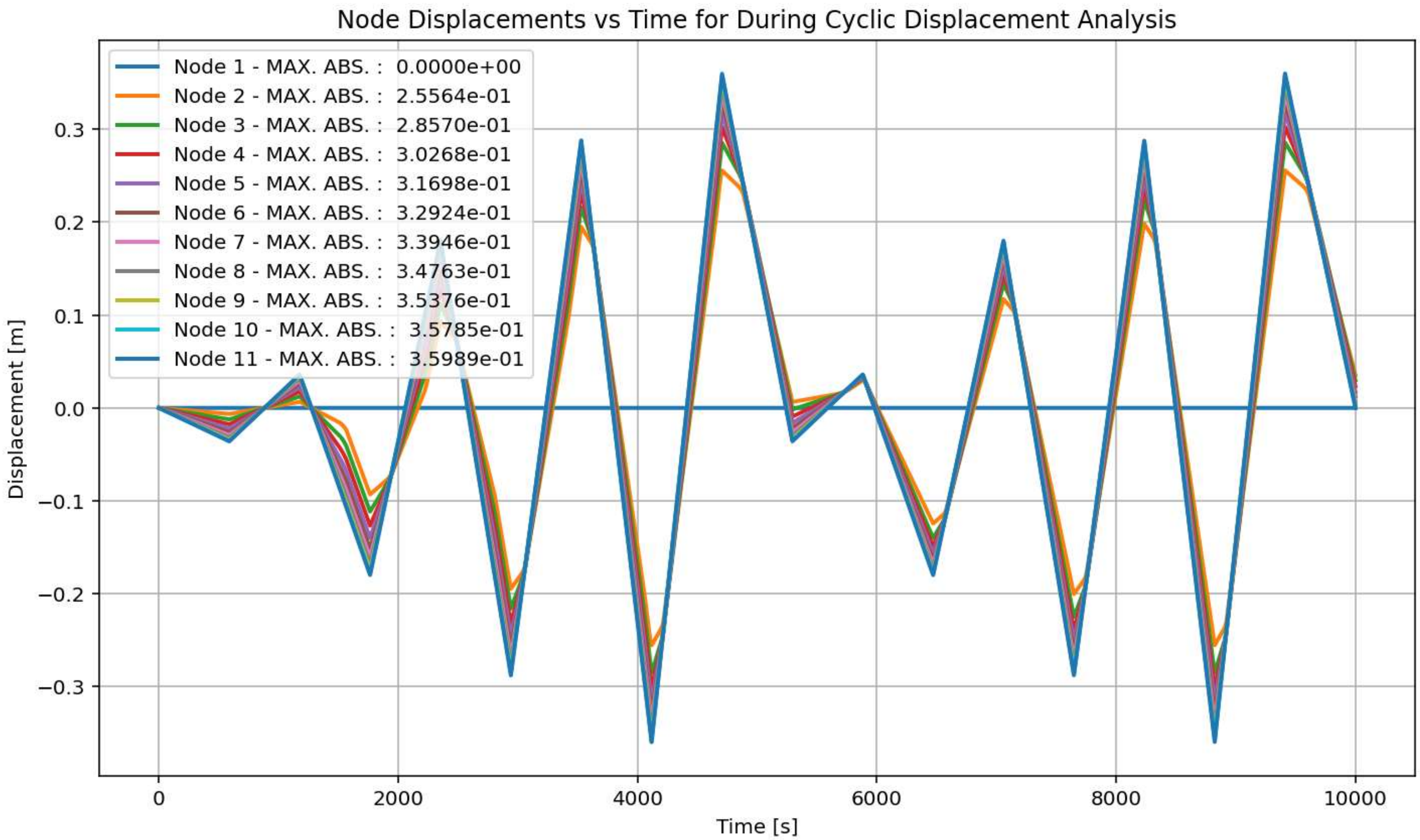
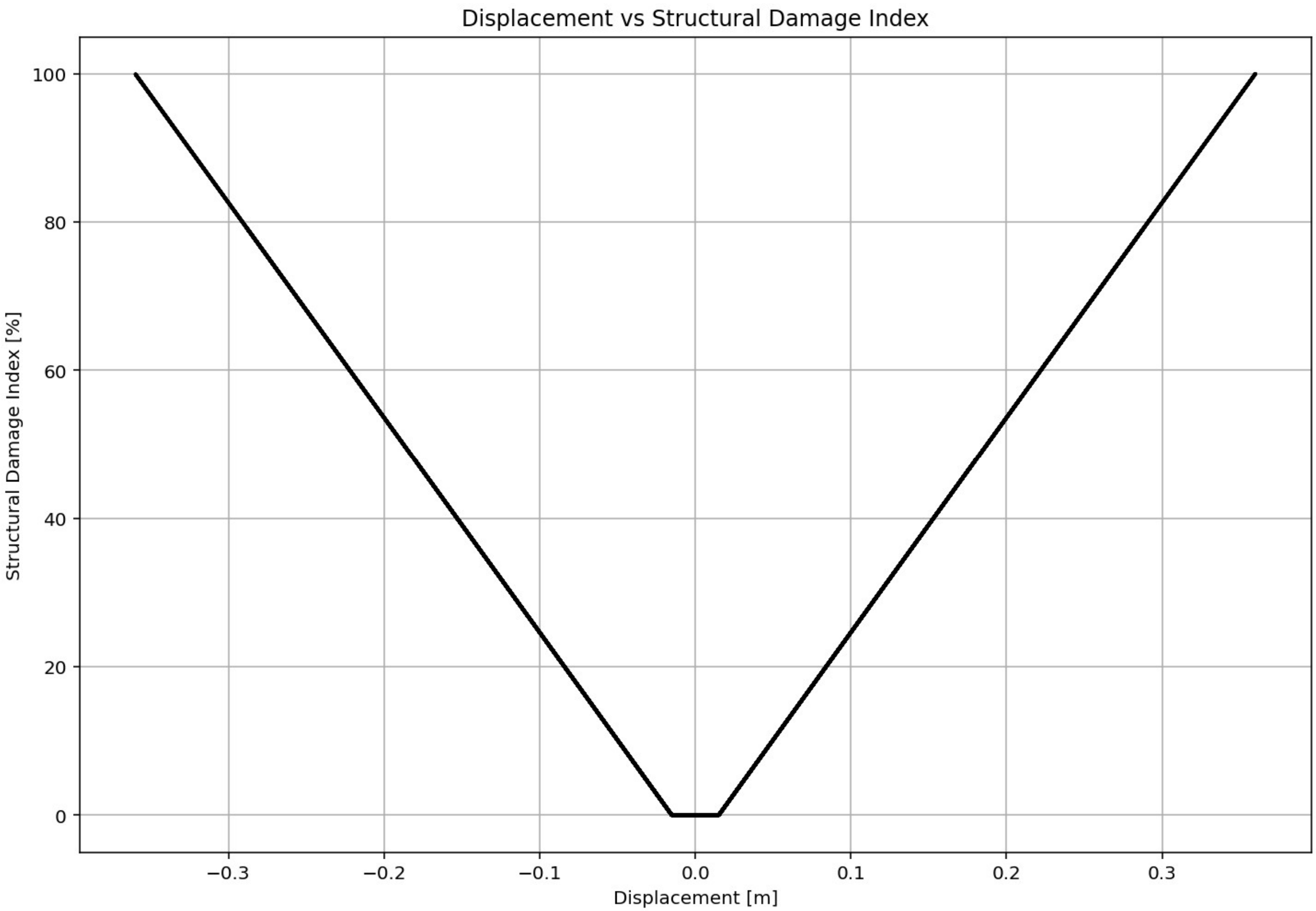
C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...MNS_8_ANA\P(t)_MDOF_10_FLOORS_04_COLUMNS_8_ANA.py
P(t)_MDOF_10_FLOORS_04_COLUMNS_8_ANA.py
# CYCLIC DISPLACEMENT ANALYSIS (STATIC TIME-HISTORY ANALYSIS)
MAT_TYPE = 'INELASTIC'   # 'ELASTIC' OR 'INELASTIC'
ANAL_TYPE = 'CYCLIC_DISPLACEMENT'
DATA = MDOF(MAT_TYPE, TOTAL_MASS, ANAL_TYPE)
(reaction_CP, disp_CP, DI_CP,
 ele_force_CP, node_displacements_CP,
 PERIOD_MIN_CP, PERIOD_MAX_CP) = DATA
XDATA = disp_CP
YDATA = reaction_CP
XLABEL = 'Displacement [m]'
YLABEL = 'Base Reaction [N]'
TITLE = 'Base Reaction and Displacement of Structure During Cyclic-Displacement Analysis'
COLOR = 'black'
SEMILOGY = False
PLOT(XDATA, YDATA, TITLE, XLABEL, YLABEL, COLOR, SEMILOGY)
# PLOT STRUCTURAL PERIOD DURING THE ANALYSIS
plt.figure(0, figsize=(12, 8))
plt.plot(disp_CP, PERIOD_MIN_CP, linewidth=3)
plt.plot(disp_CP, PERIOD_MAX_CP, linewidth=3)
plt.title('Period of Structure During Cyclic Displacement Analysis')
plt.ylabel('Structural Period [s]')
plt.xlabel('Displacement [m]')
#plt.semilogy()
plt.grid()
plt.legend([f'PERIOD - MIN VALUES: Min: {np.min(PERIOD_MIN_CP):.3f} (s) - Mean: {np.mean(PERIOD_MI
f'PERIOD - MAX VALUES: Min: {np.min(PERIOD_MAX_CP):.3f} (s) - Mean: {np.mean(PERIOD_M
])
plt.show()
26 %
Base Reaction and Displacement of Structure During Cyclic-Displacement Analysis
Displacement [m]
IPython Console Plots Variable Explorer Help Debugger History Files
Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 1, Col 1 UTF-8 CRLF RW Mem 44%

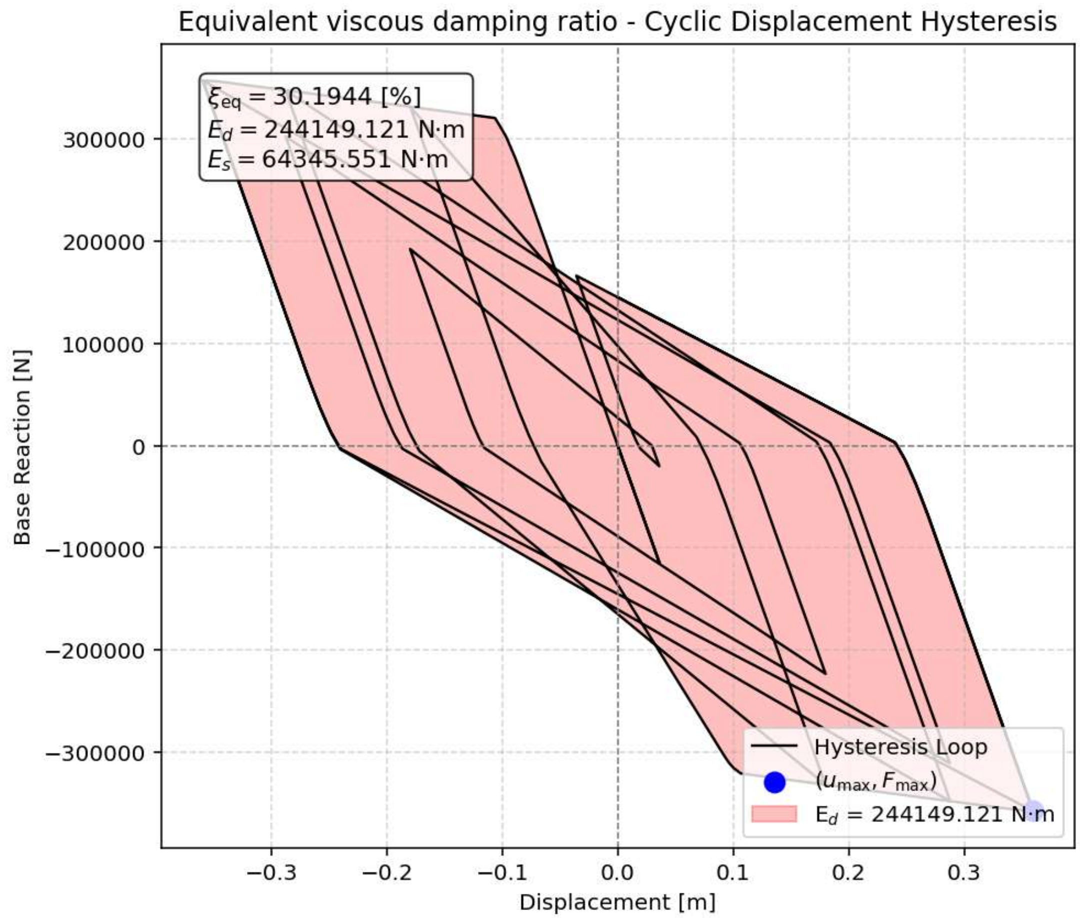

Base Reaction and Displacement of Structure During Cyclic-Displacement Analysis
300000
200000
100000
0
−100000
−200000
−300000
Base Reaction [N]
−0.3
−0.2
−0.1
0.0
0.1
0.2
0.3
Displacement [m]


Period of Structure During Cyclic Displacement Analysis
PERIOD - MIN VALUES: Min: 0.248 (s) - Mean: 0.270 (s) - Max: 0.336 (s)
PERIOD - MAX VALUES: Min: 2.034 (s) - Mean: 4.168 (s) - Max: 14.431 (s)
Structural Period [s]
Displacement [m]
0
2
4
6
8
10
12
14
−0.3
−0.2
−0.1
0.0
0.1
0.2
0.3


elements force vs Time During Cyclic Displacement Analysis
100000
50000
0
−50000
−100000
Force [N]
0
2000
4000
6000
Time [s]
Ele. 1 - MAX. ABS. : 9.7030e+04
Ele. 2 - MAX. ABS. : 1.0336e+05
Ele. 3 - MAX. ABS. : 8.7655e+04
Ele. 4 - MAX. ABS. : 6.9533e+04
Ele. 11 - MAX. ABS. : 8.5568e+04
Ele. 12 - MAX. ABS. : 9.5248e+04
Ele. 13 - MAX. ABS. : 7.5727e+04
Ele. 14 - MAX. ABS. : 6.5282e+04
Ele. 21 - MAX. ABS. : 7.6393e+04
Ele. 22 - MAX. ABS. : 6.9602e+04
Ele. 23 - MAX. ABS. : 7.5036e+04
Ele. 24 - MAX. ABS. : 6.5035e+04
Ele. 31 - MAX. ABS. : 6.4365e+04
Ele. 32 - MAX. ABS. : 5.8644e+04
Ele. 33 - MAX. ABS. : 6.5795e+04
Ele. 34 - MAX. ABS. : 6.1504e+04
Ele. 41 - MAX. ABS. : 5.5170e+04
Ele. 42 - MAX. ABS. : 5.0266e+04
Ele. 43 - MAX. ABS. : 5.6396e+04
Ele. 44 - MAX. ABS. : 5.2718e+04
Ele. 51 - MAX. ABS. : 4.5975e+04
Ele. 52 - MAX. ABS. : 4.1888e+04
Ele. 53 - MAX. ABS. : 4.6997e+04
Ele. 54 - MAX. ABS. : 4.3932e+04
Ele. 61 - MAX. ABS. : 3.6780e+04
Ele. 62 - MAX. ABS. : 3.3511e+04
Ele. 63 - MAX. ABS. : 3.7597e+04
Ele. 64 - MAX. ABS. : 3.5145e+04
Ele. 71 - MAX. ABS. : 2.7585e+04
Ele. 72 - MAX. ABS. : 2.5133e+04
Ele. 73 - MAX. ABS. : 2.8198e+04
Ele. 74 - MAX. ABS. : 2.6359e+04
Ele. 81 - MAX. ABS. : 1.8390e+04
Ele. 82 - MAX. ABS. : 1.6755e+04
Ele. 83 - MAX. ABS. : 1.8799e+04
Ele. 84 - MAX. ABS. : 1.7573e+04
Ele. 91 - MAX. ABS. : 9.1950e+03
Ele. 92 - MAX. ABS. : 8.3776e+03
Ele. 93 - MAX. ABS. : 9.3993e+03
Ele. 94 - MAX. ABS. : 8.7863e+03

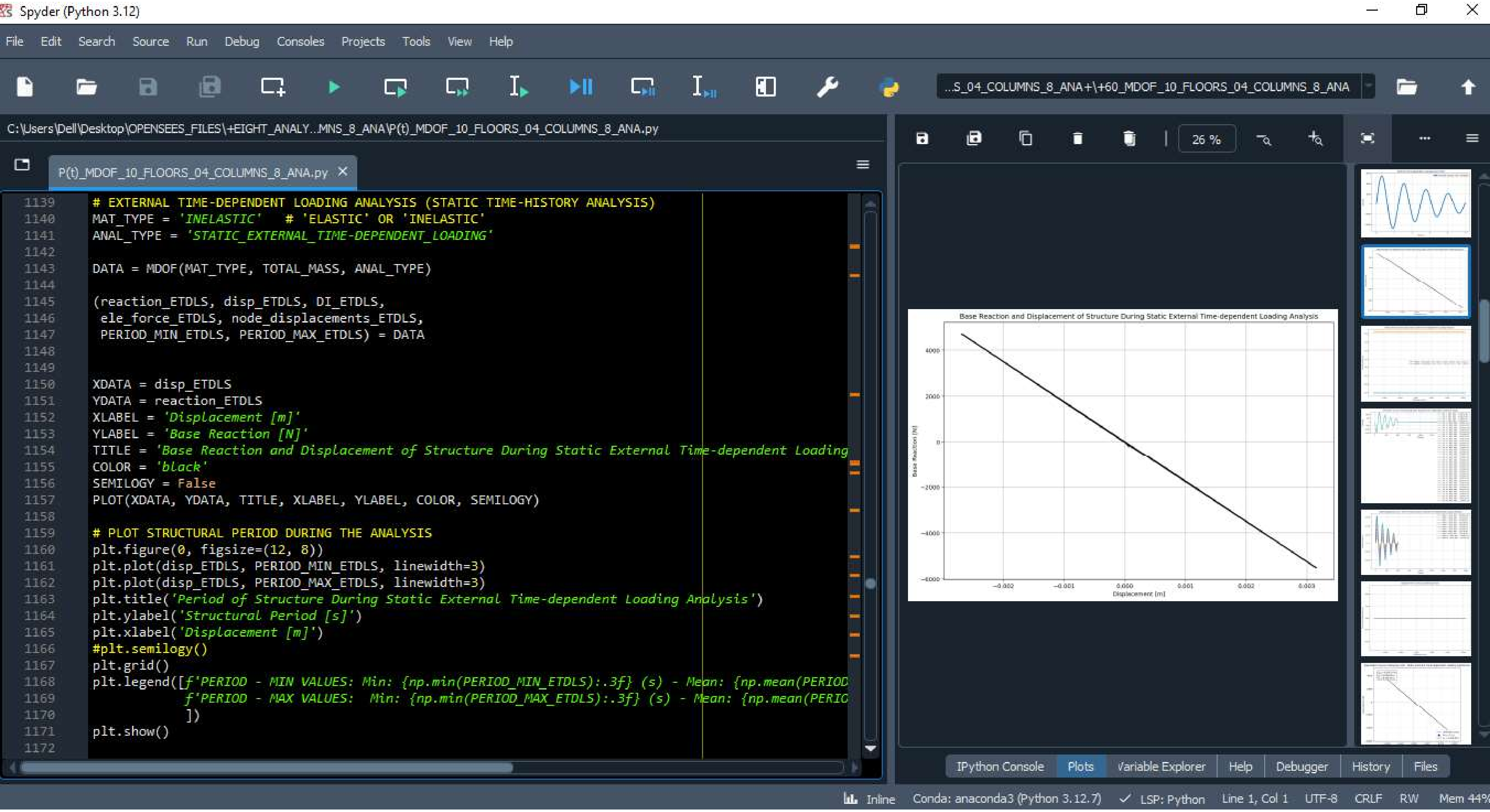

Node Displacements vs Time for During Cyclic Displacement Analysis
Node 1 - MAX. ABS. : 0.0000e+00
Node 2 - MAX. ABS. : 2.5564e-01
Node 3 - MAX. ABS. : 2.8570e-01
Node 4 - MAX. ABS. : 3.0268e-01
Node 5 - MAX. ABS. : 3.1698e-01
Node 6 - MAX. ABS. : 3.2924e-01
Node 7 - MAX. ABS. : 3.3946e-01
Node 8 - MAX. ABS. : 3.4763e-01
Node 9 - MAX. ABS. : 3.5376e-01
Node 10 - MAX. ABS. : 3.5785e-01
Node 11 - MAX. ABS. : 3.5989e-01
Displacement [m]
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
0
2000
4000
6000
8000
10000
Time [s]

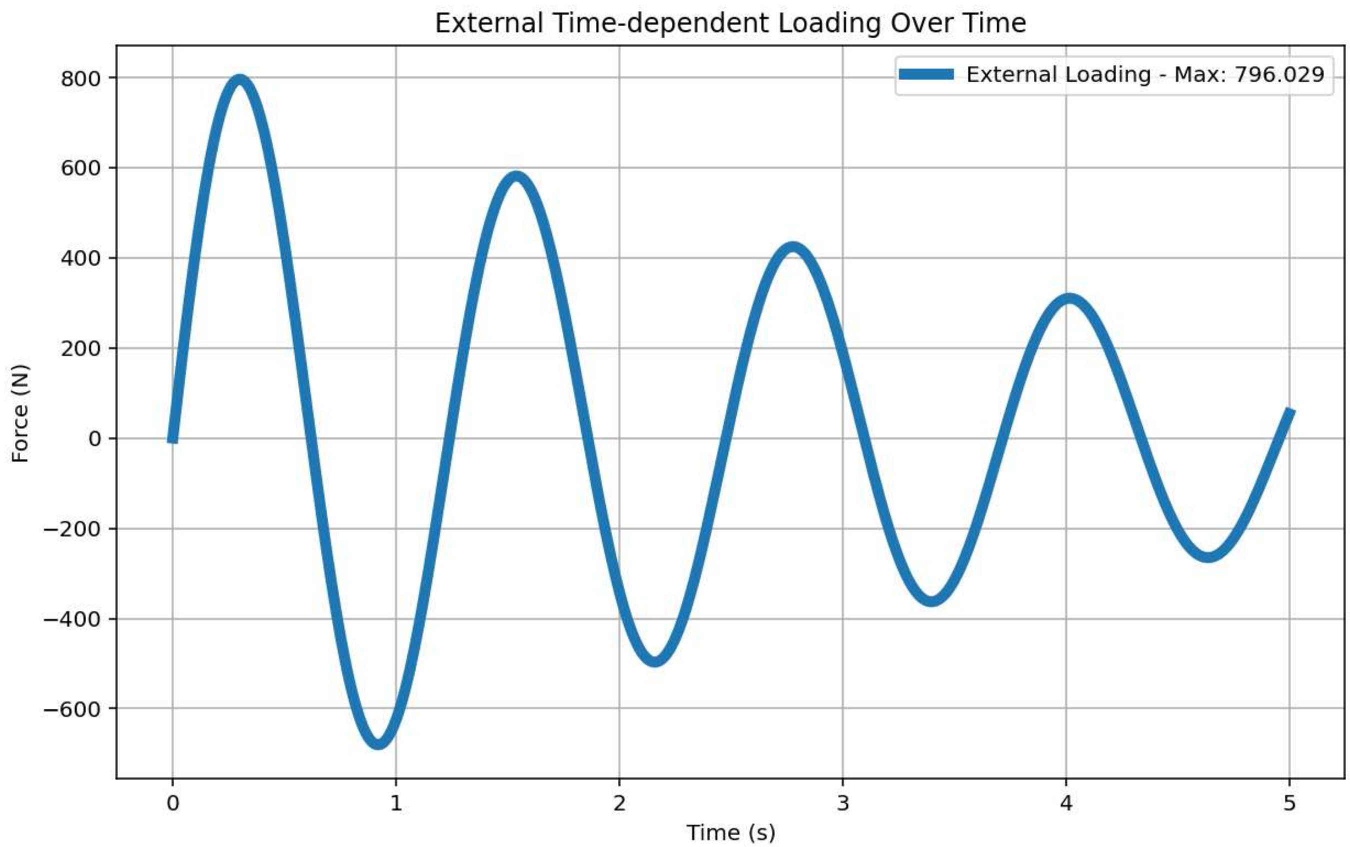

Displacement vs Structural Damage Index
Structural Damage Index [%]
Displacement [m]
0
20
40
60
80
100
−0.3
−0.2
−0.1
0.0
0.1
0.2
0.3



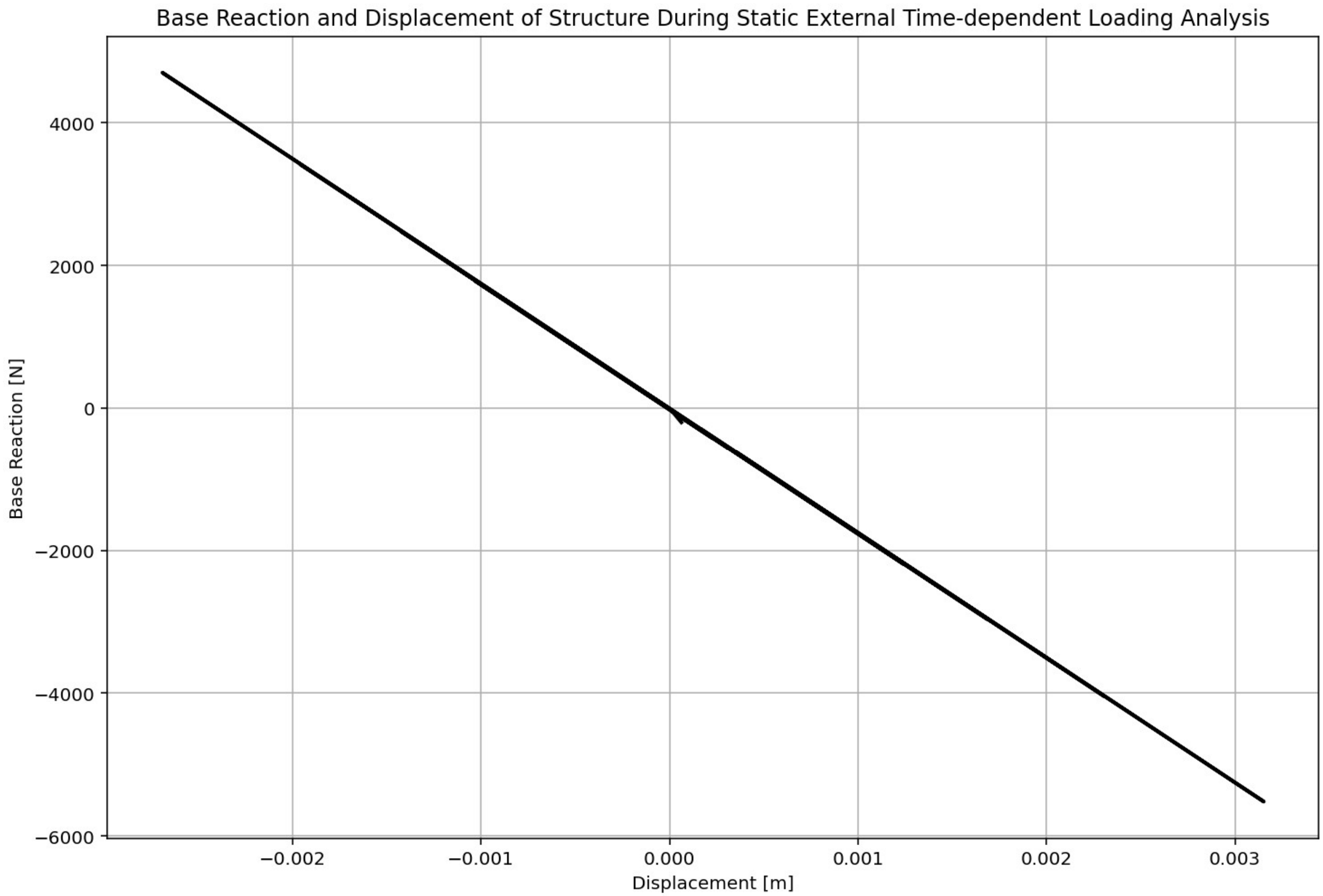
Equivalent viscous damping ratio - Cyclic Displacement Hysteresis
$\xi_{eq} = 30.1944$ [%]
$E_d = 244149.121$ N·m
$E_s = 64345.551$ N·m
Hysteresis Loop
(u_{max}, F_{max})
E_d = 244149.121 N·m
Base Reaction [N]
Displacement [m]
300000
200000
100000
0
−100000
−200000
−300000
−0.3
−0.2
−0.1
0.0
0.1
0.2
0.3

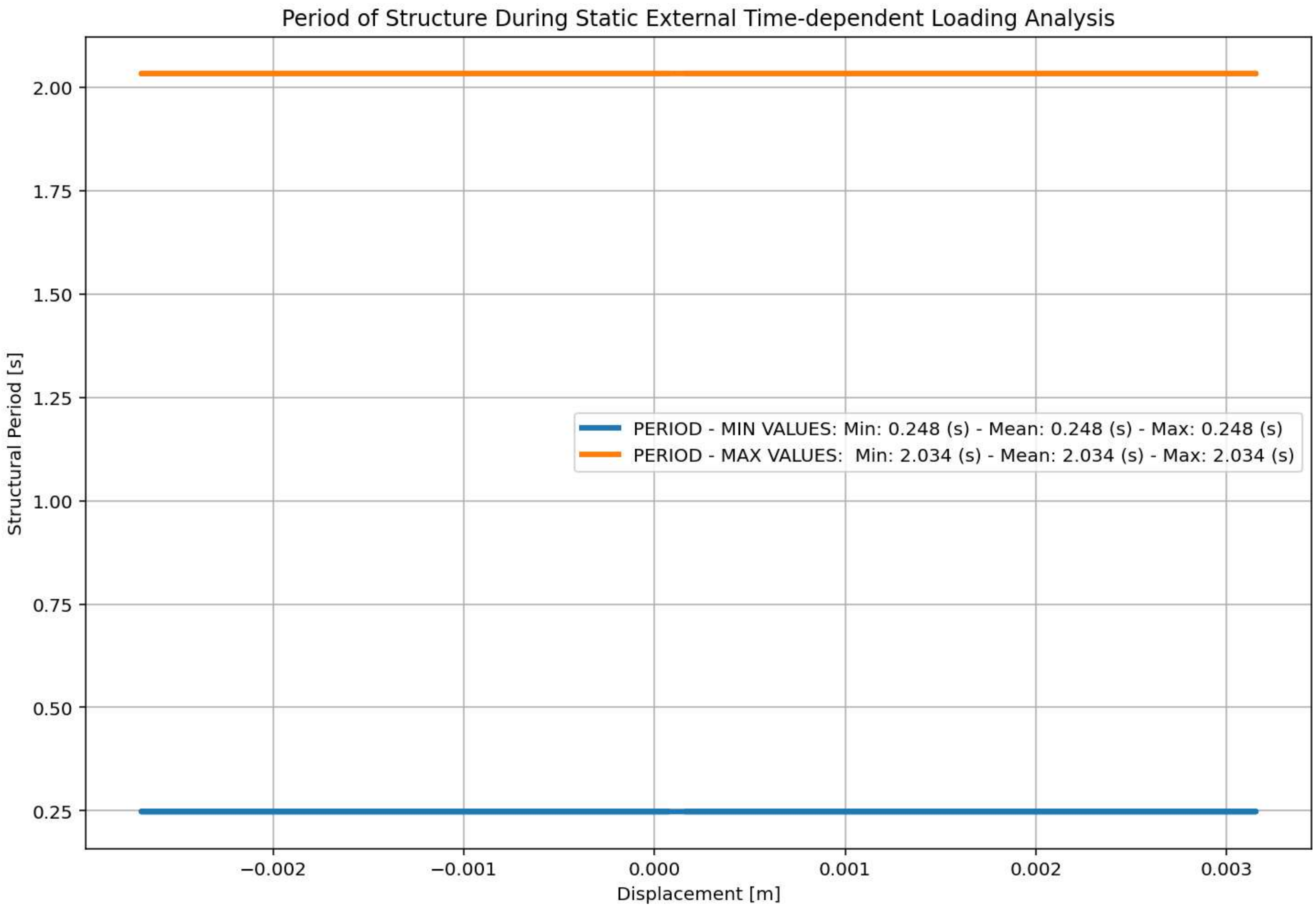
STATIC TIME-HISTORY WITH EXTERNAL TIME-DEPENDENT LOADING ANALYSIS

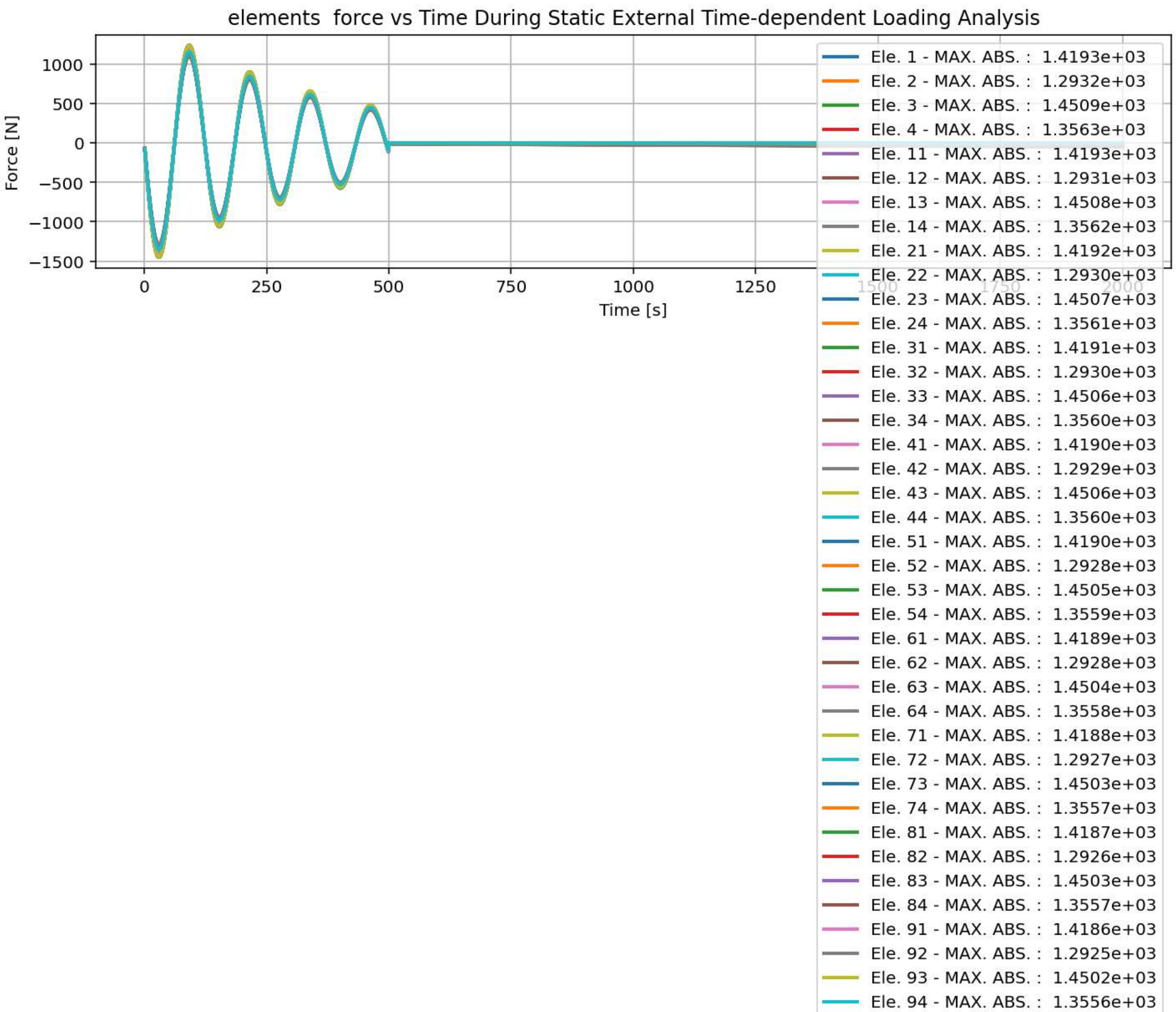

Spyder (Python 3.12)
File Edit Search Source Run Debug Consoles Projects Tools View Help
...S_04_COLUMNS_8_ANA+\+60_MDOF_10_FLOORS_04_COLUMNS_8_ANA
C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...MNS_8_ANA\P(t)_MDOF_10_FLOORS_04_COLUMNS_8_ANA.py
P(t)_MDOF_10_FLOORS_04_COLUMNS_8_ANA.py
# EXTERNAL TIME-DEPENDENT LOADING ANALYSIS (STATIC TIME-HISTORY ANALYSIS)
MAT_TYPE = 'INELASTIC'   # 'ELASTIC' OR 'INELASTIC'
ANAL_TYPE = 'STATIC_EXTERNAL_TIME-DEPENDENT_LOADING'
DATA = MDOF(MAT_TYPE, TOTAL_MASS, ANAL_TYPE)
(reaction_ETDLS, disp_ETDLS, DI_ETDLS,
 ele_force_ETDLS, node_displacements_ETDLS,
 PERIOD_MIN_ETDLS, PERIOD_MAX_ETDLS) = DATA
XDATA = disp_ETDLS
YDATA = reaction_ETDLS
XLABEL = 'Displacement [m]'
YLABEL = 'Base Reaction [N]'
TITLE = 'Base Reaction and Displacement of Structure During Static External Time-dependent Loading
COLOR = 'black'
SEMILOGY = False
PLOT(XDATA, YDATA, TITLE, XLABEL, YLABEL, COLOR, SEMILOGY)
# PLOT STRUCTURAL PERIOD DURING THE ANALYSIS
plt.figure(0, figsize=(12, 8))
plt.plot(disp_ETDLS, PERIOD_MIN_ETDLS, linewidth=3)
plt.plot(disp_ETDLS, PERIOD_MAX_ETDLS, linewidth=3)
plt.title('Period of Structure During Static External Time-dependent Loading Analysis')
plt.ylabel('Structural Period [s]')
plt.xlabel('Displacement [m]')
#plt.semilogy()
plt.grid()
plt.legend([f'PERIOD - MIN VALUES: Min: {np.min(PERIOD_MIN_ETDLS):.3f} (s) - Mean: {np.mean(PERIOD
            f'PERIOD - MAX VALUES: Min: {np.min(PERIOD_MAX_ETDLS):.3f} (s) - Mean: {np.mean(PERIO
            ])
plt.show()
26 %
Base Reaction and Displacement of Structure During Static External Time-dependent Loading Analysis
Base Reaction [N]
Displacement [m]
IPython Console Plots Variable Explorer Help Debugger History Files
Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 1, Col 1 UTF-8 CRLF RW Mem 44%



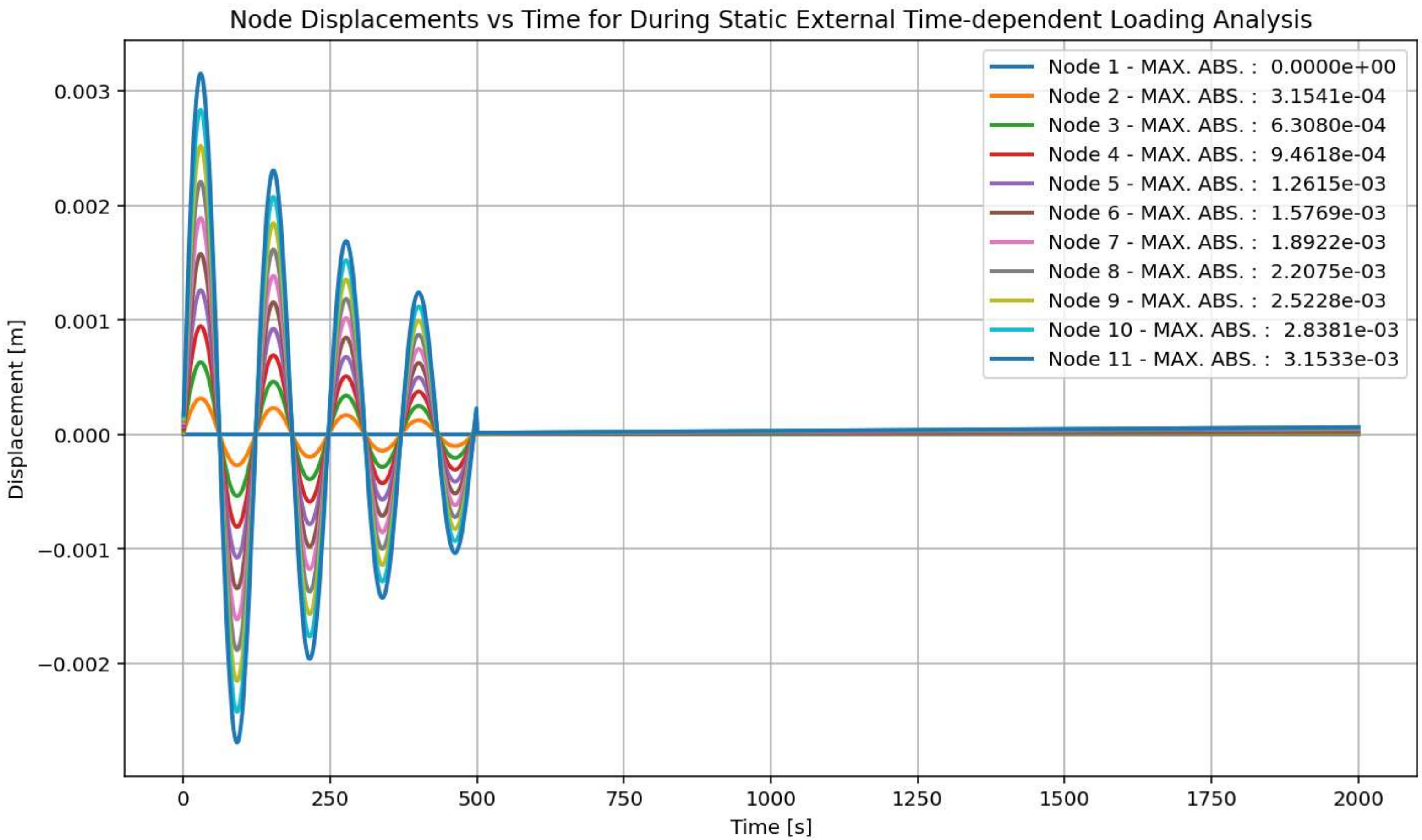
External Time-dependent Loading Over Time
External Loading - Max: 796.029
Force (N)
Time (s)
800
600
400
200
0
−200
−400
−600
0
1
2
3
4
5

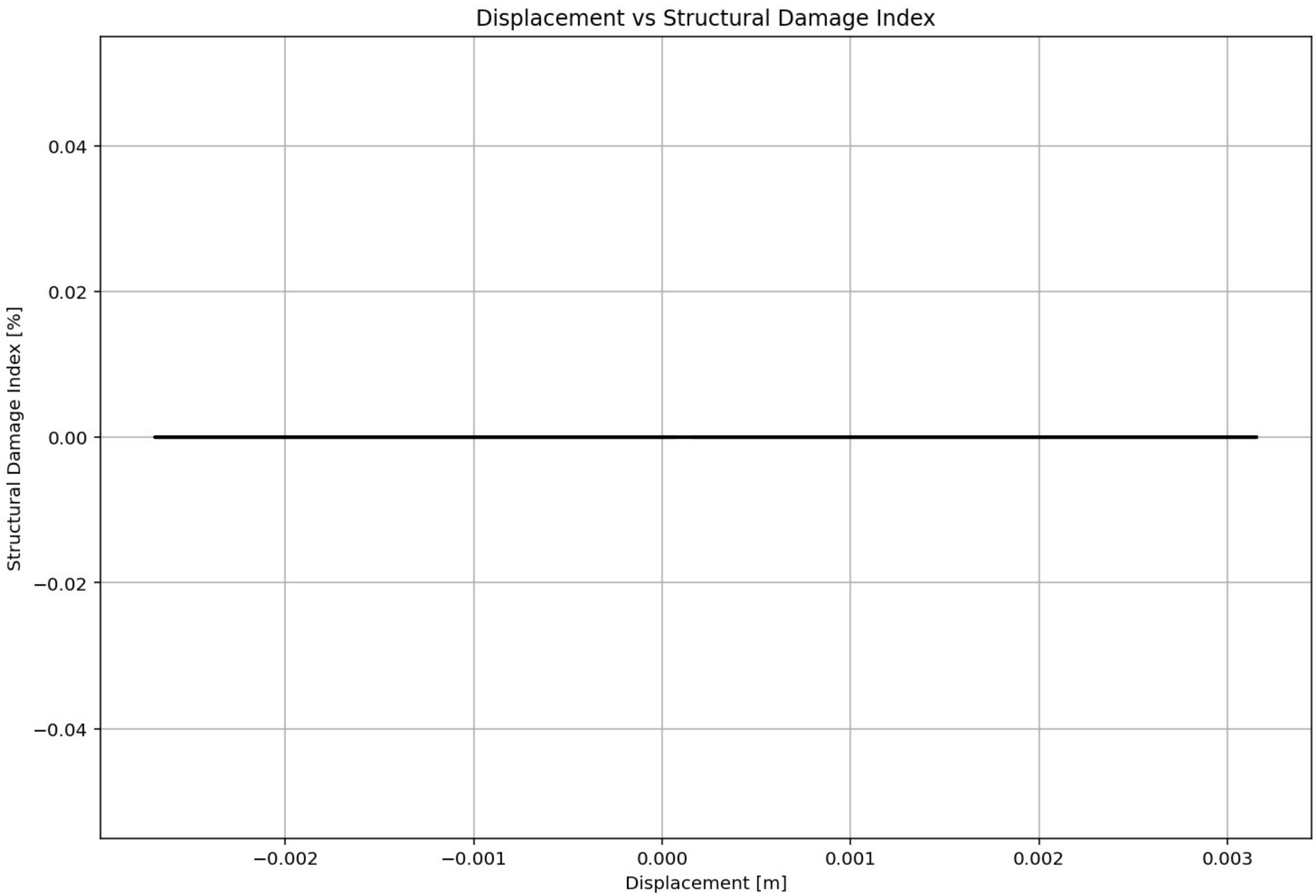

Base Reaction and Displacement of Structure During Static External Time-dependent Loading Analysis
Base Reaction [N]
4000
2000
0
−2000
−4000
−6000
−0.002
−0.001
0.000
0.001
0.002
0.003
Displacement [m]

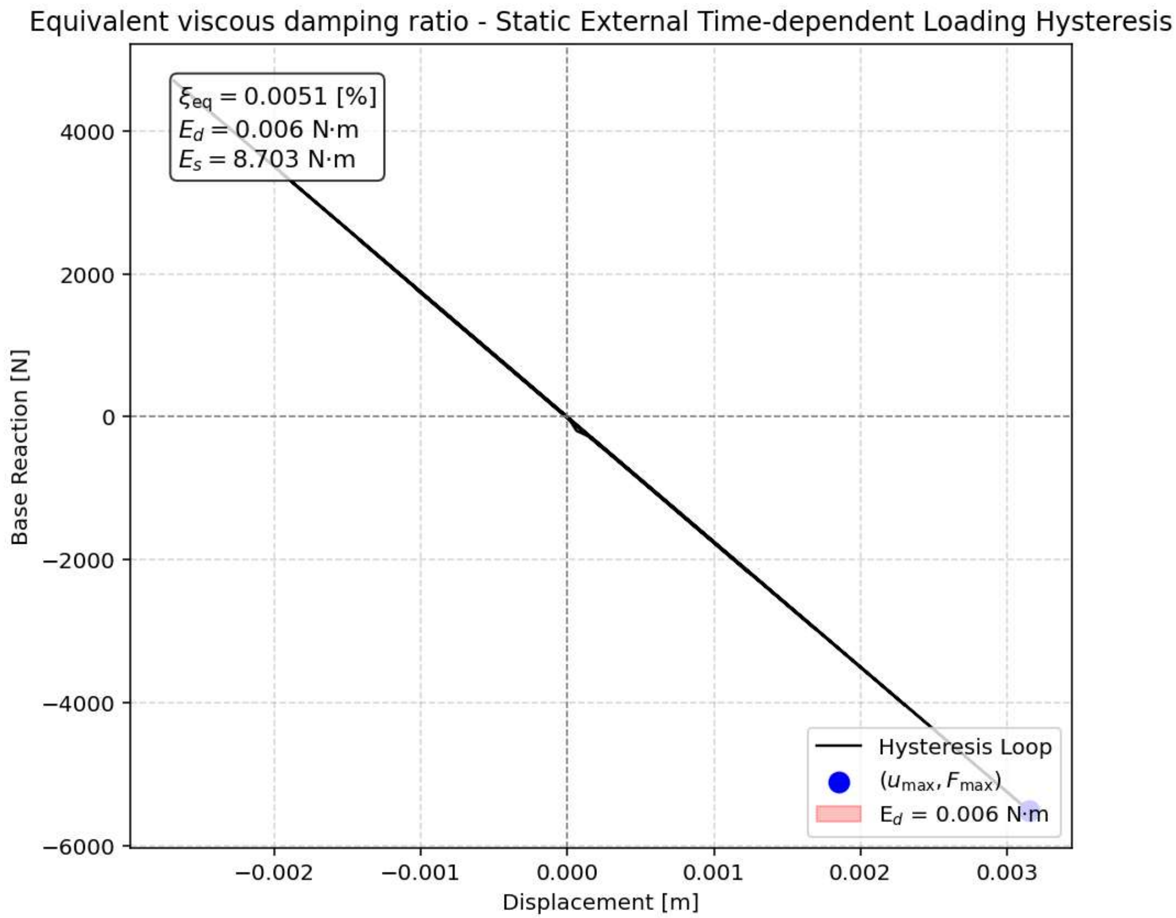

Period of Structure During Static External Time-dependent Loading Analysis
Structural Period [s]
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25
−0.002
−0.001
0.000
0.001
0.002
0.003
Displacement [m]
PERIOD - MIN VALUES: Min: 0.248 (s) - Mean: 0.248 (s) - Max: 0.248 (s)
PERIOD - MAX VALUES: Min: 2.034 (s) - Mean: 2.034 (s) - Max: 2.034 (s)



elements force vs Time During Static External Time-dependent Loading Analysis
Force [N]
Time [s]
Ele. 1 - MAX. ABS. : 1.4193e+03
Ele. 2 - MAX. ABS. : 1.2932e+03
Ele. 3 - MAX. ABS. : 1.4509e+03
Ele. 4 - MAX. ABS. : 1.3563e+03
Ele. 11 - MAX. ABS. : 1.4193e+03
Ele. 12 - MAX. ABS. : 1.2931e+03
Ele. 13 - MAX. ABS. : 1.4508e+03
Ele. 14 - MAX. ABS. : 1.3562e+03
Ele. 21 - MAX. ABS. : 1.4192e+03
Ele. 22 - MAX. ABS. : 1.2930e+03
Ele. 23 - MAX. ABS. : 1.4507e+03
Ele. 24 - MAX. ABS. : 1.3561e+03
Ele. 31 - MAX. ABS. : 1.4191e+03
Ele. 32 - MAX. ABS. : 1.2930e+03
Ele. 33 - MAX. ABS. : 1.4506e+03
Ele. 34 - MAX. ABS. : 1.3560e+03
Ele. 41 - MAX. ABS. : 1.4190e+03
Ele. 42 - MAX. ABS. : 1.2929e+03
Ele. 43 - MAX. ABS. : 1.4506e+03
Ele. 44 - MAX. ABS. : 1.3560e+03
Ele. 51 - MAX. ABS. : 1.4190e+03
Ele. 52 - MAX. ABS. : 1.2928e+03
Ele. 53 - MAX. ABS. : 1.4505e+03
Ele. 54 - MAX. ABS. : 1.3559e+03
Ele. 61 - MAX. ABS. : 1.4189e+03
Ele. 62 - MAX. ABS. : 1.2928e+03
Ele. 63 - MAX. ABS. : 1.4504e+03
Ele. 64 - MAX. ABS. : 1.3558e+03
Ele. 71 - MAX. ABS. : 1.4188e+03
Ele. 72 - MAX. ABS. : 1.2927e+03
Ele. 73 - MAX. ABS. : 1.4503e+03
Ele. 74 - MAX. ABS. : 1.3557e+03
Ele. 81 - MAX. ABS. : 1.4187e+03
Ele. 82 - MAX. ABS. : 1.2926e+03
Ele. 83 - MAX. ABS. : 1.4503e+03
Ele. 84 - MAX. ABS. : 1.3557e+03
Ele. 91 - MAX. ABS. : 1.4186e+03
Ele. 92 - MAX. ABS. : 1.2925e+03
Ele. 93 - MAX. ABS. : 1.4502e+03
Ele. 94 - MAX. ABS. : 1.3556e+03

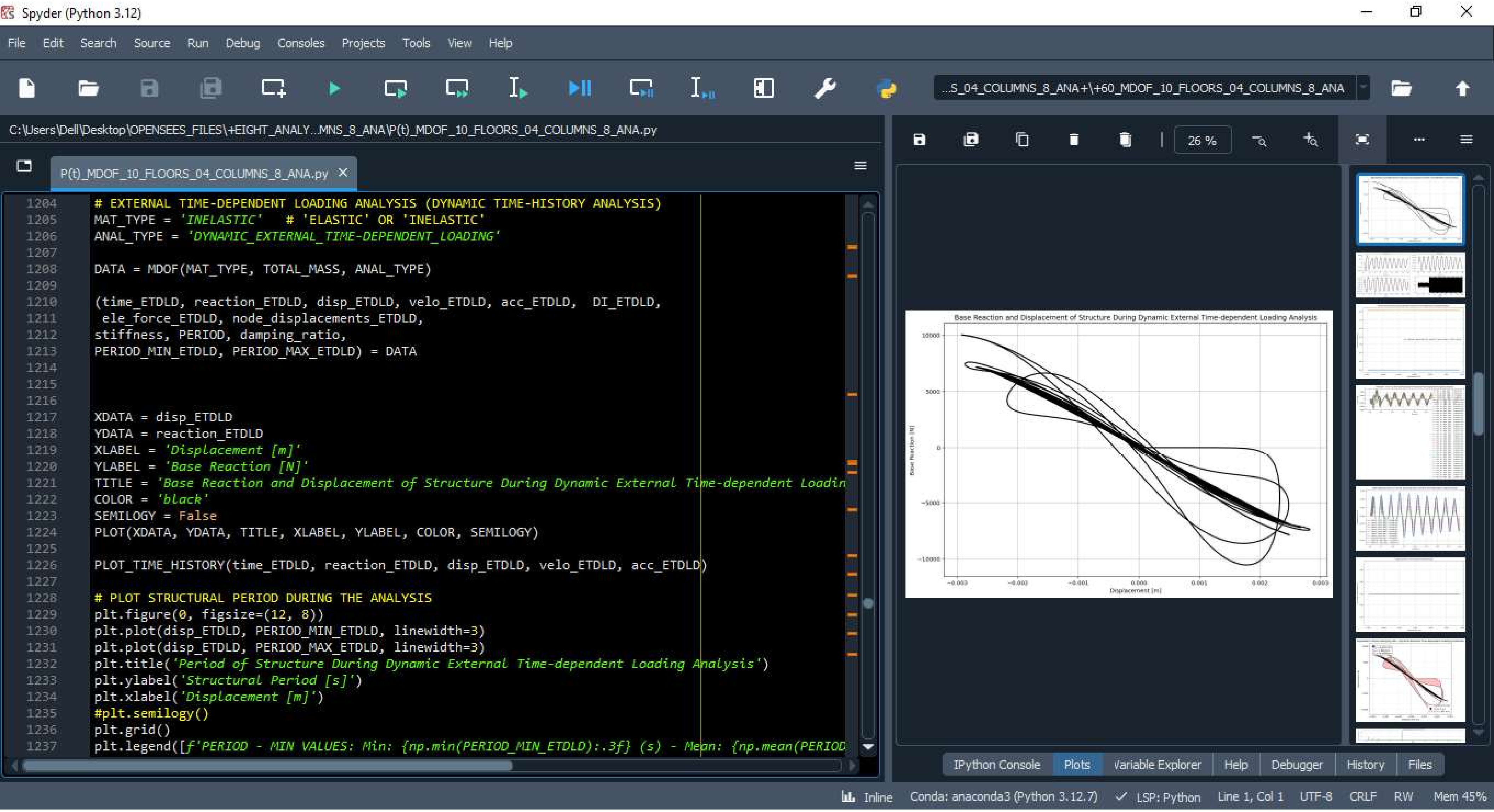

Node Displacements vs Time for During Static External Time-dependent Loading Analysis
Displacement [m]
Time [s]
0.003
0.002
0.001
0.000
−0.001
−0.002
0
250
500
750
1000
1250
1500
1750
2000
Node 1 - MAX. ABS. : 0.0000e+00
Node 2 - MAX. ABS. : 3.1541e-04
Node 3 - MAX. ABS. : 6.3080e-04
Node 4 - MAX. ABS. : 9.4618e-04
Node 5 - MAX. ABS. : 1.2615e-03
Node 6 - MAX. ABS. : 1.5769e-03
Node 7 - MAX. ABS. : 1.8922e-03
Node 8 - MAX. ABS. : 2.2075e-03
Node 9 - MAX. ABS. : 2.5228e-03
Node 10 - MAX. ABS. : 2.8381e-03
Node 11 - MAX. ABS. : 3.1533e-03


Displacement vs Structural Damage Index
Structural Damage Index [%]
0.04
0.02
0.00
−0.02
−0.04
−0.002
−0.001
0.000
0.001
0.002
0.003
Displacement [m]

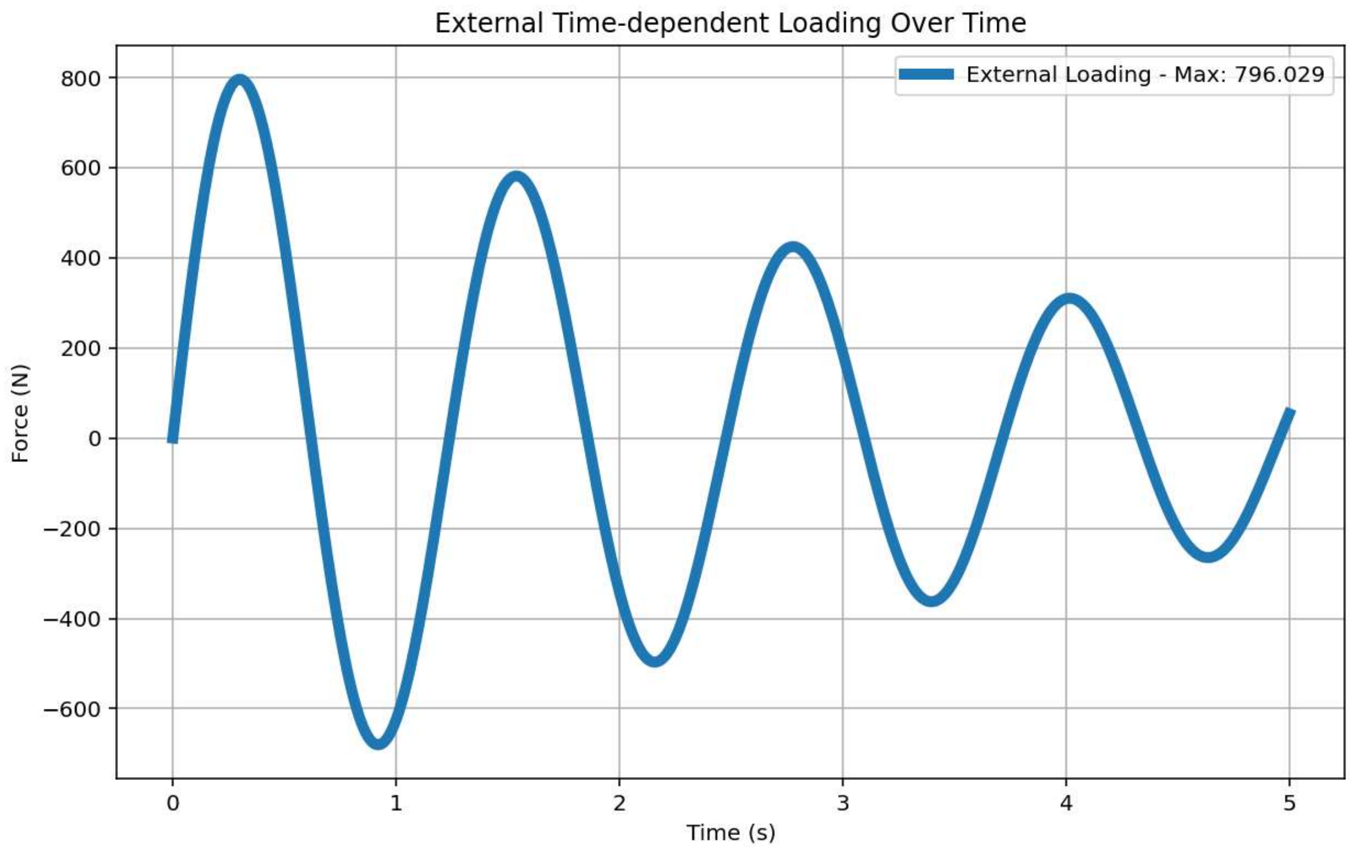


Equivalent viscous damping ratio - Static External Time-dependent Loading Hysteresis
$\xi_{eq} = 0.0051$ [%]
$E_d = 0.006$ N·m
$E_s = 8.703$ N·m
Hysteresis Loop
(u_{max}, F_{max})
E_d = 0.006 N·m
Base Reaction [N]
Displacement [m]
4000
2000
0
−2000
−4000
−6000
−0.002
−0.001
0.000
0.001
0.002
0.003

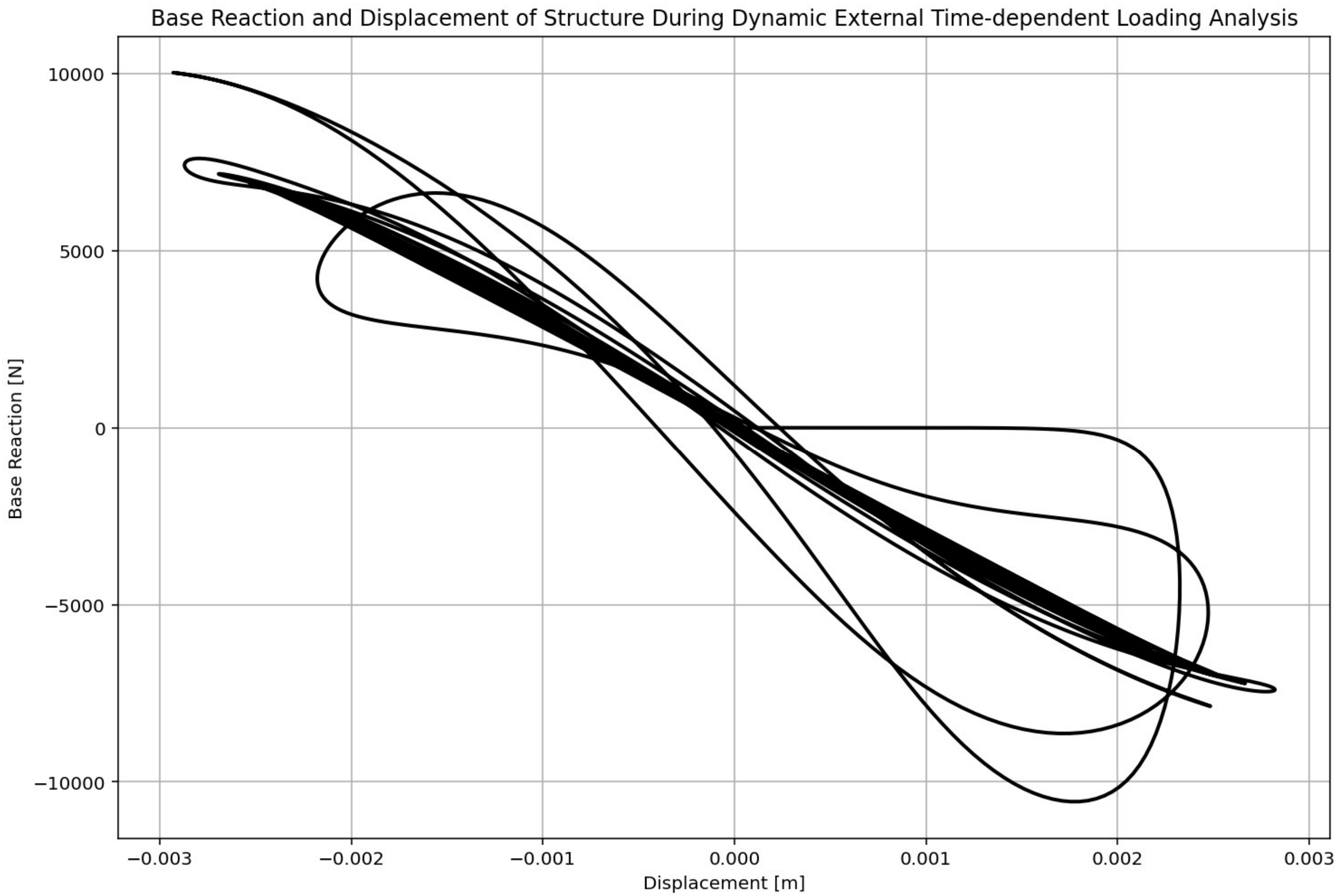
DYNAMIC TIME-HISTORY WITH EXTERNAL TIME-DEPENDENT LOADING ANALYSIS



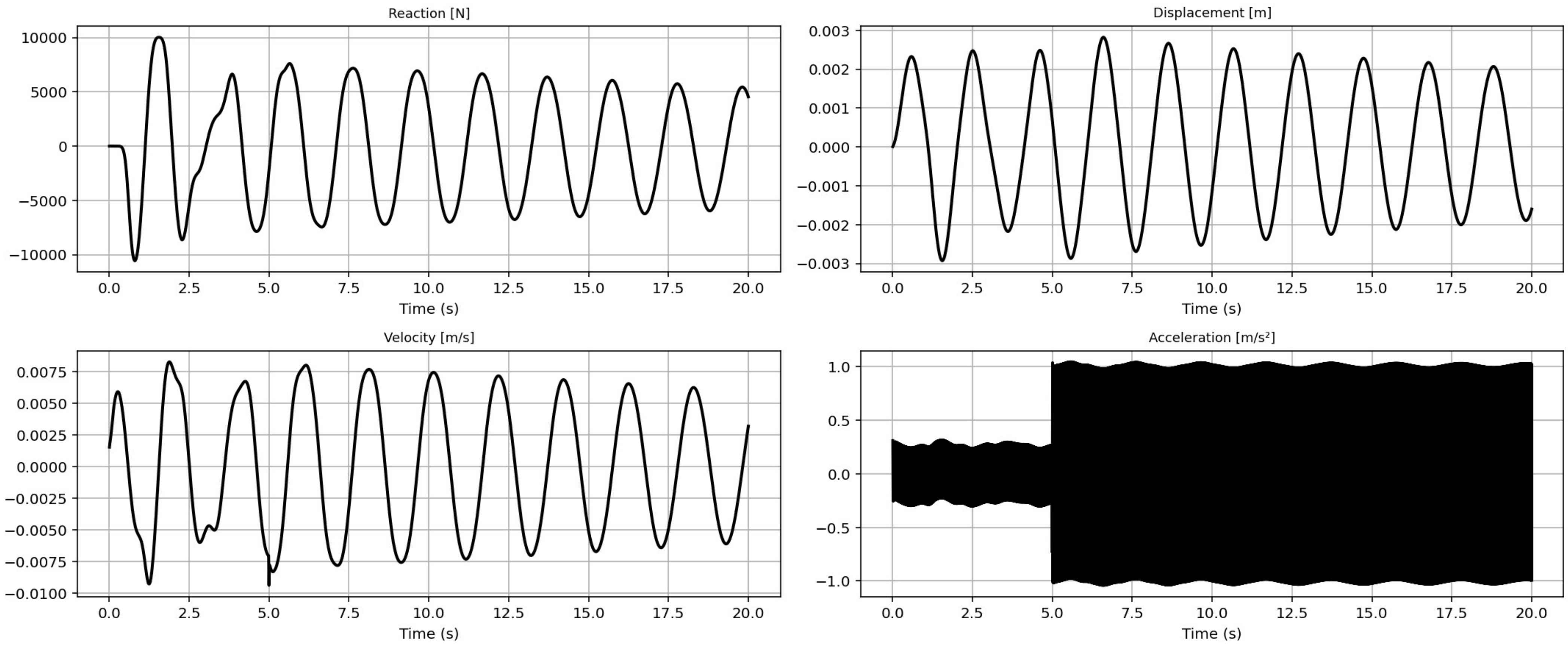
Spyder (Python 3.12)
File Edit Search Source Run Debug Consoles Projects Tools View Help
...S_04_COLUMNS_8_ANA+\+60_MDOF_10_FLOORS_04_COLUMNS_8_ANA
C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...MNS_8_ANA\P(t)_MDOF_10_FLOORS_04_COLUMNS_8_ANA.py
P(t)_MDOF_10_FLOORS_04_COLUMNS_8_ANA.py
# EXTERNAL TIME-DEPENDENT LOADING ANALYSIS (DYNAMIC TIME-HISTORY ANALYSIS)
MAT_TYPE = 'INELASTIC'   # 'ELASTIC' OR 'INELASTIC'
ANAL_TYPE = 'DYNAMIC_EXTERNAL_TIME-DEPENDENT_LOADING'
DATA = MDOF(MAT_TYPE, TOTAL_MASS, ANAL_TYPE)
(time_ETDLD, reaction_ETDLD, disp_ETDLD, velo_ETDLD, acc_ETDLD, DI_ETDLD,
 ele_force_ETDLD, node_displacements_ETDLD,
stiffness, PERIOD, damping_ratio,
PERIOD_MIN_ETDLD, PERIOD_MAX_ETDLD) = DATA
XDATA = disp_ETDLD
YDATA = reaction_ETDLD
XLABEL = 'Displacement [m]'
YLABEL = 'Base Reaction [N]'
TITLE = 'Base Reaction and Displacement of Structure During Dynamic External Time-dependent Loadin
COLOR = 'black'
SEMILOGY = False
PLOT(XDATA, YDATA, TITLE, XLABEL, YLABEL, COLOR, SEMILOGY)
PLOT_TIME_HISTORY(time_ETDLD, reaction_ETDLD, disp_ETDLD, velo_ETDLD, acc_ETDLD)
# PLOT STRUCTURAL PERIOD DURING THE ANALYSIS
plt.figure(0, figsize=(12, 8))
plt.plot(disp_ETDLD, PERIOD_MIN_ETDLD, linewidth=3)
plt.plot(disp_ETDLD, PERIOD_MAX_ETDLD, linewidth=3)
plt.title('Period of Structure During Dynamic External Time-dependent Loading Analysis')
plt.ylabel('Structural Period [s]')
plt.xlabel('Displacement [m]')
#plt.semilogy()
plt.grid()
plt.legend([f'PERIOD - MIN VALUES: Min: {np.min(PERIOD_MIN_ETDLD):.3f} (s) - Mean: {np.mean(PERIOD
26 %
Base Reaction and Displacement of Structure During Dynamic External Time-dependent Loading Analysis
Base Reaction [N]
Displacement [m]
IPython Console Plots Variable Explorer Help Debugger History Files
Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 1, Col 1 UTF-8 CRLF RW Mem 45%



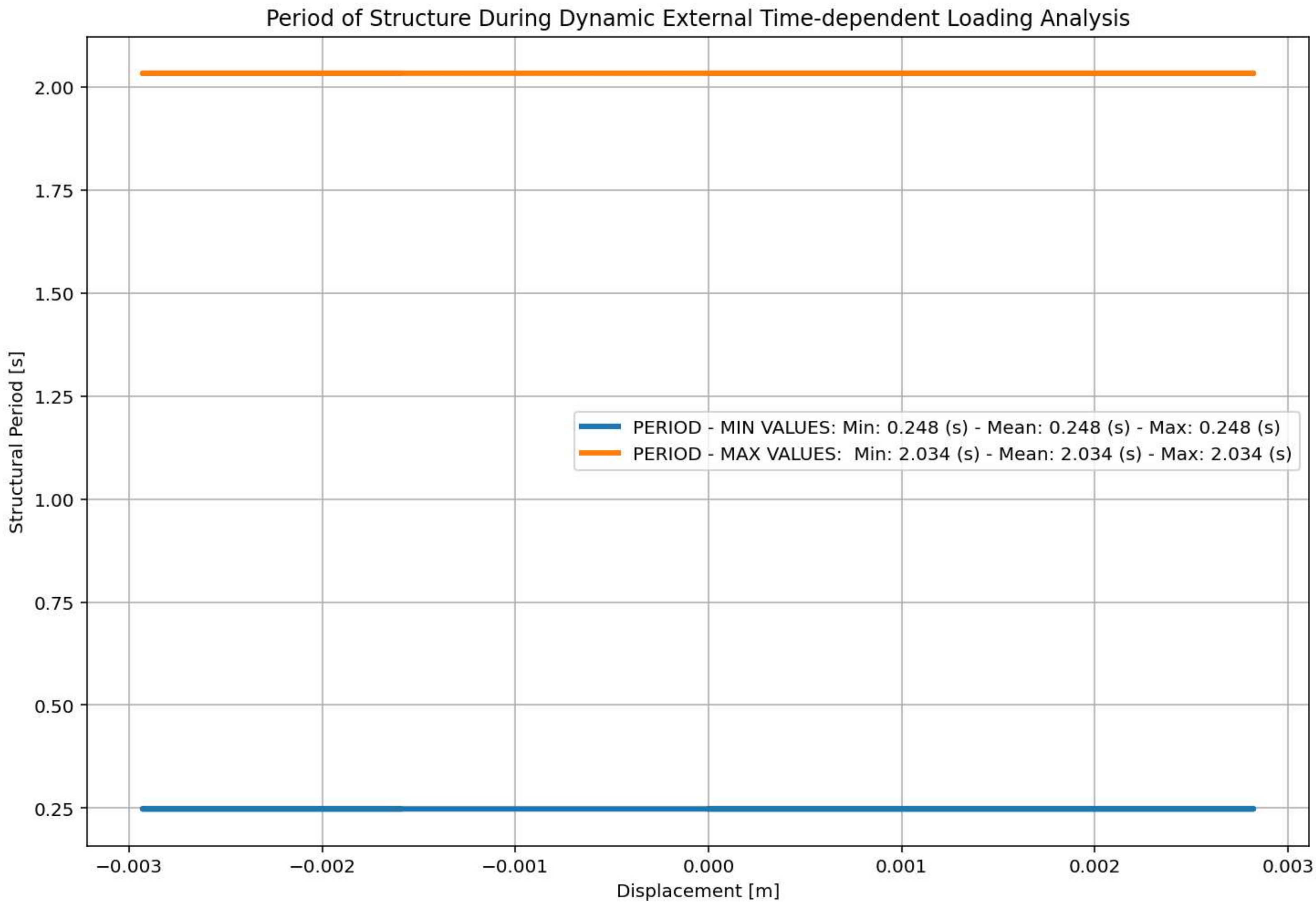
External Time-dependent Loading Over Time
External Loading - Max: 796.029
800
600
400
200
0
−200
−400
−600
Force (N)
0
1
2
3
4
5
Time (s)

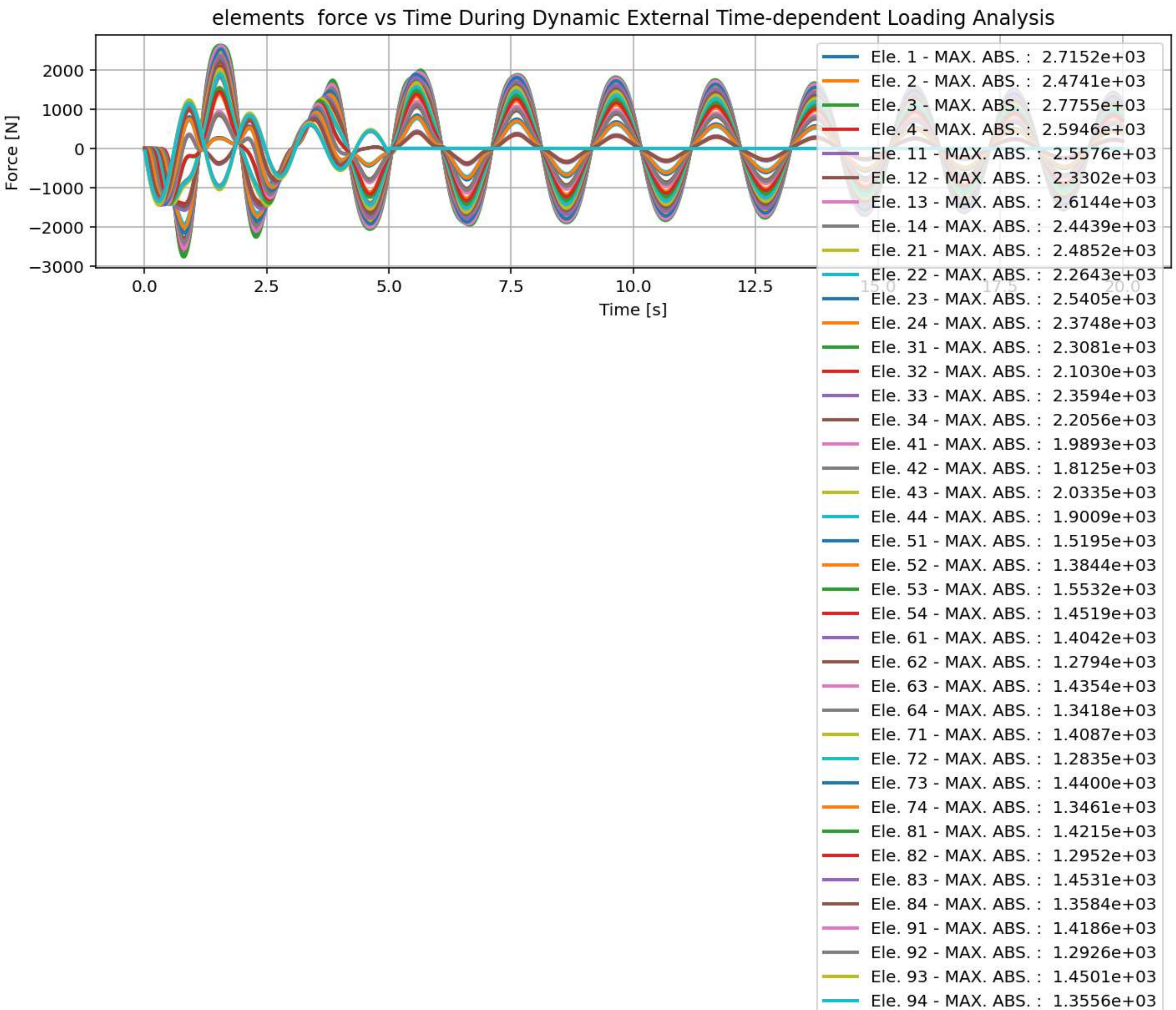


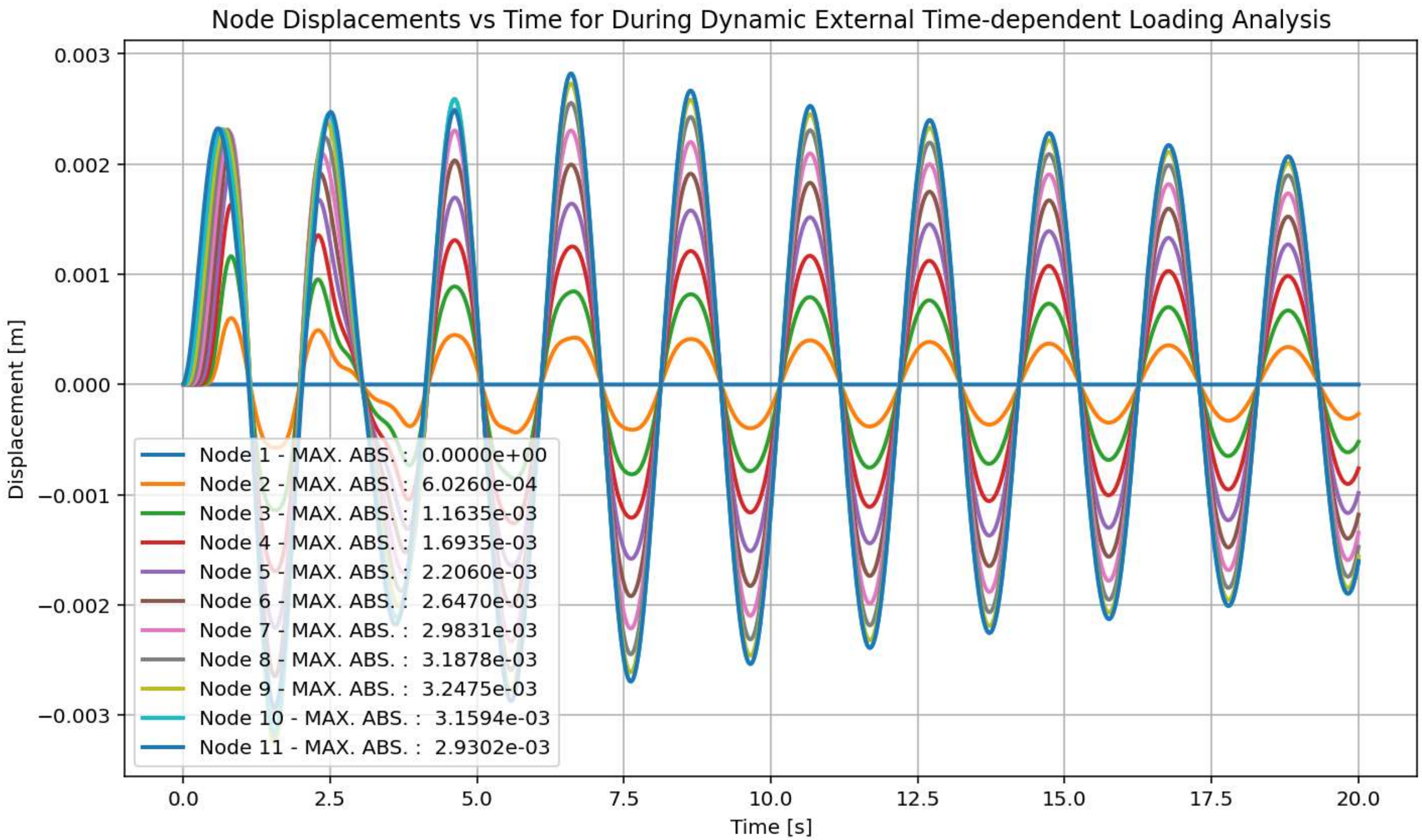
Base Reaction and Displacement of Structure During Dynamic External Time-dependent Loading Analysis
10000
5000
0
−5000
−10000
Base Reaction [N]
−0.003
−0.002
−0.001
0.000
0.001
0.002
0.003
Displacement [m]

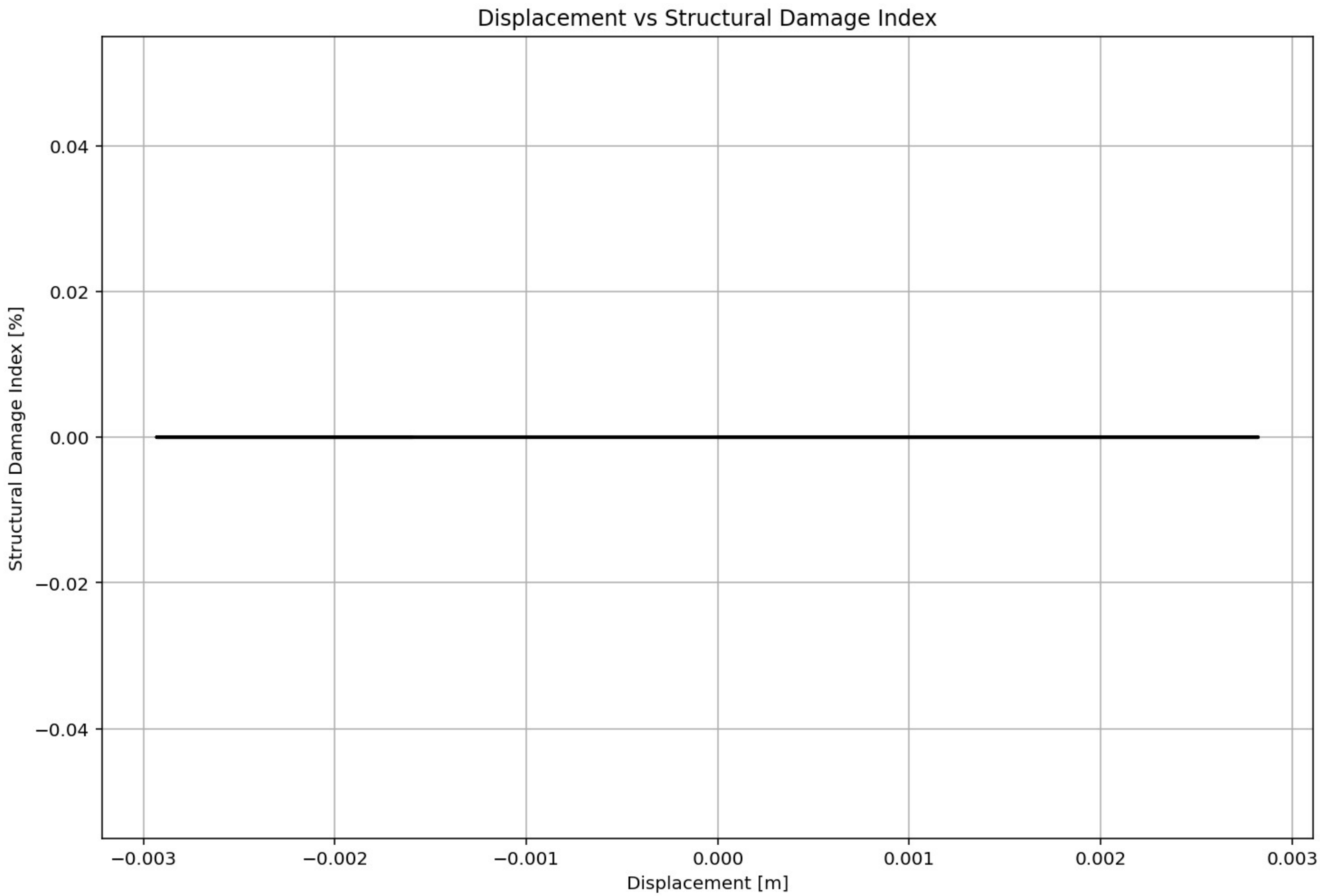


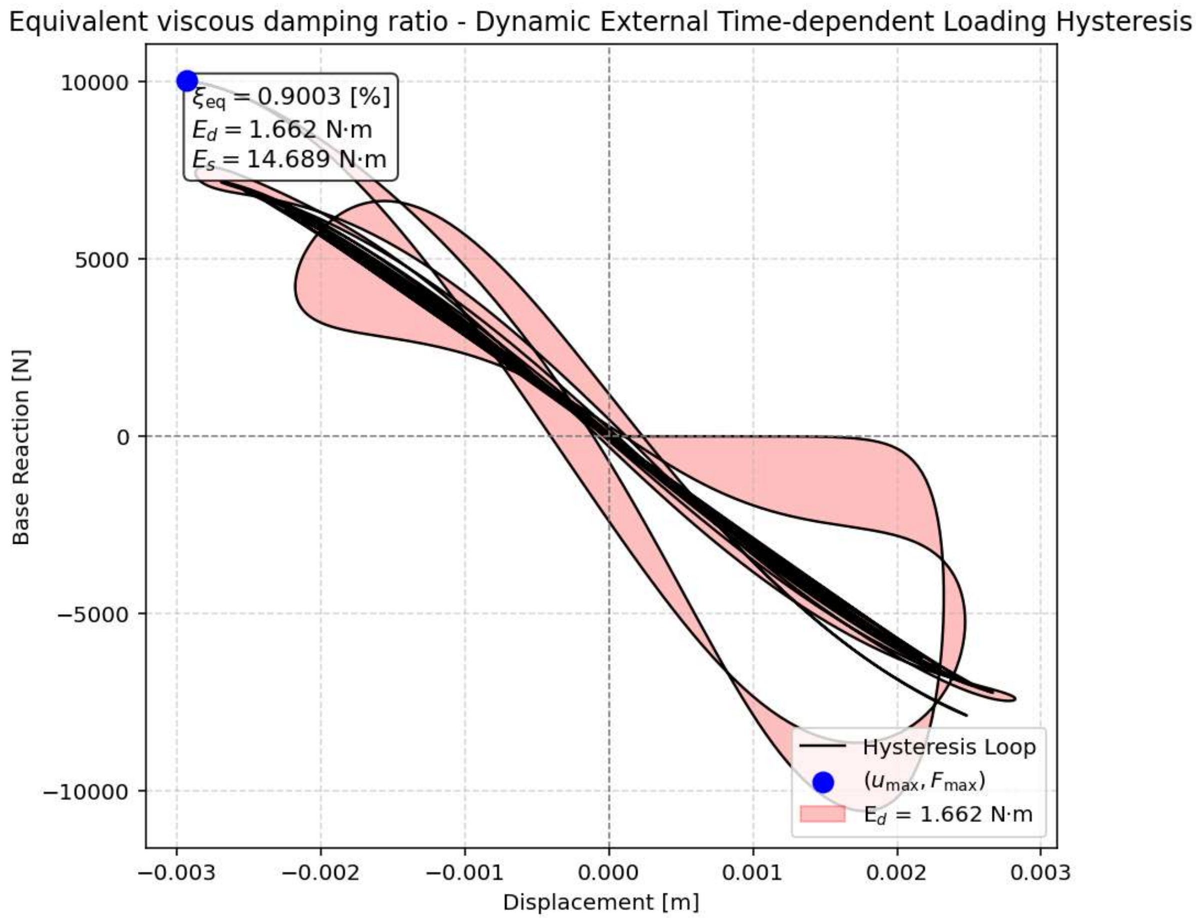
Reaction [N]
Displacement [m]
Velocity [m/s]
Acceleration [m/s²]
Time (s)


Period of Structure During Dynamic External Time-dependent Loading Analysis
PERIOD - MIN VALUES: Min: 0.248 (s) - Mean: 0.248 (s) - Max: 0.248 (s)
PERIOD - MAX VALUES: Min: 2.034 (s) - Mean: 2.034 (s) - Max: 2.034 (s)
Structural Period [s]
Displacement [m]
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25
−0.003
−0.002
−0.001
0.000
0.001
0.002
0.003

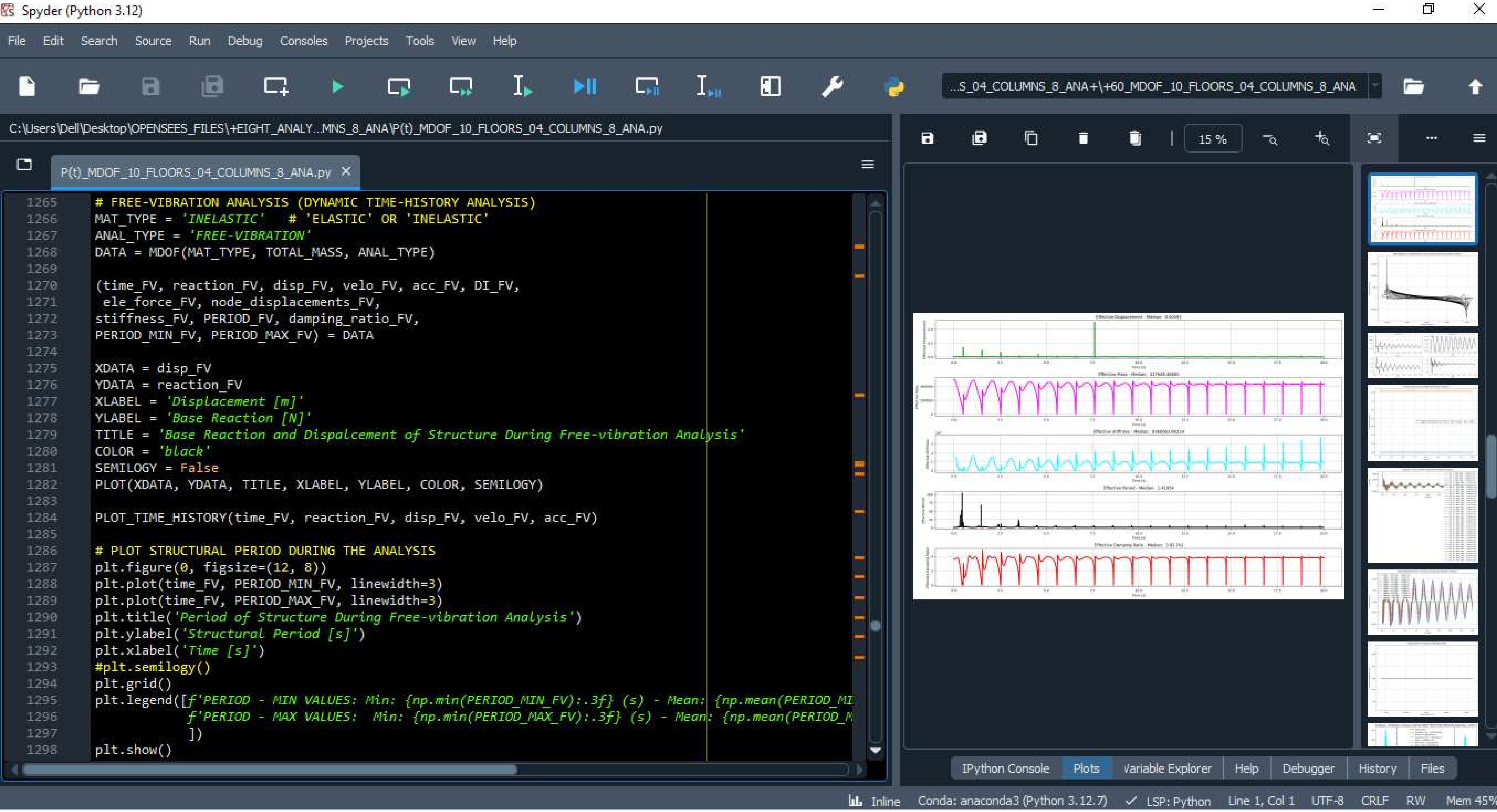

elements force vs Time During Dynamic External Time-dependent Loading Analysis
Force [N]
2000
1000
0
−1000
−2000
−3000
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time [s]
Ele. 1 - MAX. ABS. : 2.7152e+03
Ele. 2 - MAX. ABS. : 2.4741e+03
Ele. 3 - MAX. ABS. : 2.7755e+03
Ele. 4 - MAX. ABS. : 2.5946e+03
Ele. 11 - MAX. ABS. : 2.5576e+03
Ele. 12 - MAX. ABS. : 2.3302e+03
Ele. 13 - MAX. ABS. : 2.6144e+03
Ele. 14 - MAX. ABS. : 2.4439e+03
Ele. 21 - MAX. ABS. : 2.4852e+03
Ele. 22 - MAX. ABS. : 2.2643e+03
Ele. 23 - MAX. ABS. : 2.5405e+03
Ele. 24 - MAX. ABS. : 2.3748e+03
Ele. 31 - MAX. ABS. : 2.3081e+03
Ele. 32 - MAX. ABS. : 2.1030e+03
Ele. 33 - MAX. ABS. : 2.3594e+03
Ele. 34 - MAX. ABS. : 2.2056e+03
Ele. 41 - MAX. ABS. : 1.9893e+03
Ele. 42 - MAX. ABS. : 1.8125e+03
Ele. 43 - MAX. ABS. : 2.0335e+03
Ele. 44 - MAX. ABS. : 1.9009e+03
Ele. 51 - MAX. ABS. : 1.5195e+03
Ele. 52 - MAX. ABS. : 1.3844e+03
Ele. 53 - MAX. ABS. : 1.5532e+03
Ele. 54 - MAX. ABS. : 1.4519e+03
Ele. 61 - MAX. ABS. : 1.4042e+03
Ele. 62 - MAX. ABS. : 1.2794e+03
Ele. 63 - MAX. ABS. : 1.4354e+03
Ele. 64 - MAX. ABS. : 1.3418e+03
Ele. 71 - MAX. ABS. : 1.4087e+03
Ele. 72 - MAX. ABS. : 1.2835e+03
Ele. 73 - MAX. ABS. : 1.4400e+03
Ele. 74 - MAX. ABS. : 1.3461e+03
Ele. 81 - MAX. ABS. : 1.4215e+03
Ele. 82 - MAX. ABS. : 1.2952e+03
Ele. 83 - MAX. ABS. : 1.4531e+03
Ele. 84 - MAX. ABS. : 1.3584e+03
Ele. 91 - MAX. ABS. : 1.4186e+03
Ele. 92 - MAX. ABS. : 1.2926e+03
Ele. 93 - MAX. ABS. : 1.4501e+03
Ele. 94 - MAX. ABS. : 1.3556e+03

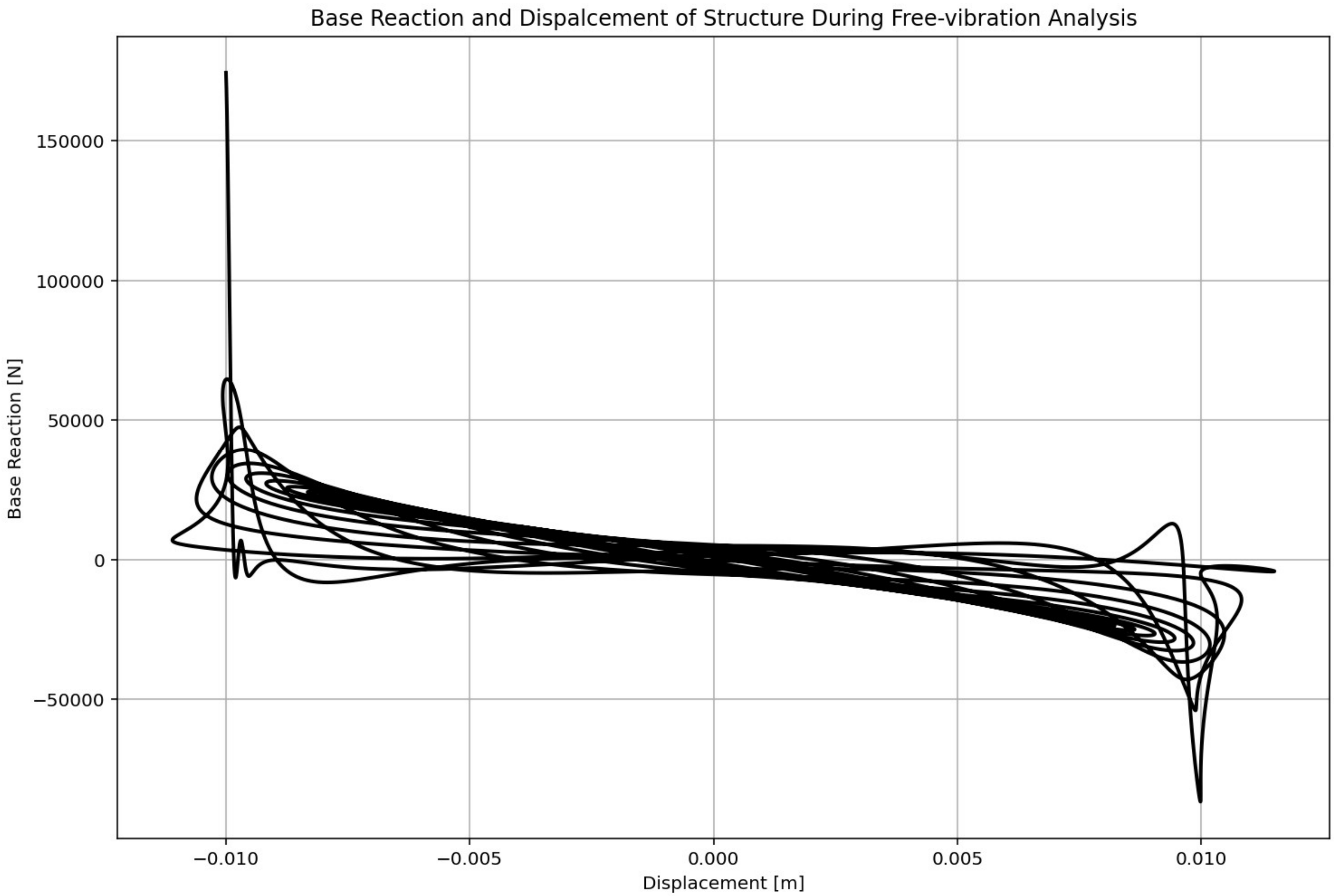

Node Displacements vs Time for During Dynamic External Time-dependent Loading Analysis
0.003
0.002
0.001
0.000
−0.001
−0.002
−0.003
Displacement [m]
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time [s]
Node 1 - MAX. ABS. : 0.0000e+00
Node 2 - MAX. ABS. : 6.0260e-04
Node 3 - MAX. ABS. : 1.1635e-03
Node 4 - MAX. ABS. : 1.6935e-03
Node 5 - MAX. ABS. : 2.2060e-03
Node 6 - MAX. ABS. : 2.6470e-03
Node 7 - MAX. ABS. : 2.9831e-03
Node 8 - MAX. ABS. : 3.1878e-03
Node 9 - MAX. ABS. : 3.2475e-03
Node 10 - MAX. ABS. : 3.1594e-03
Node 11 - MAX. ABS. : 2.9302e-03

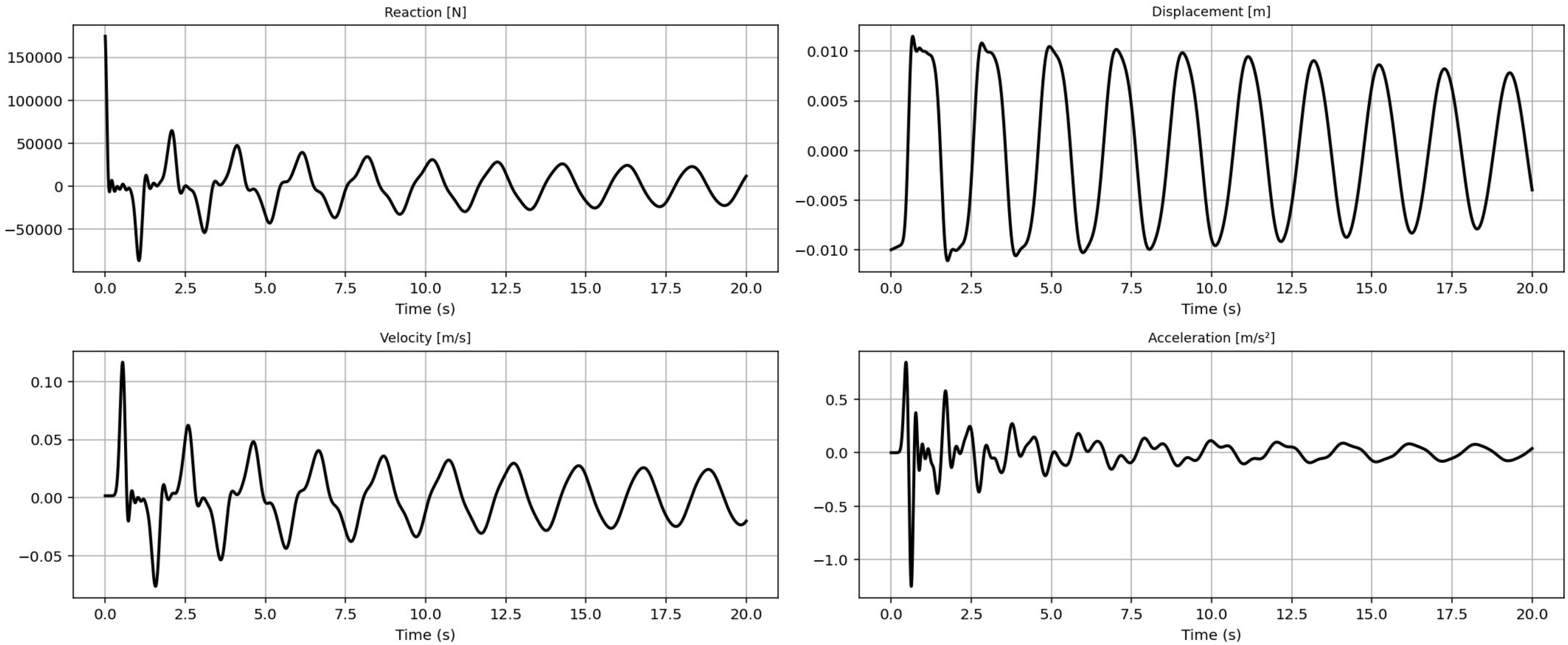

Displacement vs Structural Damage Index
Structural Damage Index [%]
0.04
0.02
0.00
−0.02
−0.04
−0.003
−0.002
−0.001
0.000
0.001
0.002
0.003
Displacement [m]

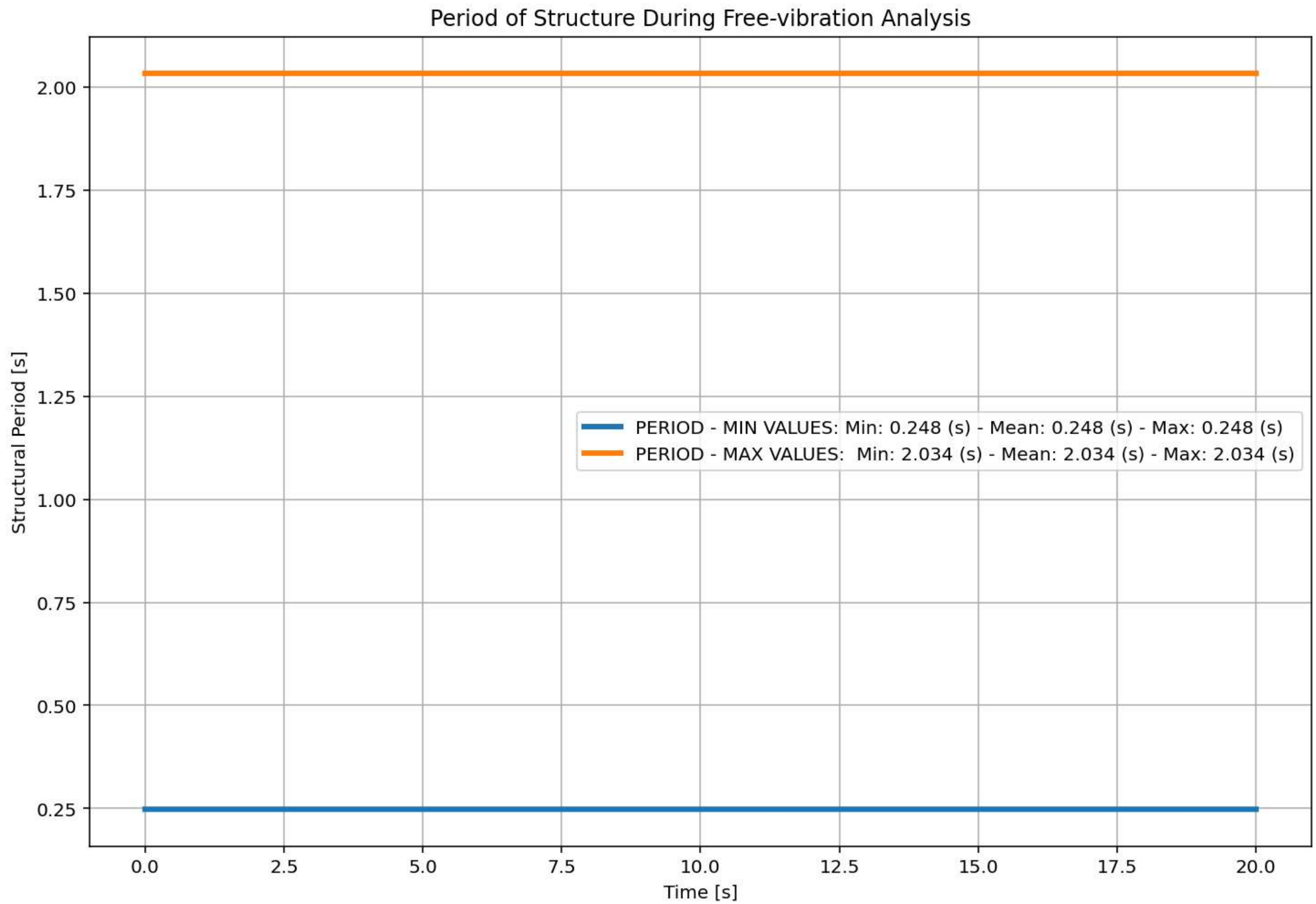

Equivalent viscous damping ratio - Dynamic External Time-dependent Loading Hysteresis
$\xi_{eq} = 0.9003$ [%]
$E_d = 1.662$ N·m
$E_s = 14.689$ N·m
Base Reaction [N]
Displacement [m]
10000
5000
0
−5000
−10000
−0.003
−0.002
−0.001
0.000
0.001
0.002
0.003
Hysteresis Loop
(u_{max}, F_{max})
E_d = 1.662 N·m

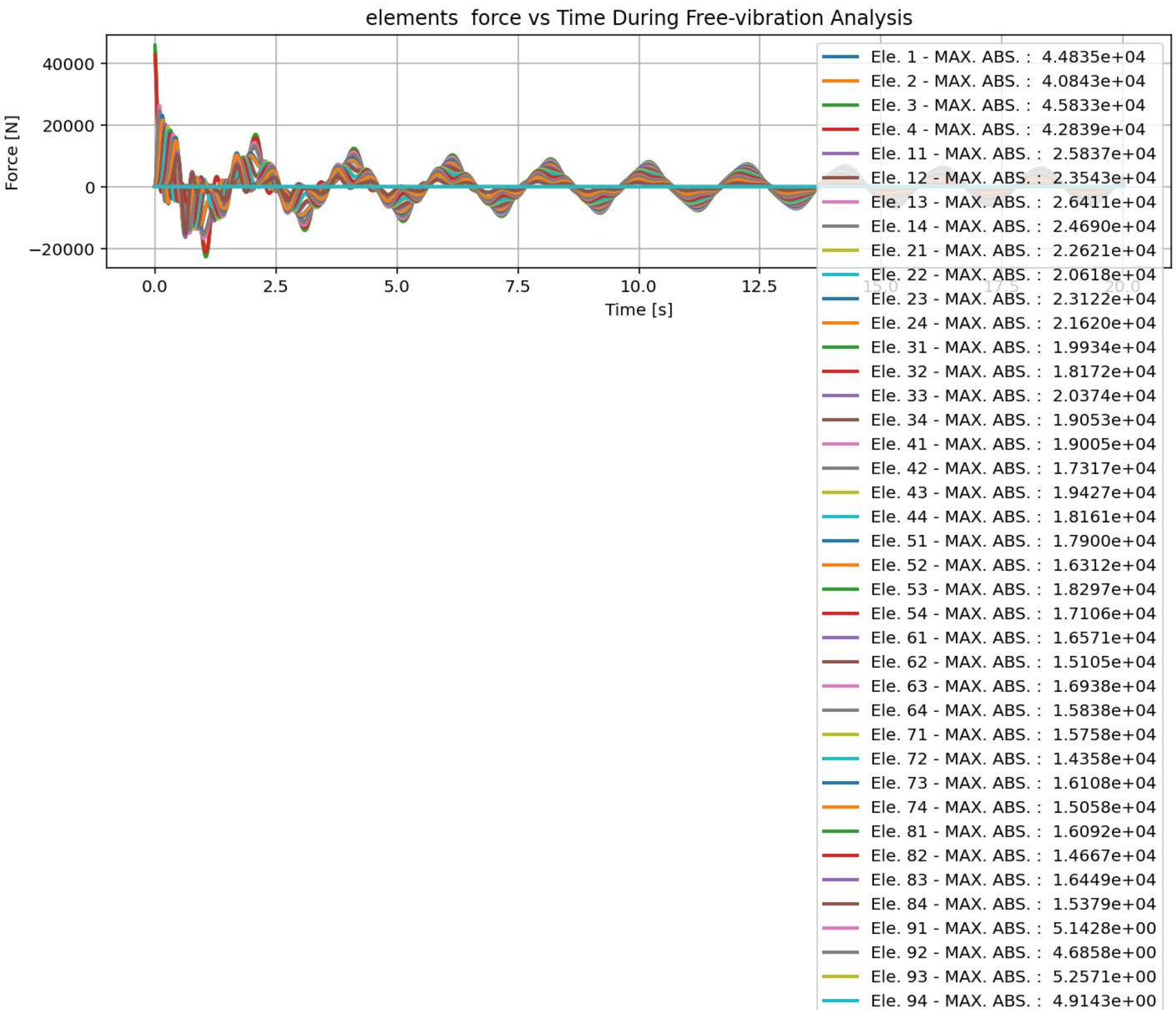
FREE-VIBRATION ANALYSIS

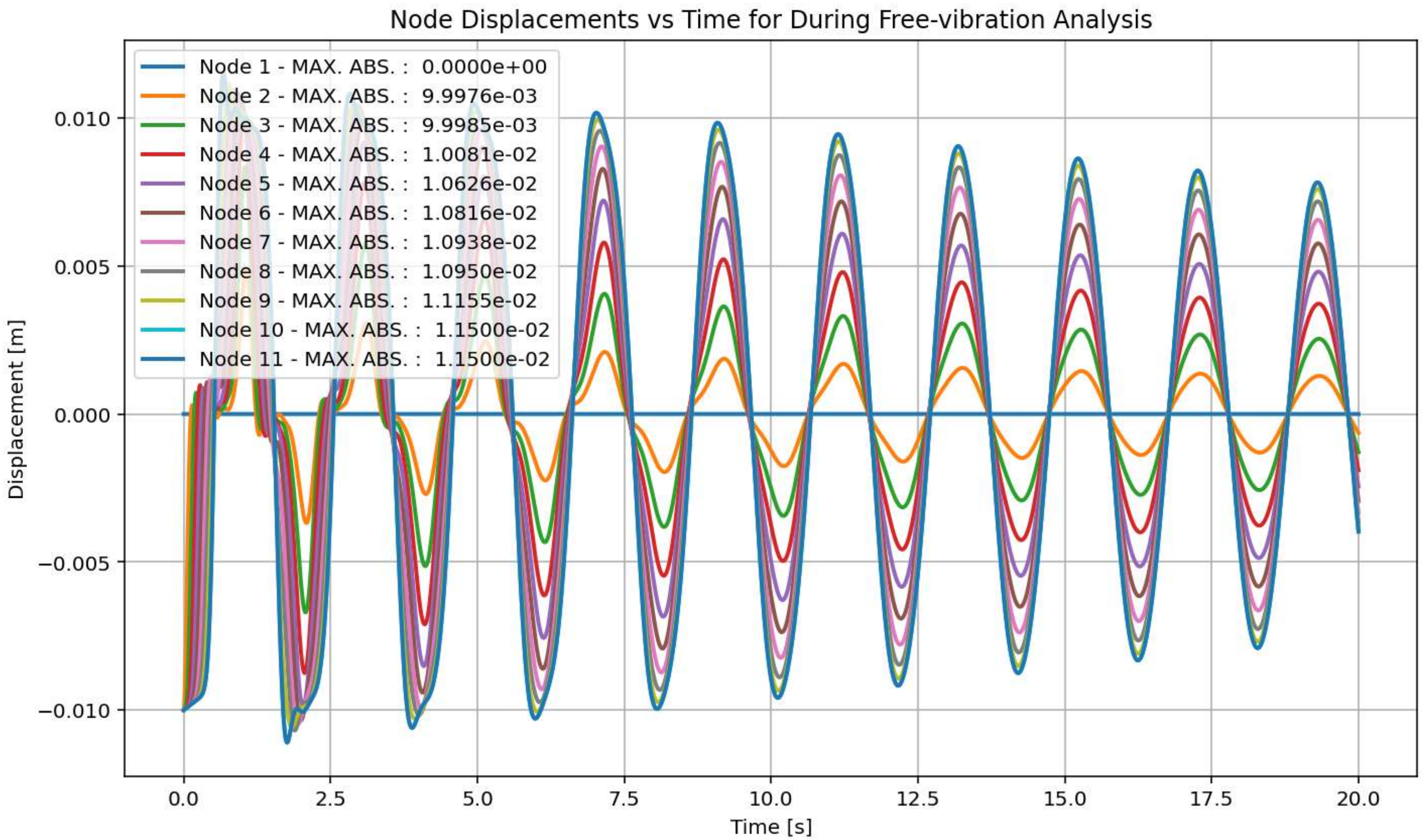

Spyder (Python 3.12)
File Edit Search Source Run Debug Consoles Projects Tools View Help
...S_04_COLUMNS_8_ANA+\+60_MDOF_10_FLOORS_04_COLUMNS_8_ANA
C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...MNS_8_ANA\P(t)_MDOF_10_FLOORS_04_COLUMNS_8_ANA.py
P(t)_MDOF_10_FLOORS_04_COLUMNS_8_ANA.py
# FREE-VIBRATION ANALYSIS (DYNAMIC TIME-HISTORY ANALYSIS)
MAT_TYPE = 'INELASTIC'   # 'ELASTIC' OR 'INELASTIC'
ANAL_TYPE = 'FREE-VIBRATION'
DATA = MDOF(MAT_TYPE, TOTAL_MASS, ANAL_TYPE)
(time_FV, reaction_FV, disp_FV, velo_FV, acc_FV, DI_FV,
 ele_force_FV, node_displacements_FV,
stiffness_FV, PERIOD_FV, damping_ratio_FV,
PERIOD_MIN_FV, PERIOD_MAX_FV) = DATA
XDATA = disp_FV
YDATA = reaction_FV
XLABEL = 'Displacement [m]'
YLABEL = 'Base Reaction [N]'
TITLE = 'Base Reaction and Dispalcement of Structure During Free-vibration Analysis'
COLOR = 'black'
SEMILOGY = False
PLOT(XDATA, YDATA, TITLE, XLABEL, YLABEL, COLOR, SEMILOGY)
PLOT_TIME_HISTORY(time_FV, reaction_FV, disp_FV, velo_FV, acc_FV)
# PLOT STRUCTURAL PERIOD DURING THE ANALYSIS
plt.figure(0, figsize=(12, 8))
plt.plot(time_FV, PERIOD_MIN_FV, linewidth=3)
plt.plot(time_FV, PERIOD_MAX_FV, linewidth=3)
plt.title('Period of Structure During Free-vibration Analysis')
plt.ylabel('Structural Period [s]')
plt.xlabel('Time [s]')
#plt.semilogy()
plt.grid()
plt.legend([f'PERIOD - MIN VALUES: Min: {np.min(PERIOD_MIN_FV):.3f} (s) - Mean: {np.mean(PERIOD_MI
            f'PERIOD - MAX VALUES: Min: {np.min(PERIOD_MAX_FV):.3f} (s) - Mean: {np.mean(PERIOD_M
            ])
plt.show()
15 %
IPython Console Plots Variable Explorer Help Debugger History Files
Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 1, Col 1 UTF-8 CRLF RW Mem 45%


Base Reaction and Dispalcement of Structure During Free-vibration Analysis
Base Reaction [N]
150000
100000
50000
0
−50000
−0.010
−0.005
0.000
0.005
0.010
Displacement [m]

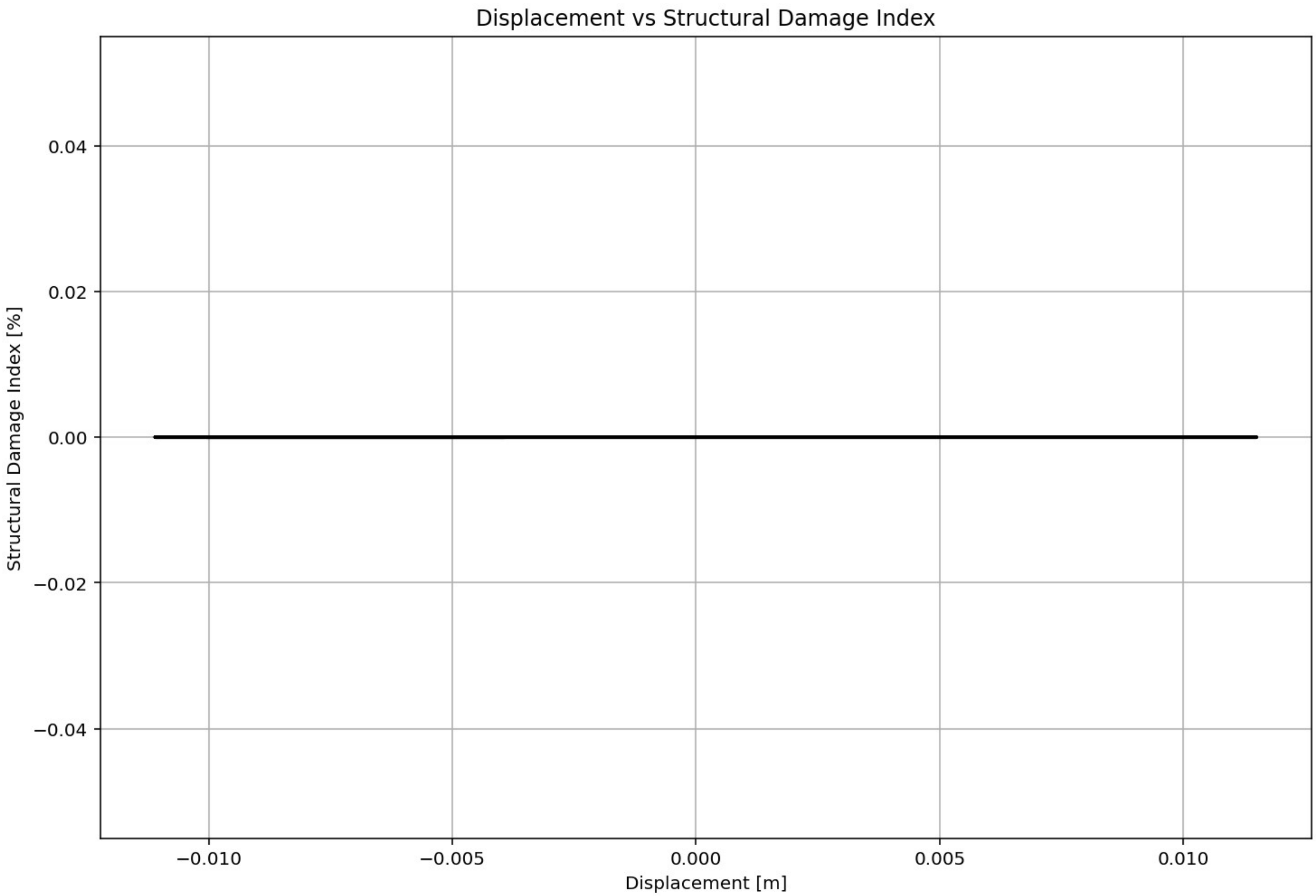

Reaction [N]
Time (s)
Displacement [m]
Time (s)
Velocity [m/s]
Time (s)
Acceleration [m/s²]
Time (s)

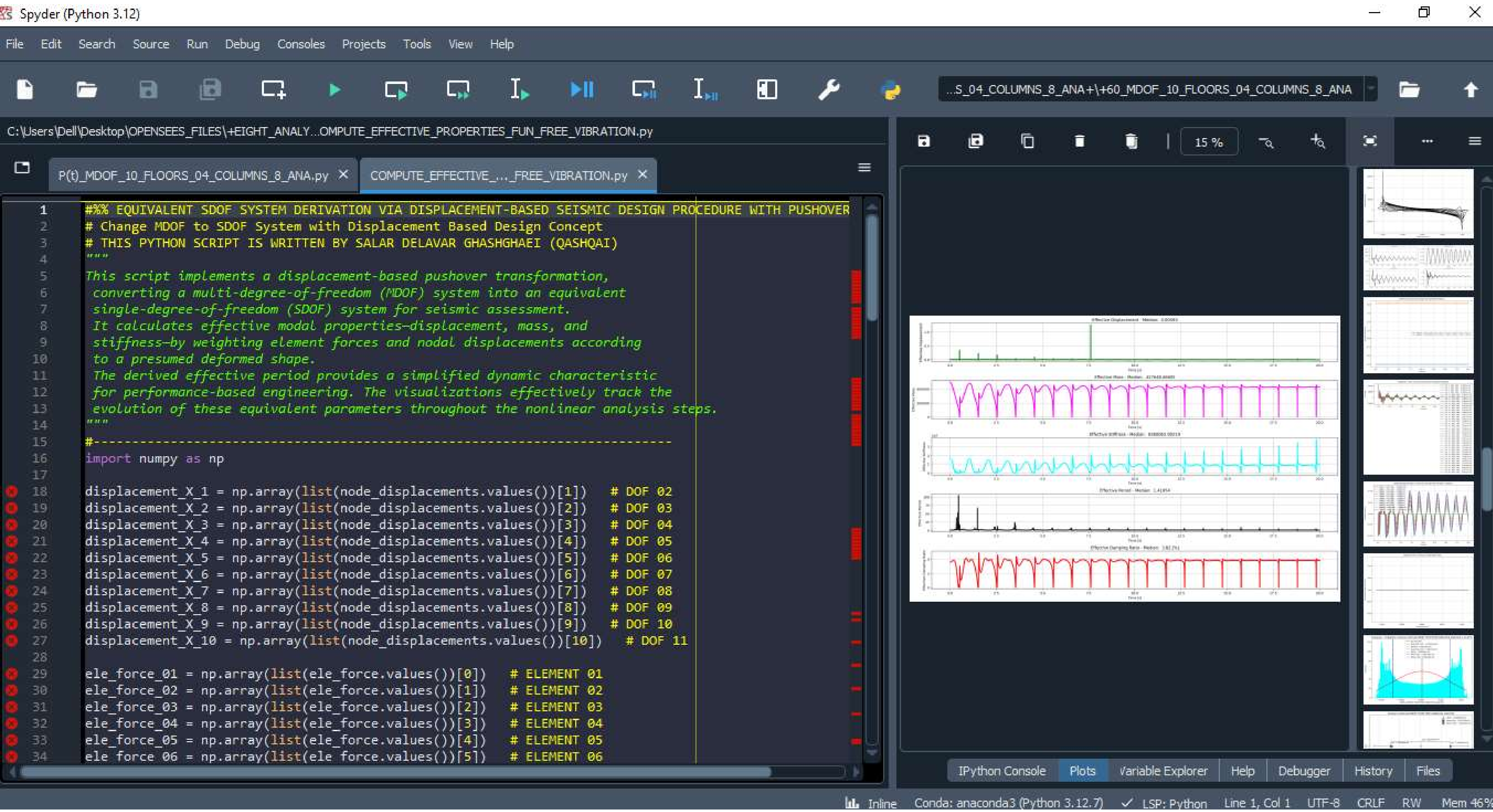

Period of Structure During Free-vibration Analysis
Structural Period [s]
Time [s]
PERIOD - MIN VALUES: Min: 0.248 (s) - Mean: 0.248 (s) - Max: 0.248 (s)
PERIOD - MAX VALUES: Min: 2.034 (s) - Mean: 2.034 (s) - Max: 2.034 (s)
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0


elements force vs Time During Free-vibration Analysis
Force [N]
Time [s]
40000
20000
0
−20000
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Ele. 1 - MAX. ABS. : 4.4835e+04
Ele. 2 - MAX. ABS. : 4.0843e+04
Ele. 3 - MAX. ABS. : 4.5833e+04
Ele. 4 - MAX. ABS. : 4.2839e+04
Ele. 11 - MAX. ABS. : 2.5837e+04
Ele. 12 - MAX. ABS. : 2.3543e+04
Ele. 13 - MAX. ABS. : 2.6411e+04
Ele. 14 - MAX. ABS. : 2.4690e+04
Ele. 21 - MAX. ABS. : 2.2621e+04
Ele. 22 - MAX. ABS. : 2.0618e+04
Ele. 23 - MAX. ABS. : 2.3122e+04
Ele. 24 - MAX. ABS. : 2.1620e+04
Ele. 31 - MAX. ABS. : 1.9934e+04
Ele. 32 - MAX. ABS. : 1.8172e+04
Ele. 33 - MAX. ABS. : 2.0374e+04
Ele. 34 - MAX. ABS. : 1.9053e+04
Ele. 41 - MAX. ABS. : 1.9005e+04
Ele. 42 - MAX. ABS. : 1.7317e+04
Ele. 43 - MAX. ABS. : 1.9427e+04
Ele. 44 - MAX. ABS. : 1.8161e+04
Ele. 51 - MAX. ABS. : 1.7900e+04
Ele. 52 - MAX. ABS. : 1.6312e+04
Ele. 53 - MAX. ABS. : 1.8297e+04
Ele. 54 - MAX. ABS. : 1.7106e+04
Ele. 61 - MAX. ABS. : 1.6571e+04
Ele. 62 - MAX. ABS. : 1.5105e+04
Ele. 63 - MAX. ABS. : 1.6938e+04
Ele. 64 - MAX. ABS. : 1.5838e+04
Ele. 71 - MAX. ABS. : 1.5758e+04
Ele. 72 - MAX. ABS. : 1.4358e+04
Ele. 73 - MAX. ABS. : 1.6108e+04
Ele. 74 - MAX. ABS. : 1.5058e+04
Ele. 81 - MAX. ABS. : 1.6092e+04
Ele. 82 - MAX. ABS. : 1.4667e+04
Ele. 83 - MAX. ABS. : 1.6449e+04
Ele. 84 - MAX. ABS. : 1.5379e+04
Ele. 91 - MAX. ABS. : 5.1428e+00
Ele. 92 - MAX. ABS. : 4.6858e+00
Ele. 93 - MAX. ABS. : 5.2571e+00
Ele. 94 - MAX. ABS. : 4.9143e+00

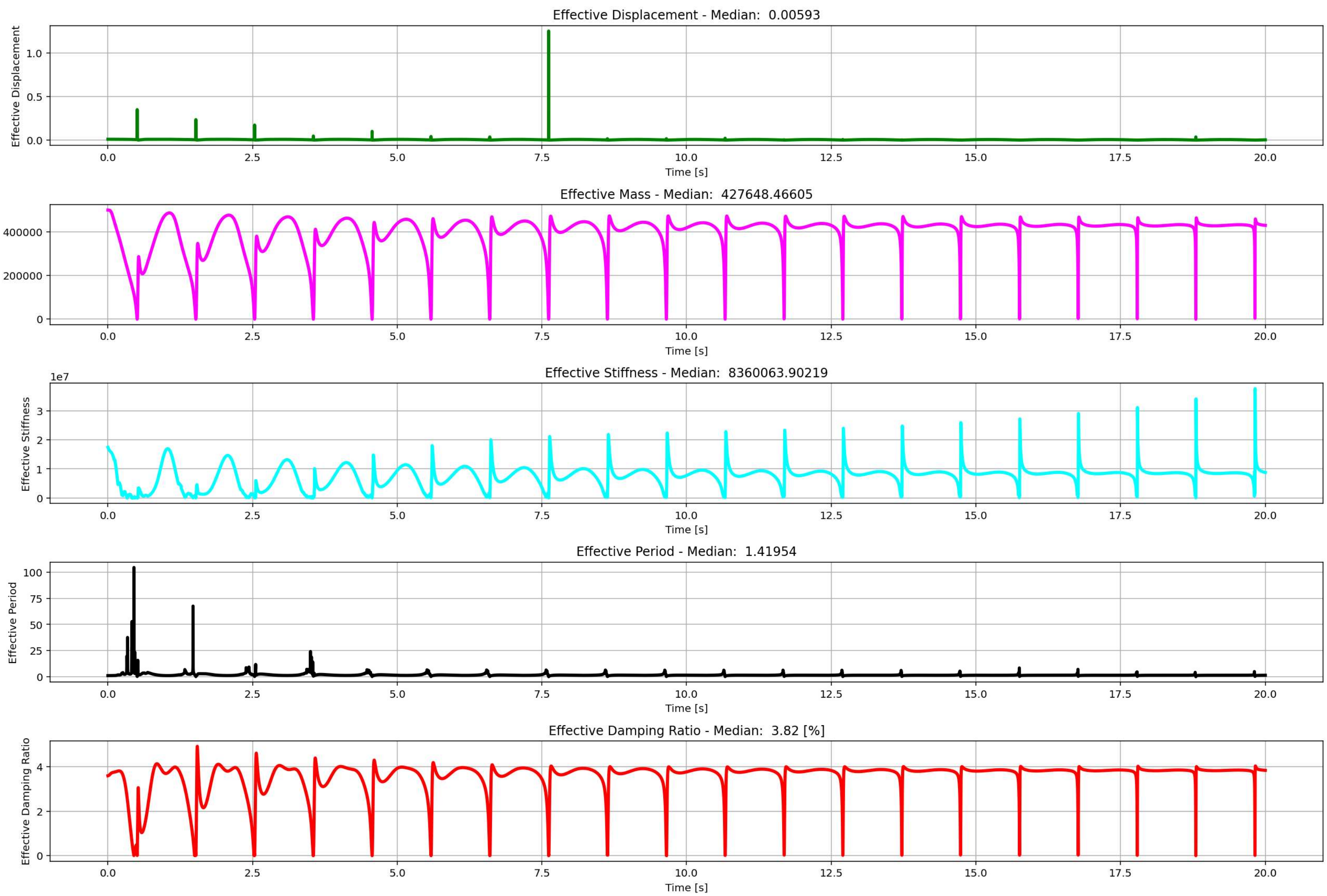


Node Displacements vs Time for During Free-vibration Analysis
Node 1 - MAX. ABS. : 0.0000e+00
Node 2 - MAX. ABS. : 9.9976e-03
Node 3 - MAX. ABS. : 9.9985e-03
Node 4 - MAX. ABS. : 1.0081e-02
Node 5 - MAX. ABS. : 1.0626e-02
Node 6 - MAX. ABS. : 1.0816e-02
Node 7 - MAX. ABS. : 1.0938e-02
Node 8 - MAX. ABS. : 1.0950e-02
Node 9 - MAX. ABS. : 1.1155e-02
Node 10 - MAX. ABS. : 1.1500e-02
Node 11 - MAX. ABS. : 1.1500e-02
Displacement [m]
Time [s]
0.010
0.005
0.000
−0.005
−0.010
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0


Displacement vs Structural Damage Index
Structural Damage Index [%]
0.04
0.02
0.00
−0.02
−0.04
−0.010
−0.005
0.000
0.005
0.010
Displacement [m]



Spyder (Python 3.12)
File Edit Search Source Run Debug Consoles Projects Tools View Help
...S_04_COLUMNS_8_ANA+\+60_MDOF_10_FLOORS_04_COLUMNS_8_ANA
C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...OMPUTE_EFFECTIVE_PROPERTIES_FUN_FREE_VIBRATION.py
P(t)_MDOF_10_FLOORS_04_COLUMNS_8_ANA.py
COMPUTE_EFFECTIVE_..._FREE_VIBRATION.py
15 %
#%% EQUIVALENT SDOF SYSTEM DERIVATION VIA DISPLACEMENT-BASED SEISMIC DESIGN PROCEDURE WITH PUSHOVER
Change MDOF to SDOF System with Displacement Based Design Concept
THIS PYTHON SCRIPT IS WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)
"""
This script implements a displacement-based pushover transformation,
converting a multi-degree-of-freedom (MDOF) system into an equivalent
single-degree-of-freedom (SDOF) system for seismic assessment.
It calculates effective modal properties—displacement, mass, and
stiffness—by weighting element forces and nodal displacements according
to a presumed deformed shape.
The derived effective period provides a simplified dynamic characteristic
for performance-based engineering. The visualizations effectively track the
evolution of these equivalent parameters throughout the nonlinear analysis steps.
"""
#--
import numpy as np
displacement_X_1 = np.array(list(node_displacements.values())[1])   # DOF 02
displacement_X_2 = np.array(list(node_displacements.values())[2])   # DOF 03
displacement_X_3 = np.array(list(node_displacements.values())[3])   # DOF 04
displacement_X_4 = np.array(list(node_displacements.values())[4])   # DOF 05
displacement_X_5 = np.array(list(node_displacements.values())[5])   # DOF 06
displacement_X_6 = np.array(list(node_displacements.values())[6])   # DOF 07
displacement_X_7 = np.array(list(node_displacements.values())[7])   # DOF 08
displacement_X_8 = np.array(list(node_displacements.values())[8])   # DOF 09
displacement_X_9 = np.array(list(node_displacements.values())[9])   # DOF 10
displacement_X_10 = np.array(list(node_displacements.values())[10])   # DOF 11
ele_force_01 = np.array(list(ele_force.values())[0])   # ELEMENT 01
ele_force_02 = np.array(list(ele_force.values())[1])   # ELEMENT 02
ele_force_03 = np.array(list(ele_force.values())[2])   # ELEMENT 03
ele_force_04 = np.array(list(ele_force.values())[3])   # ELEMENT 04
ele_force_05 = np.array(list(ele_force.values())[4])   # ELEMENT 05
ele_force_06 = np.array(list(ele_force.values())[5])   # ELEMENT 06
IPython Console Plots Variable Explorer Help Debugger History Files
Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 1, Col 1 UTF-8 CRLF RW Mem 46%



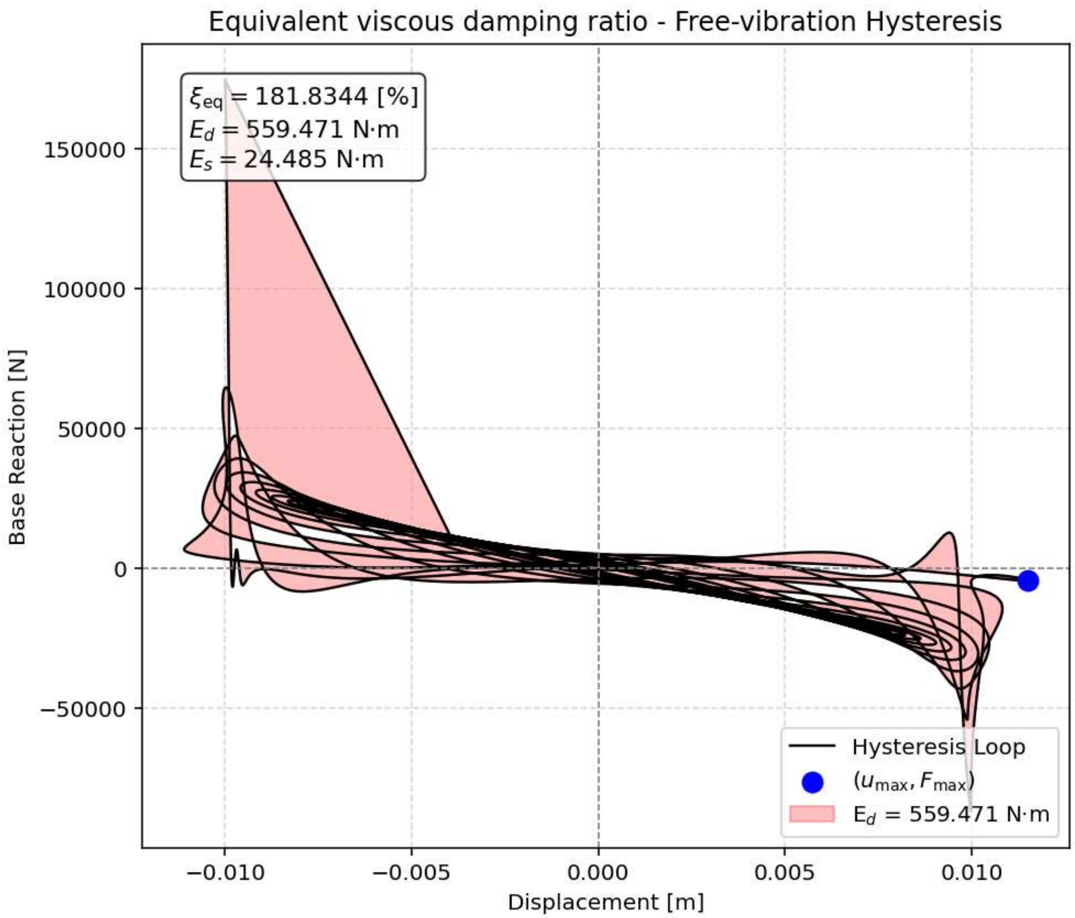
$$\Delta_d = \frac{\sum_{i=1}^{n} m_i \Delta_i^2}{\sum_{i=1}^{n} m_i \Delta_i}$$

$$m_{eff} = \frac{\sum_{i=1}^{n} m_i \Delta_i}{\Delta_d}$$

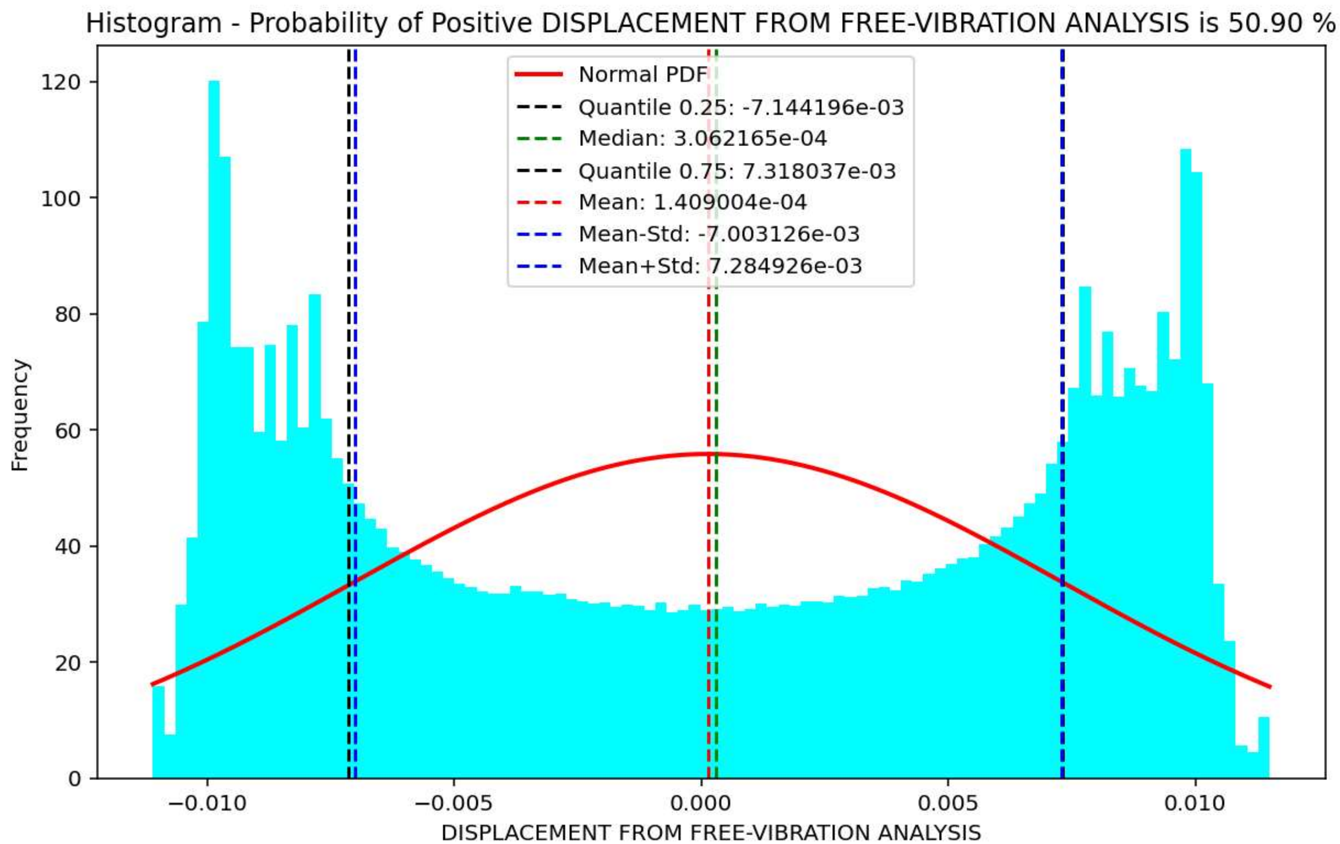
$$k_{eff} = \frac{\sum_{i=1}^{n} k_i \Delta_i}{\Delta_d}$$

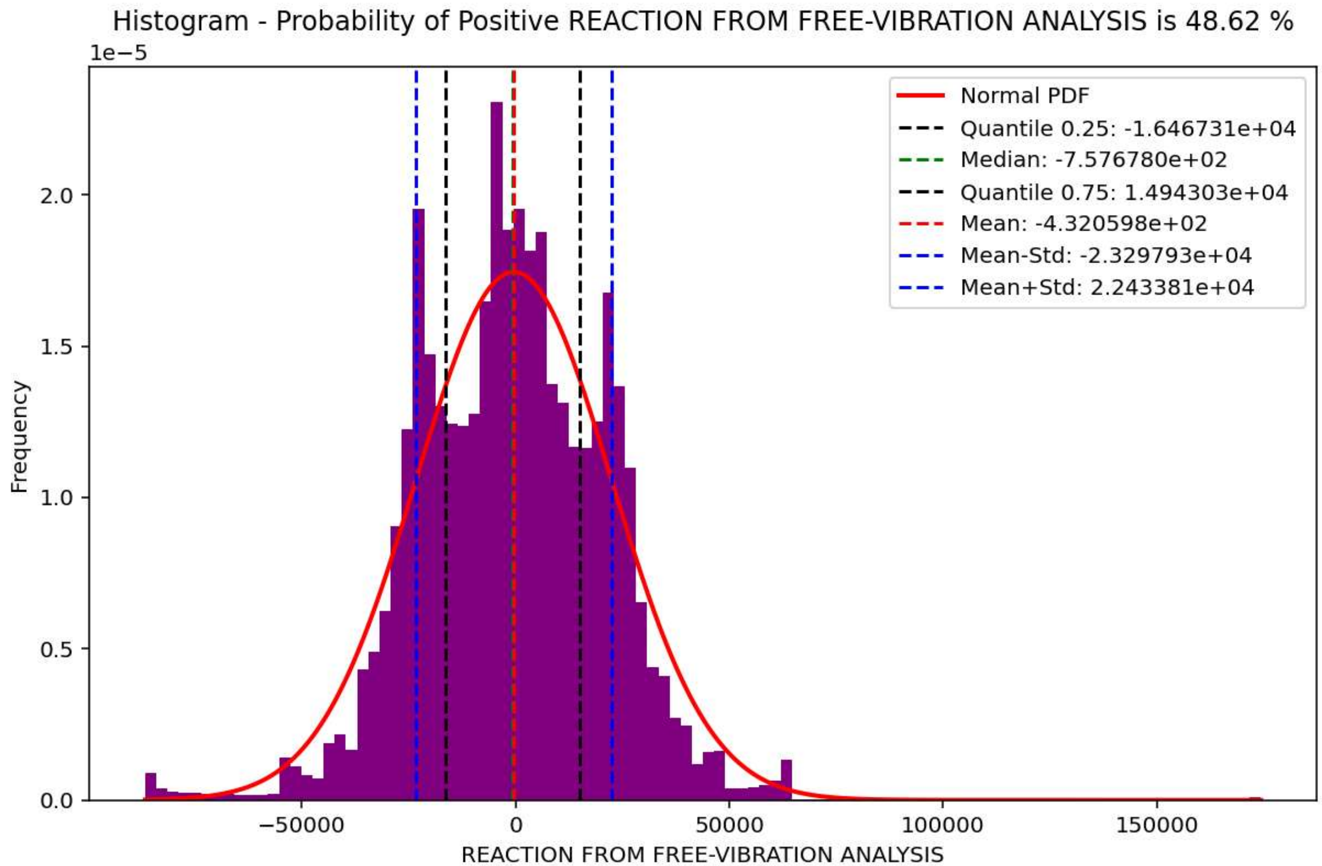
$$T_{eff} = \frac{2\pi}{\sqrt{\frac{k_{eff}}{m_{eff}}}}$$

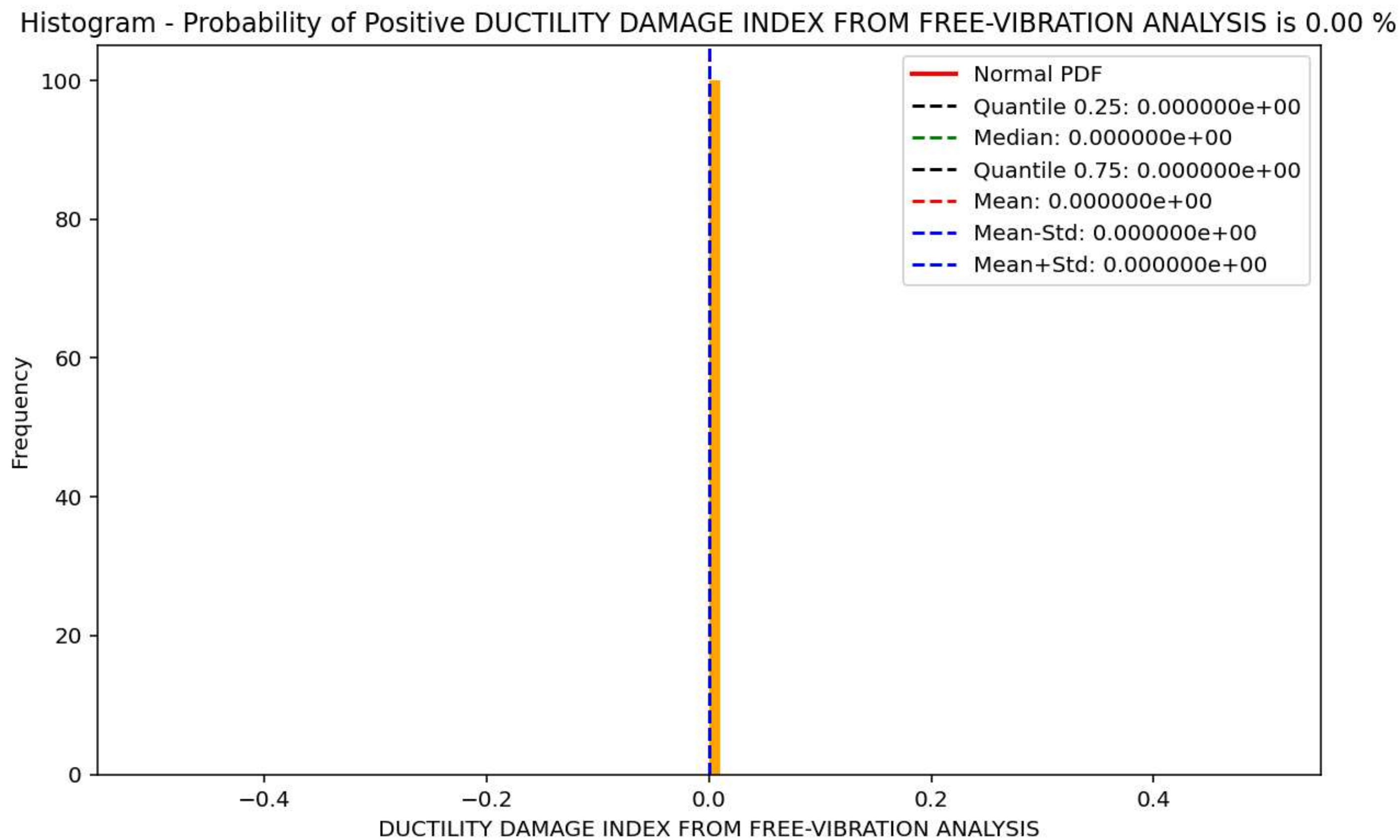
$$\xi_{eff} = \frac{\sum_{i=1}^{n} \xi_i \Delta_i}{\Delta_d}$$



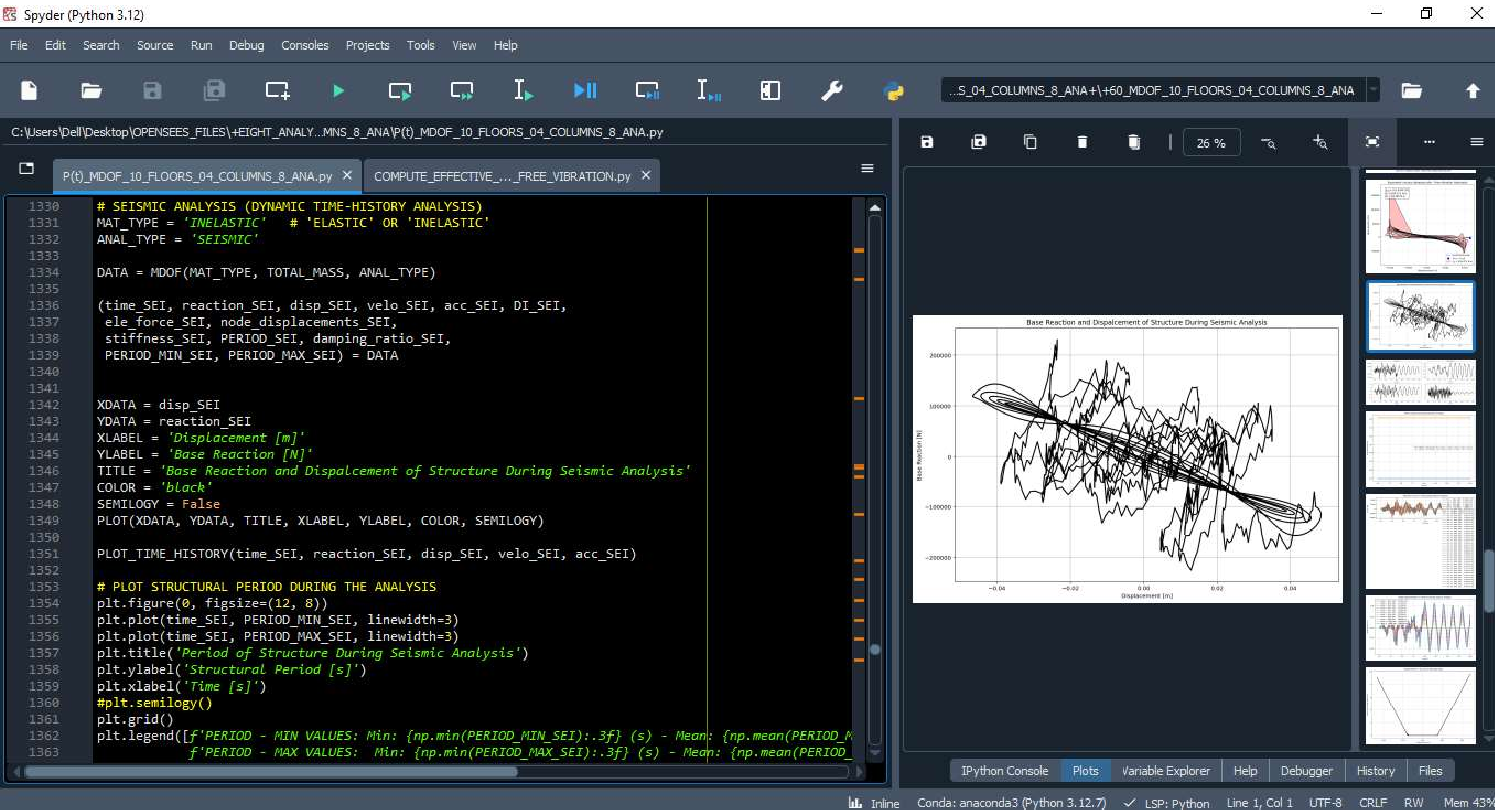
Equivalent viscous damping ratio - Free-vibration Hysteresis
$\xi_{eq} = 181.8344$ [%]
$E_d = 559.471$ N·m
$E_s = 24.485$ N·m
150000
100000
50000
0
−50000
Base Reaction [N]
−0.010
−0.005
0.000
0.005
0.010
Displacement [m]
Hysteresis Loop
(u_{max}, F_{max})
E_d = 559.471 N·m

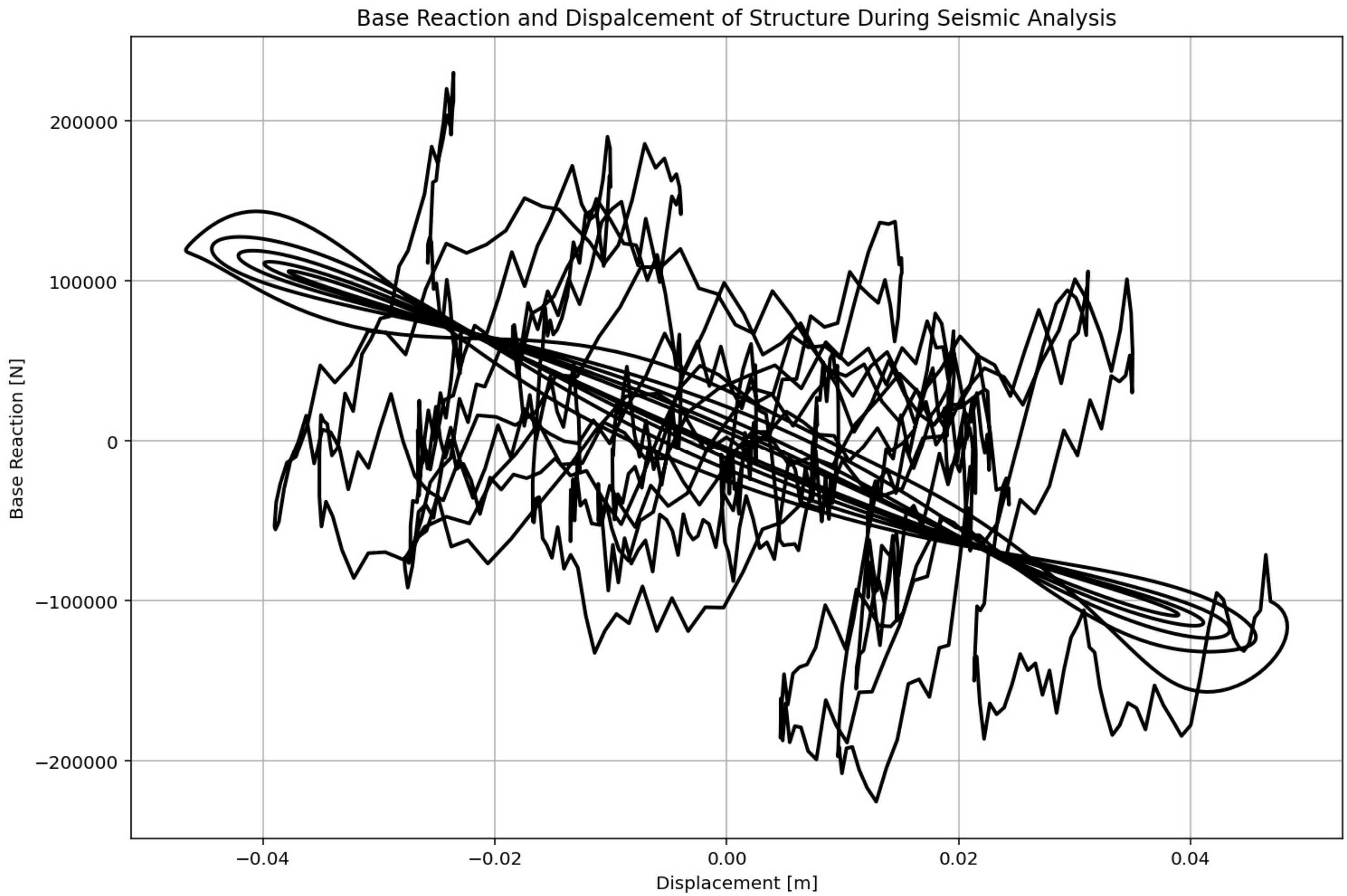

Histogram - Probability of Positive DISPLACEMENT FROM FREE-VIBRATION ANALYSIS is 50.90 %
Normal PDF
Quantile 0.25: -7.144196e-03
Median: 3.062165e-04
Quantile 0.75: 7.318037e-03
Mean: 1.409004e-04
Mean-Std: -7.003126e-03
Mean+Std: 7.284926e-03
Frequency
0
20
40
60
80
100
120
−0.010
−0.005
0.000
0.005
0.010
DISPLACEMENT FROM FREE-VIBRATION ANALYSIS

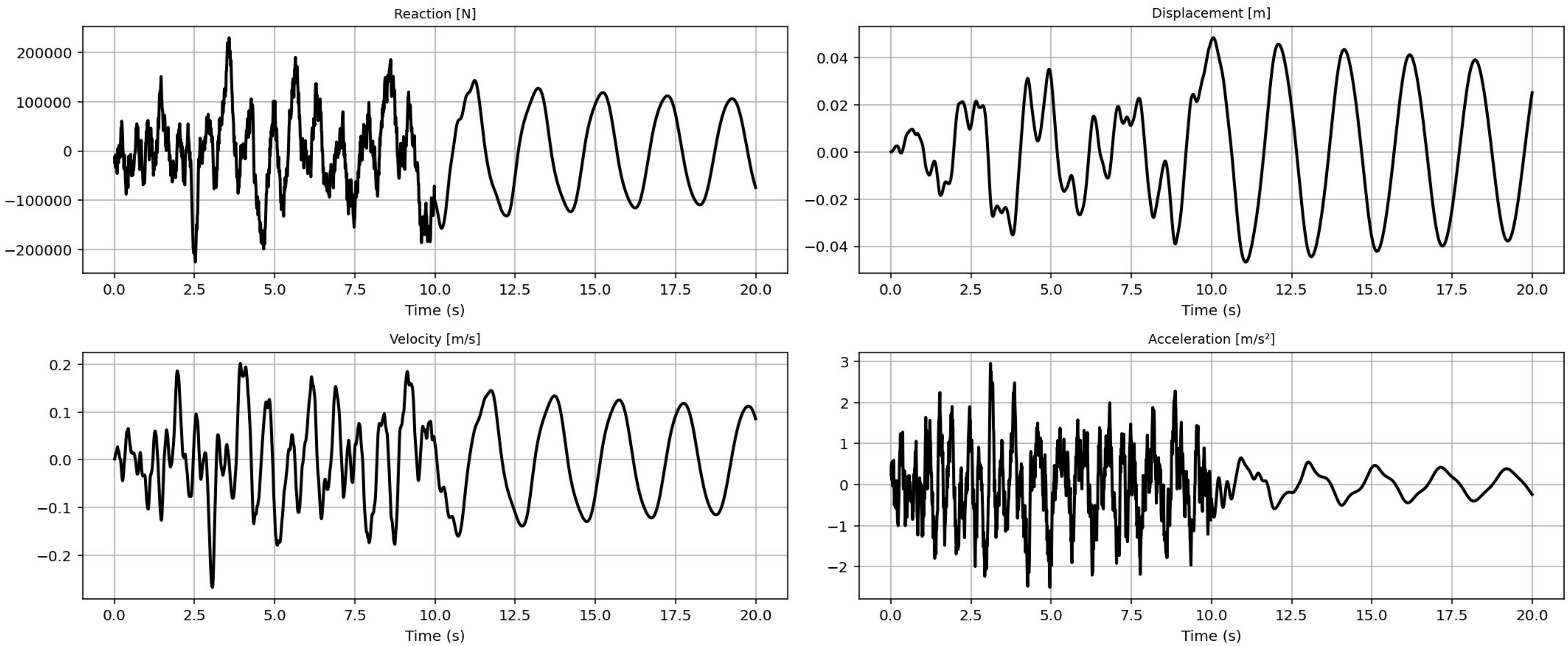

Histogram - Probability of Positive REACTION FROM FREE-VIBRATION ANALYSIS is 48.62 %
1e−5
Normal PDF
Quantile 0.25: -1.646731e+04
Median: -7.576780e+02
Quantile 0.75: 1.494303e+04
Mean: -4.320598e+02
Mean-Std: -2.329793e+04
Mean+Std: 2.243381e+04
Frequency
0.0
0.5
1.0
1.5
2.0
−50000
0
50000
100000
150000
REACTION FROM FREE-VIBRATION ANALYSIS

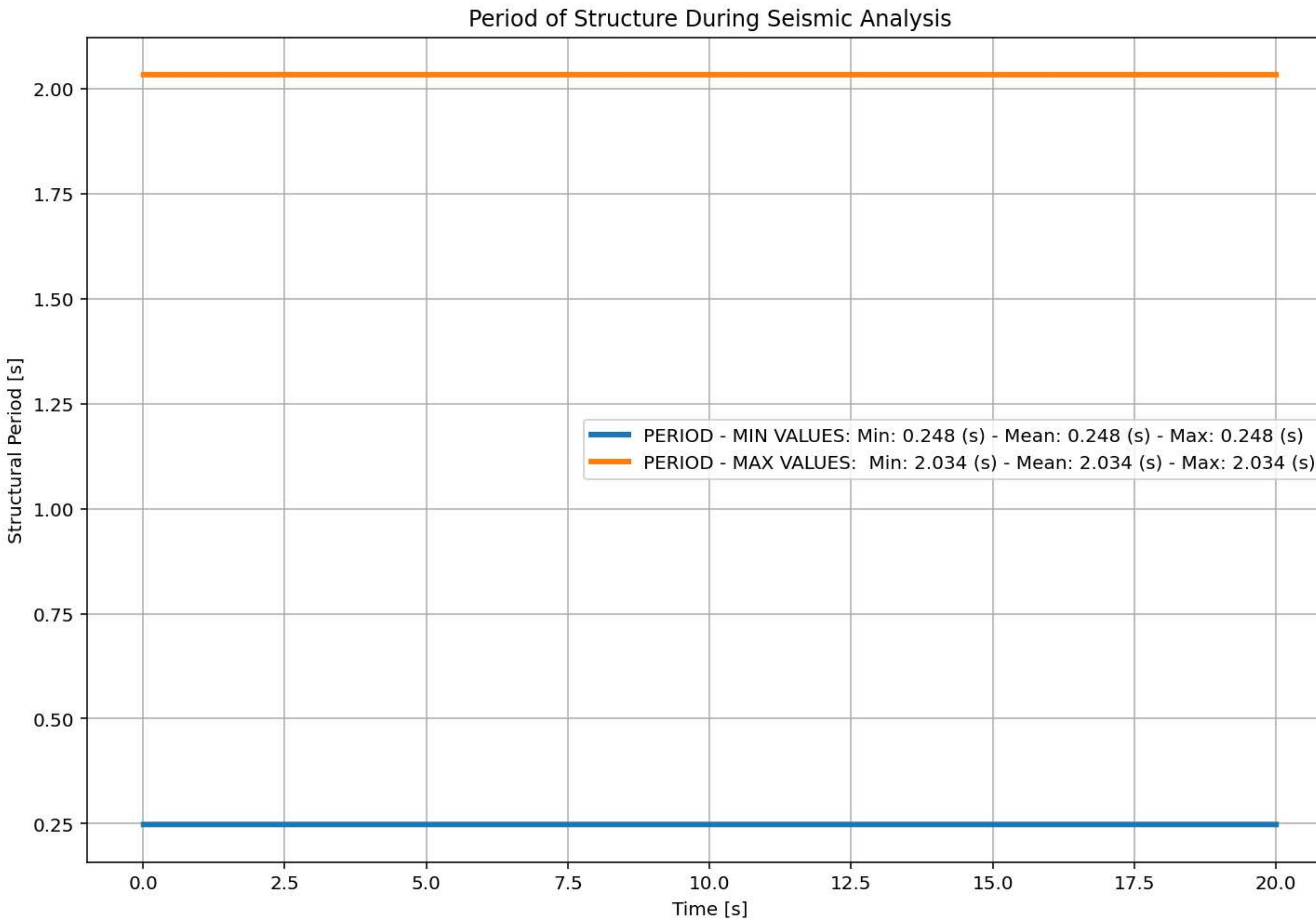

Histogram - Probability of Positive DUCTILITY DAMAGE INDEX FROM FREE-VIBRATION ANALYSIS is 0.00 %
Normal PDF
Quantile 0.25: 0.000000e+00
Median: 0.000000e+00
Quantile 0.75: 0.000000e+00
Mean: 0.000000e+00
Mean-Std: 0.000000e+00
Mean+Std: 0.000000e+00
Frequency
0
20
40
60
80
100
−0.4
−0.2
0.0
0.2
0.4
DUCTILITY DAMAGE INDEX FROM FREE-VIBRATION ANALYSIS

SEISMIC ANALYSIS

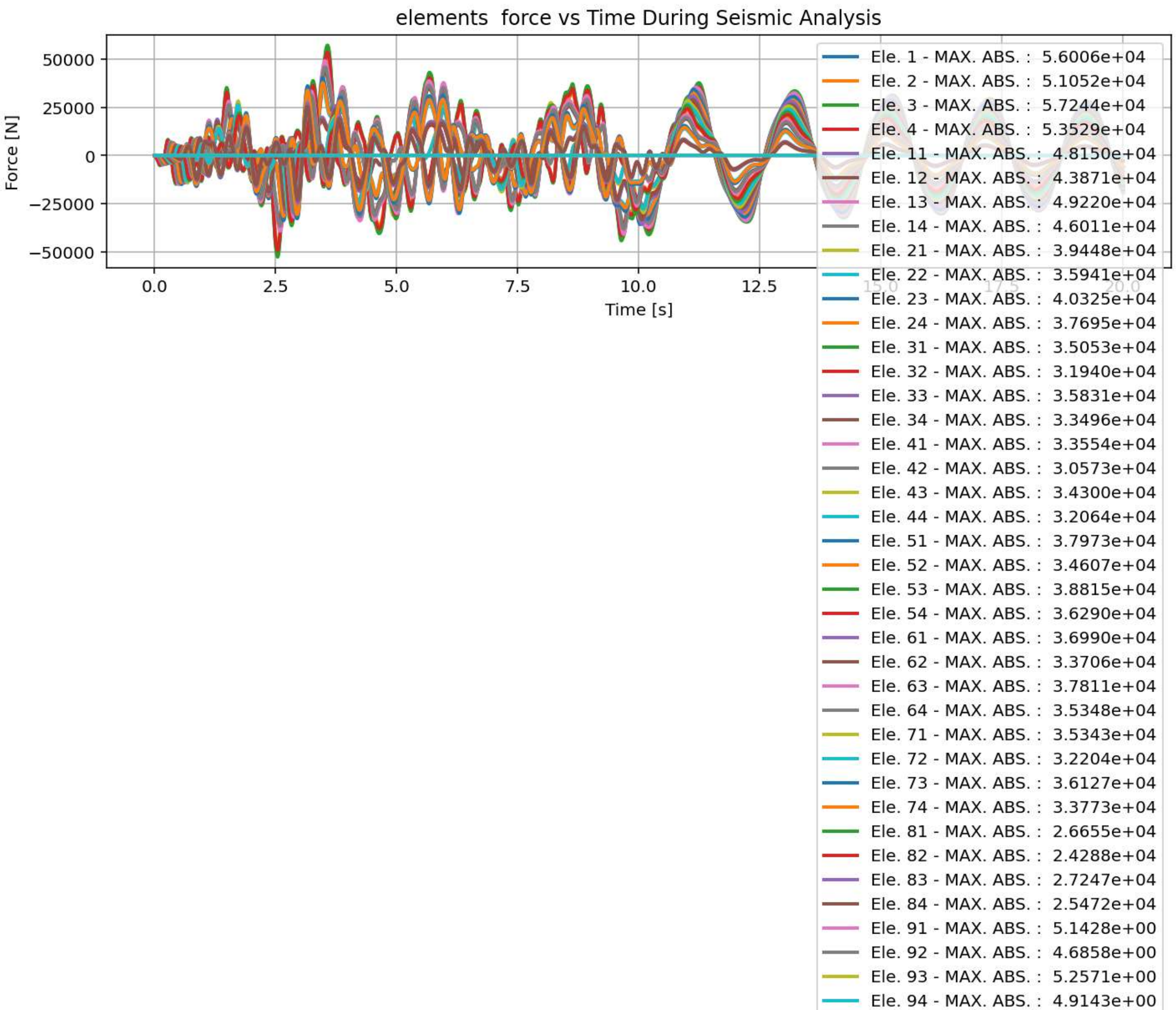

Spyder (Python 3.12)
File Edit Search Source Run Debug Consoles Projects Tools View Help
...S_04_COLUMNS_8_ANA+\+60_MDOF_10_FLOORS_04_COLUMNS_8_ANA
C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...MNS_8_ANA\P(t)_MDOF_10_FLOORS_04_COLUMNS_8_ANA.py
P(t)_MDOF_10_FLOORS_04_COLUMNS_8_ANA.py
COMPUTE_EFFECTIVE_..._FREE_VIBRATION.py
# SEISMIC ANALYSIS (DYNAMIC TIME-HISTORY ANALYSIS)
MAT_TYPE = 'INELASTIC'   # 'ELASTIC' OR 'INELASTIC'
ANAL_TYPE = 'SEISMIC'
DATA = MDOF(MAT_TYPE, TOTAL_MASS, ANAL_TYPE)
(time_SEI, reaction_SEI, disp_SEI, velo_SEI, acc_SEI, DI_SEI,
 ele_force_SEI, node_displacements_SEI,
 stiffness_SEI, PERIOD_SEI, damping_ratio_SEI,
 PERIOD_MIN_SEI, PERIOD_MAX_SEI) = DATA
XDATA = disp_SEI
YDATA = reaction_SEI
XLABEL = 'Displacement [m]'
YLABEL = 'Base Reaction [N]'
TITLE = 'Base Reaction and Dispalcement of Structure During Seismic Analysis'
COLOR = 'black'
SEMILOGY = False
PLOT(XDATA, YDATA, TITLE, XLABEL, YLABEL, COLOR, SEMILOGY)
PLOT_TIME_HISTORY(time_SEI, reaction_SEI, disp_SEI, velo_SEI, acc_SEI)
# PLOT STRUCTURAL PERIOD DURING THE ANALYSIS
plt.figure(0, figsize=(12, 8))
plt.plot(time_SEI, PERIOD_MIN_SEI, linewidth=3)
plt.plot(time_SEI, PERIOD_MAX_SEI, linewidth=3)
plt.title('Period of Structure During Seismic Analysis')
plt.ylabel('Structural Period [s]')
plt.xlabel('Time [s]')
#plt.semilogy()
plt.grid()
plt.legend([f'PERIOD - MIN VALUES: Min: {np.min(PERIOD_MIN_SEI):.3f} (s) - Mean: {np.mean(PERIOD_M
            f'PERIOD - MAX VALUES: Min: {np.min(PERIOD_MAX_SEI):.3f} (s) - Mean: {np.mean(PERIOD_
26 %
Base Reaction and Dispalcement of Structure During Seismic Analysis
200000
100000
0
−100000
−200000
−0.04
−0.02
0.00
0.02
0.04
Displacement [m]
Base Reaction [N]
IPython Console
Plots
Variable Explorer
Help
Debugger
History
Files
Inline
Conda: anaconda3 (Python 3.12.7)
LSP: Python
Line 1, Col 1
UTF-8
CRLF
RW
Mem 43%

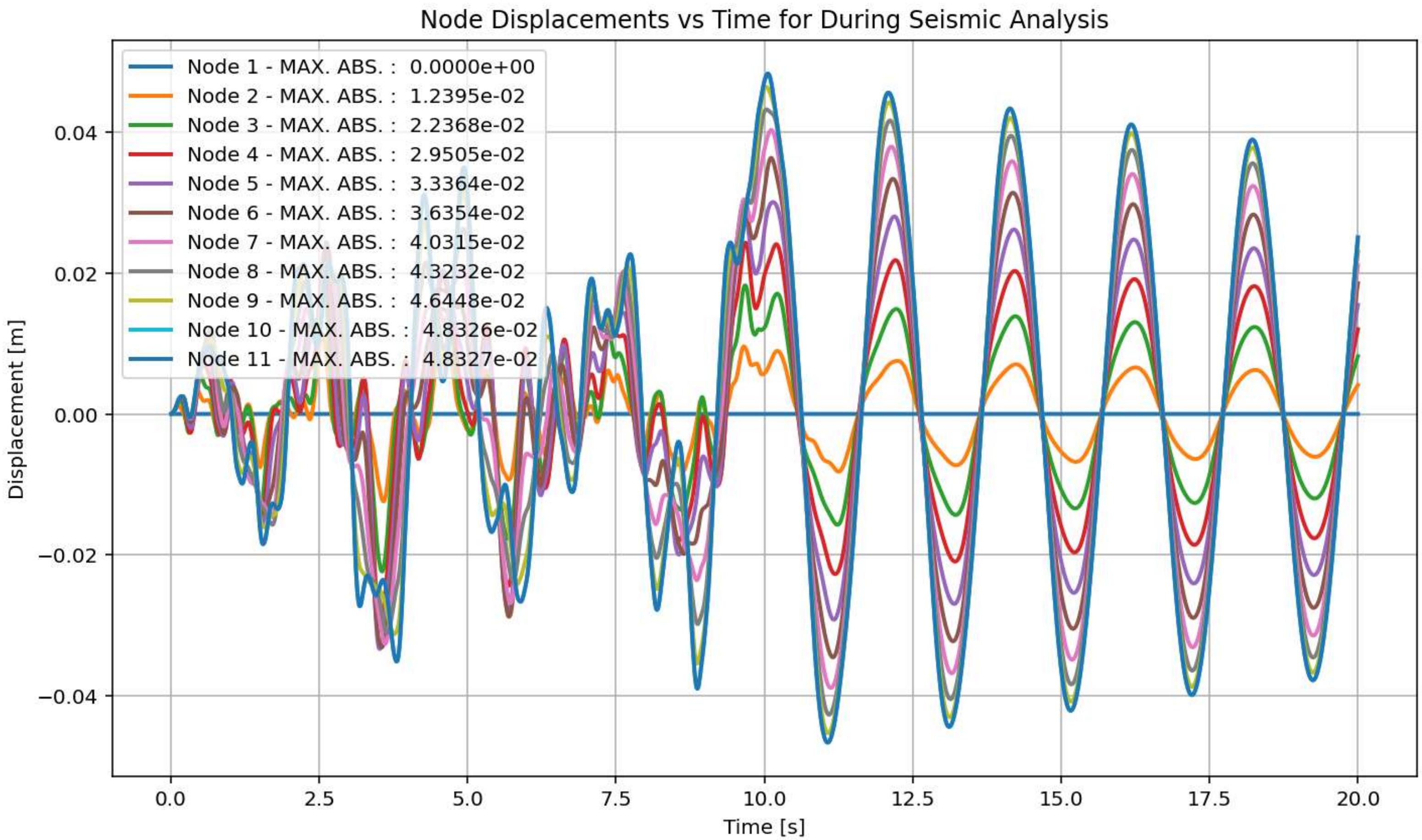

Base Reaction and Dispalcement of Structure During Seismic Analysis
Base Reaction [N]
200000
100000
0
−100000
−200000
−0.04
−0.02
0.00
0.02
0.04
Displacement [m]


Reaction [N]
200000
100000
0
−100000
−200000
Displacement [m]
0.04
0.02
0.00
−0.02
−0.04
Velocity [m/s]
0.2
0.1
0.0
−0.1
−0.2
Acceleration [m/s²]
3
2
1
0
−1
−2
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time (s)

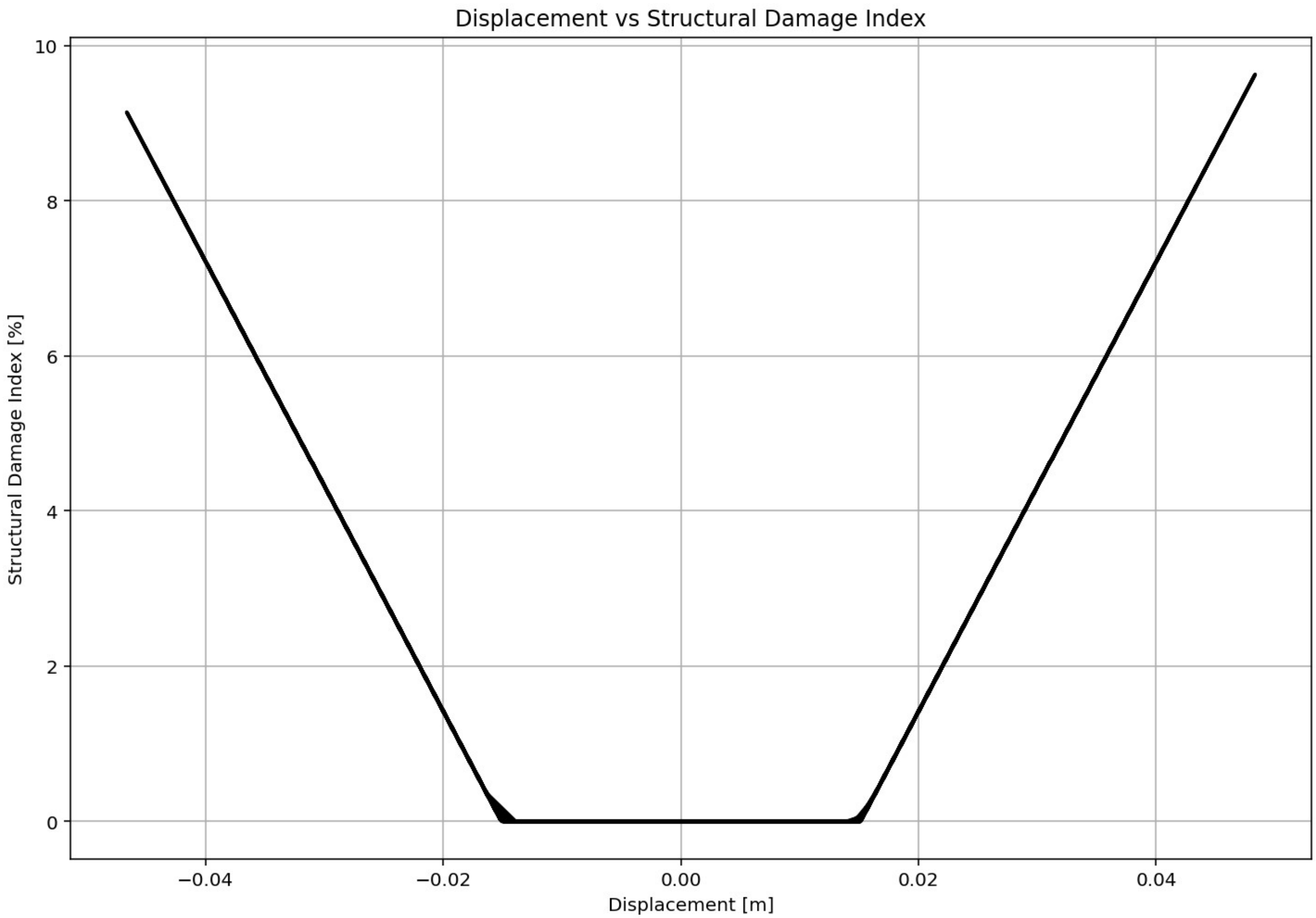


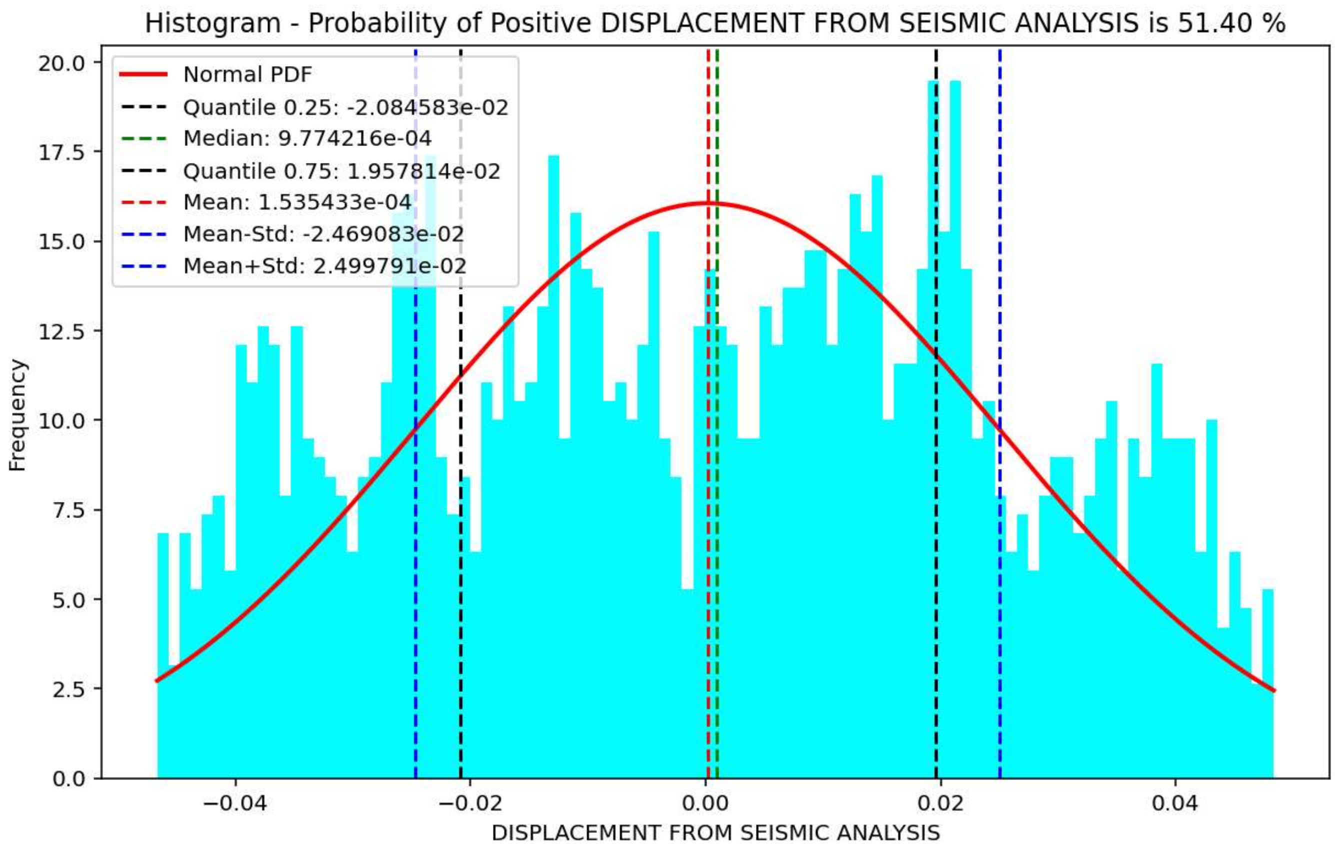
Period of Structure During Seismic Analysis
PERIOD - MIN VALUES: Min: 0.248 (s) - Mean: 0.248 (s) - Max: 0.248 (s)
PERIOD - MAX VALUES: Min: 2.034 (s) - Mean: 2.034 (s) - Max: 2.034 (s)
Structural Period [s]
Time [s]
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0

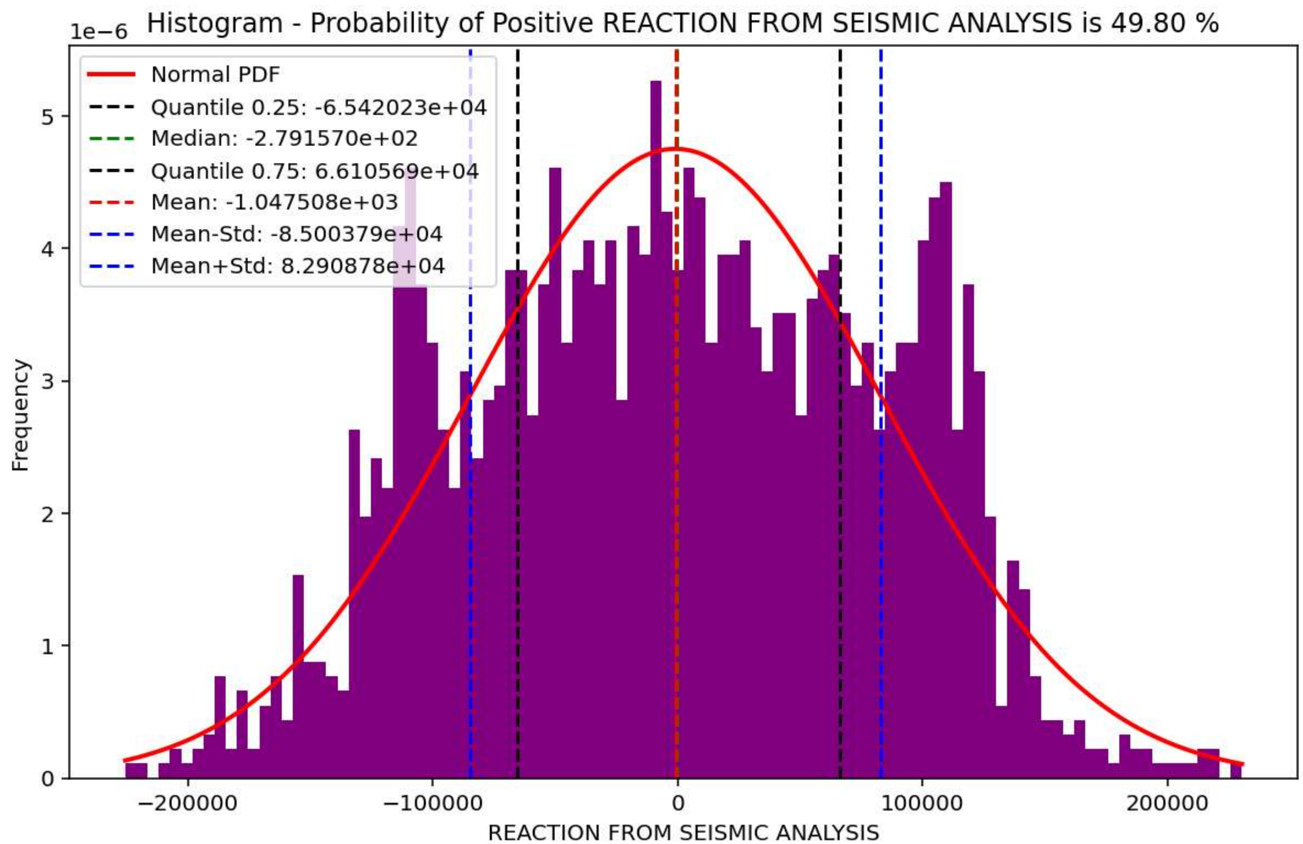

elements force vs Time During Seismic Analysis
Force [N]
Time [s]
50000
25000
0
−25000
−50000
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Ele. 1 - MAX. ABS. : 5.6006e+04
Ele. 2 - MAX. ABS. : 5.1052e+04
Ele. 3 - MAX. ABS. : 5.7244e+04
Ele. 4 - MAX. ABS. : 5.3529e+04
Ele. 11 - MAX. ABS. : 4.8150e+04
Ele. 12 - MAX. ABS. : 4.3871e+04
Ele. 13 - MAX. ABS. : 4.9220e+04
Ele. 14 - MAX. ABS. : 4.6011e+04
Ele. 21 - MAX. ABS. : 3.9448e+04
Ele. 22 - MAX. ABS. : 3.5941e+04
Ele. 23 - MAX. ABS. : 4.0325e+04
Ele. 24 - MAX. ABS. : 3.7695e+04
Ele. 31 - MAX. ABS. : 3.5053e+04
Ele. 32 - MAX. ABS. : 3.1940e+04
Ele. 33 - MAX. ABS. : 3.5831e+04
Ele. 34 - MAX. ABS. : 3.3496e+04
Ele. 41 - MAX. ABS. : 3.3554e+04
Ele. 42 - MAX. ABS. : 3.0573e+04
Ele. 43 - MAX. ABS. : 3.4300e+04
Ele. 44 - MAX. ABS. : 3.2064e+04
Ele. 51 - MAX. ABS. : 3.7973e+04
Ele. 52 - MAX. ABS. : 3.4607e+04
Ele. 53 - MAX. ABS. : 3.8815e+04
Ele. 54 - MAX. ABS. : 3.6290e+04
Ele. 61 - MAX. ABS. : 3.6990e+04
Ele. 62 - MAX. ABS. : 3.3706e+04
Ele. 63 - MAX. ABS. : 3.7811e+04
Ele. 64 - MAX. ABS. : 3.5348e+04
Ele. 71 - MAX. ABS. : 3.5343e+04
Ele. 72 - MAX. ABS. : 3.2204e+04
Ele. 73 - MAX. ABS. : 3.6127e+04
Ele. 74 - MAX. ABS. : 3.3773e+04
Ele. 81 - MAX. ABS. : 2.6655e+04
Ele. 82 - MAX. ABS. : 2.4288e+04
Ele. 83 - MAX. ABS. : 2.7247e+04
Ele. 84 - MAX. ABS. : 2.5472e+04
Ele. 91 - MAX. ABS. : 5.1428e+00
Ele. 92 - MAX. ABS. : 4.6858e+00
Ele. 93 - MAX. ABS. : 5.2571e+00
Ele. 94 - MAX. ABS. : 4.9143e+00



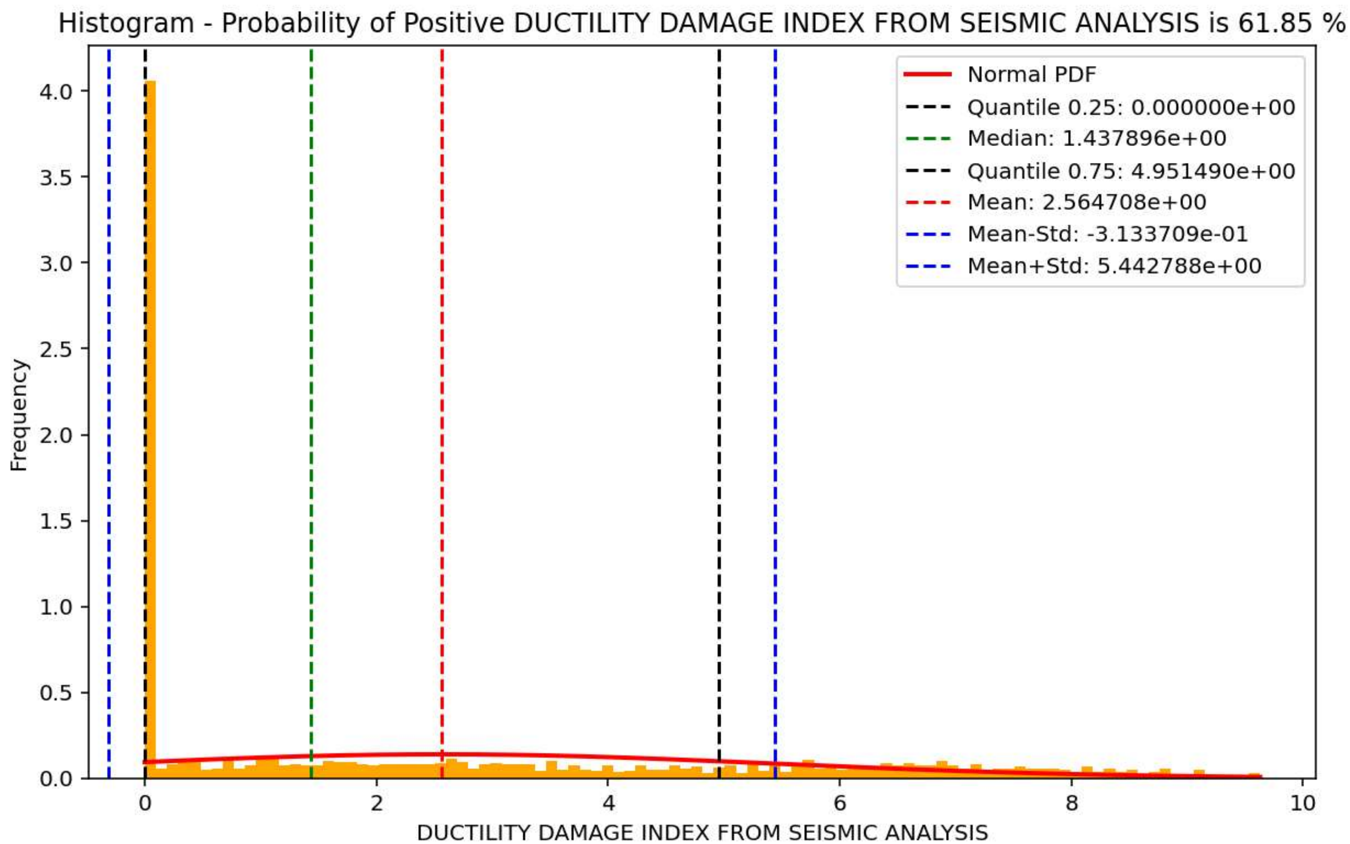
Node Displacements vs Time for During Seismic Analysis
Node 1 - MAX. ABS. : 0.0000e+00
Node 2 - MAX. ABS. : 1.2395e-02
Node 3 - MAX. ABS. : 2.2368e-02
Node 4 - MAX. ABS. : 2.9505e-02
Node 5 - MAX. ABS. : 3.3364e-02
Node 6 - MAX. ABS. : 3.6354e-02
Node 7 - MAX. ABS. : 4.0315e-02
Node 8 - MAX. ABS. : 4.3232e-02
Node 9 - MAX. ABS. : 4.6448e-02
Node 10 - MAX. ABS. : 4.8326e-02
Node 11 - MAX. ABS. : 4.8327e-02
Displacement [m]
0.04
0.02
0.00
−0.02
−0.04
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time [s]


Displacement vs Structural Damage Index
Structural Damage Index [%]
Displacement [m]
10
8
6
4
2
0
−0.04
−0.02
0.00
0.02
0.04

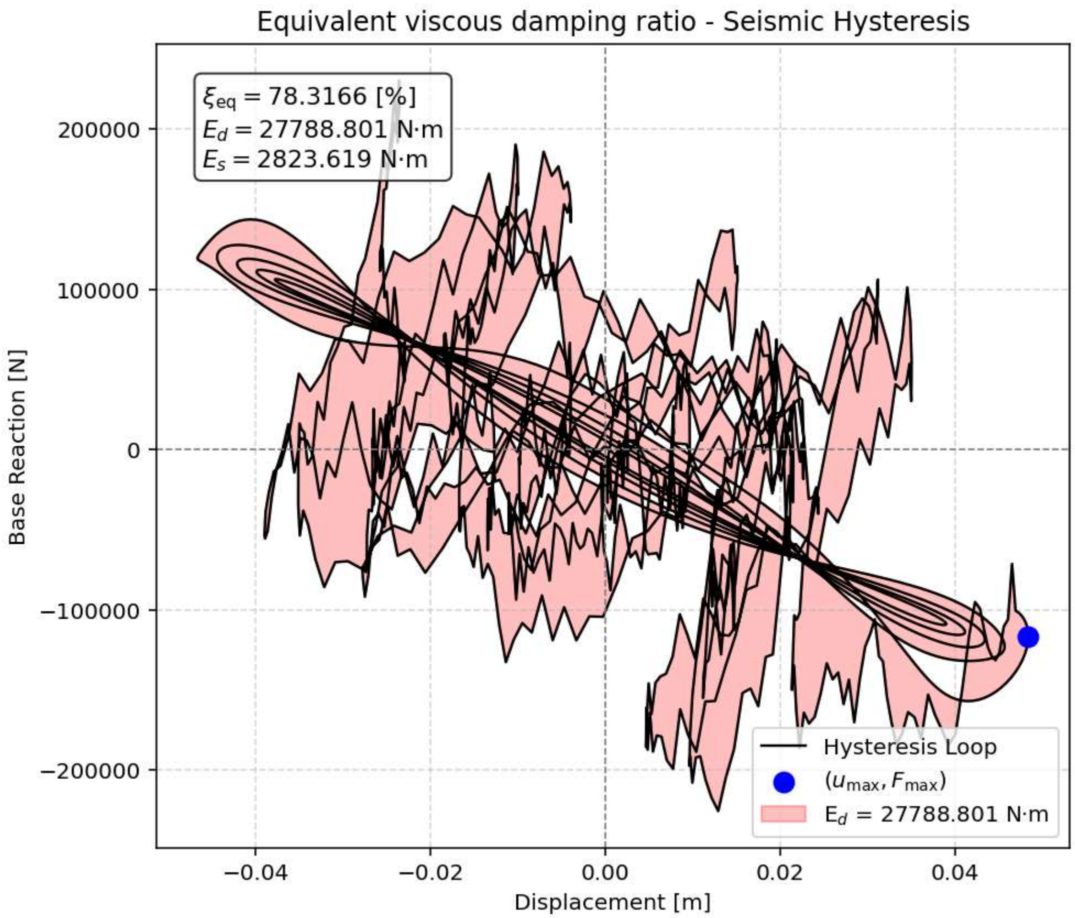

Histogram - Probability of Positive DISPLACEMENT FROM SEISMIC ANALYSIS is 51.40 %
Normal PDF
Quantile 0.25: -2.084583e-02
Median: 9.774216e-04
Quantile 0.75: 1.957814e-02
Mean: 1.535433e-04
Mean-Std: -2.469083e-02
Mean+Std: 2.499791e-02
Frequency
DISPLACEMENT FROM SEISMIC ANALYSIS
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
−0.04
−0.02
0.00
0.02
0.04

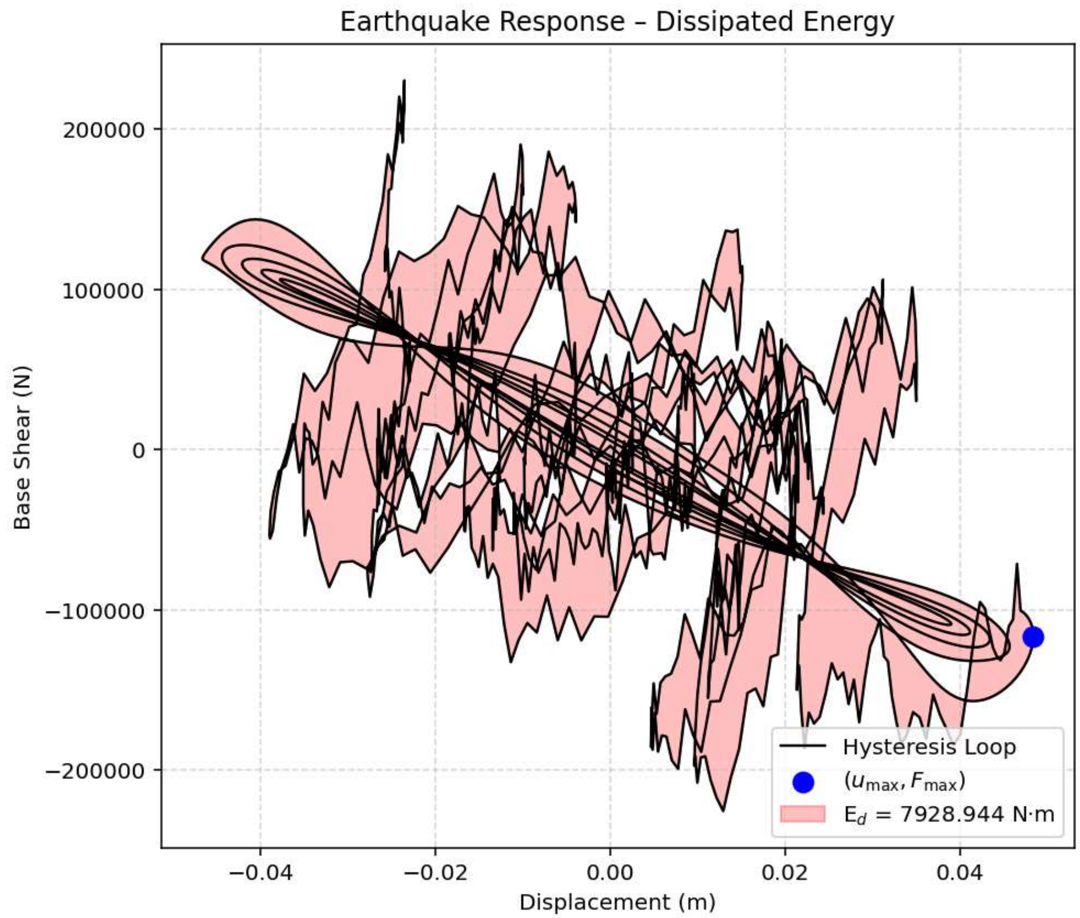


Histogram - Probability of Positive REACTION FROM SEISMIC ANALYSIS is 49.80 %
1e−6
Normal PDF
Quantile 0.25: -6.542023e+04
Median: -2.791570e+02
Quantile 0.75: 6.610569e+04
Mean: -1.047508e+03
Mean-Std: -8.500379e+04
Mean+Std: 8.290878e+04
Frequency
0
1
2
3
4
5
−200000
−100000
0
100000
200000
REACTION FROM SEISMIC ANALYSIS

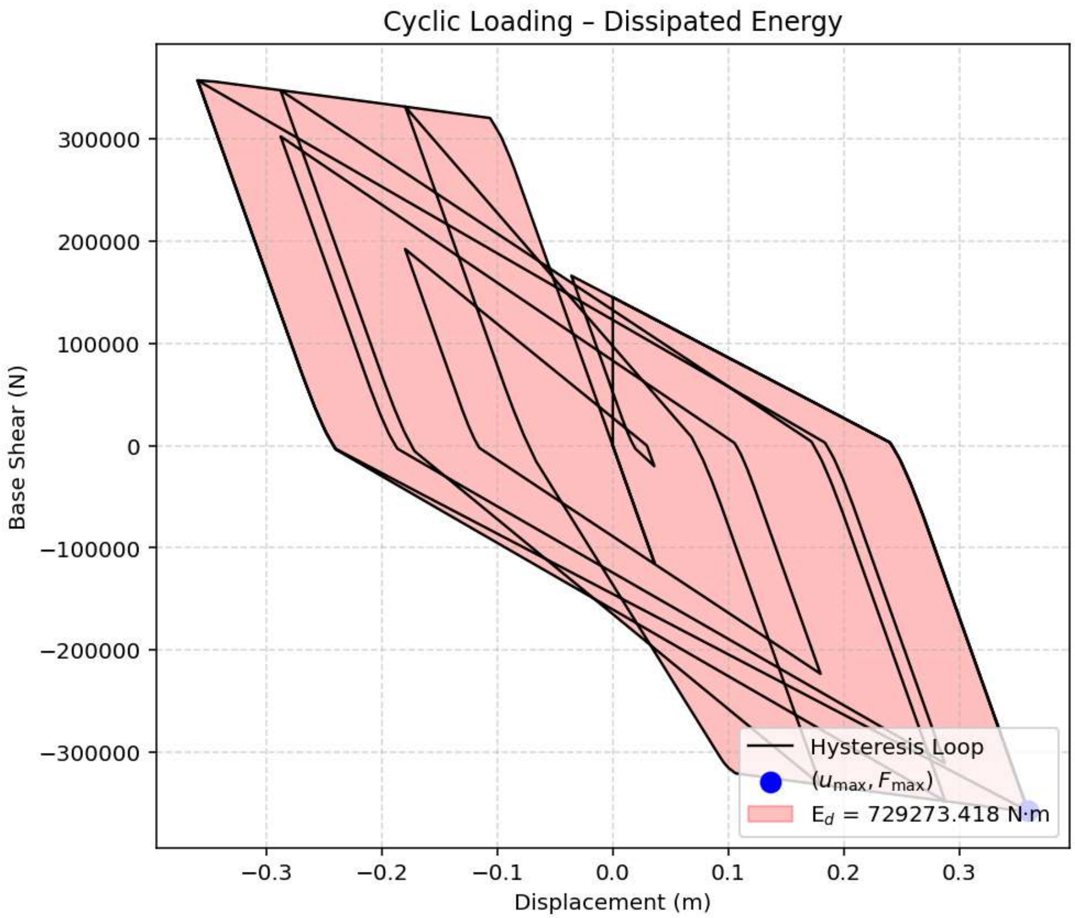


Histogram - Probability of Positive DUCTILITY DAMAGE INDEX FROM SEISMIC ANALYSIS is 61.85 %
Normal PDF
Quantile 0.25: 0.000000e+00
Median: 1.437896e+00
Quantile 0.75: 4.951490e+00
Mean: 2.564708e+00
Mean-Std: -3.133709e-01
Mean+Std: 5.442788e+00
Frequency
DUCTILITY DAMAGE INDEX FROM SEISMIC ANALYSIS


Equivalent viscous damping ratio - Seismic Hysteresis
$\xi_{eq} = 78.3166$ [%]
$E_d = 27788.801$ N·m
$E_s = 2823.619$ N·m
Hysteresis Loop
(u_{max}, F_{max})
E$_d$ = 27788.801 N·m
Base Reaction [N]
Displacement [m]
200000
100000
0
−100000
−200000
−0.04
−0.02
0.00
0.02
0.04



Earthquake Response – Dissipated Energy
Base Shear (N)
200000
100000
0
−100000
−200000
−0.04
−0.02
0.00
0.02
0.04
Displacement (m)
Hysteresis Loop
$(u_{\max}, F_{\max})$
E_d = 7928.944 N·m

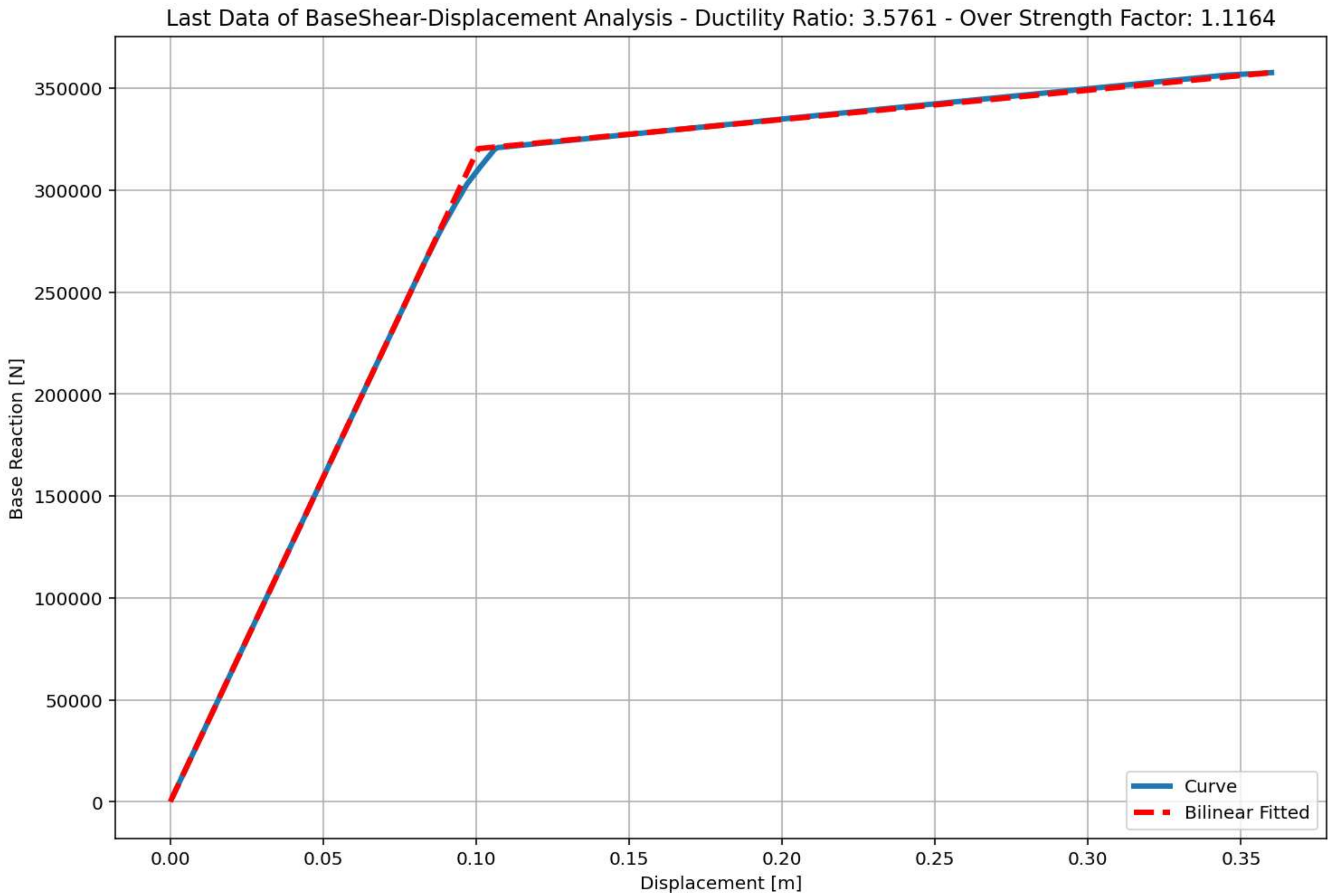

Cyclic Loading – Dissipated Energy
Base Shear (N)
Displacement (m)
300000
200000
100000
0
−100000
−200000
−300000
−0.3
−0.2
−0.1
0.0
0.1
0.2
0.3
Hysteresis Loop
$(u_{\max}, F_{\max})$
E_d = 729273.418 N·m

Dissipated Energy from Earthquake= 7928.94 N·m

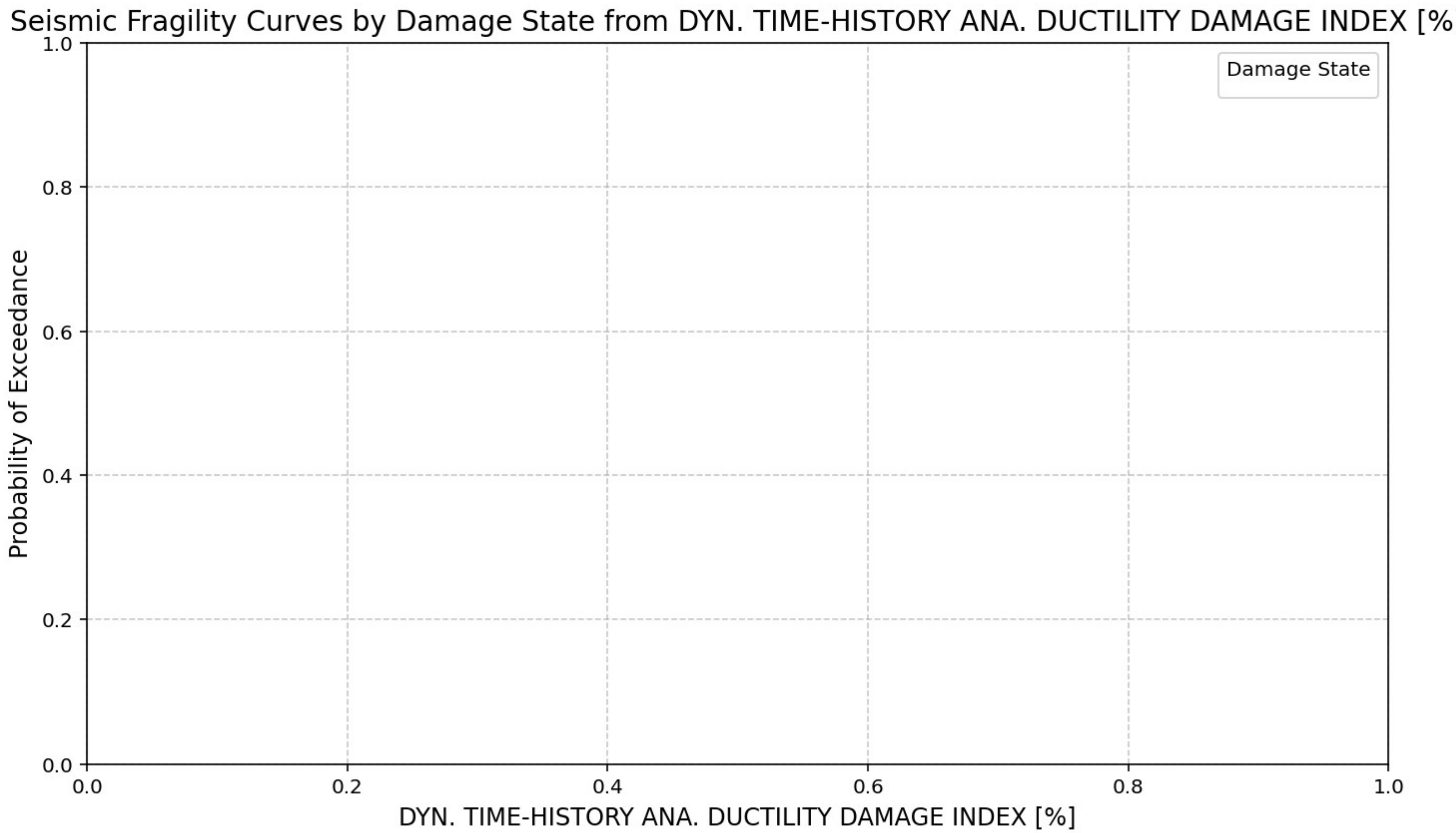
Dissipated Energy from Cyclic Displacement= 729273.42 N·m

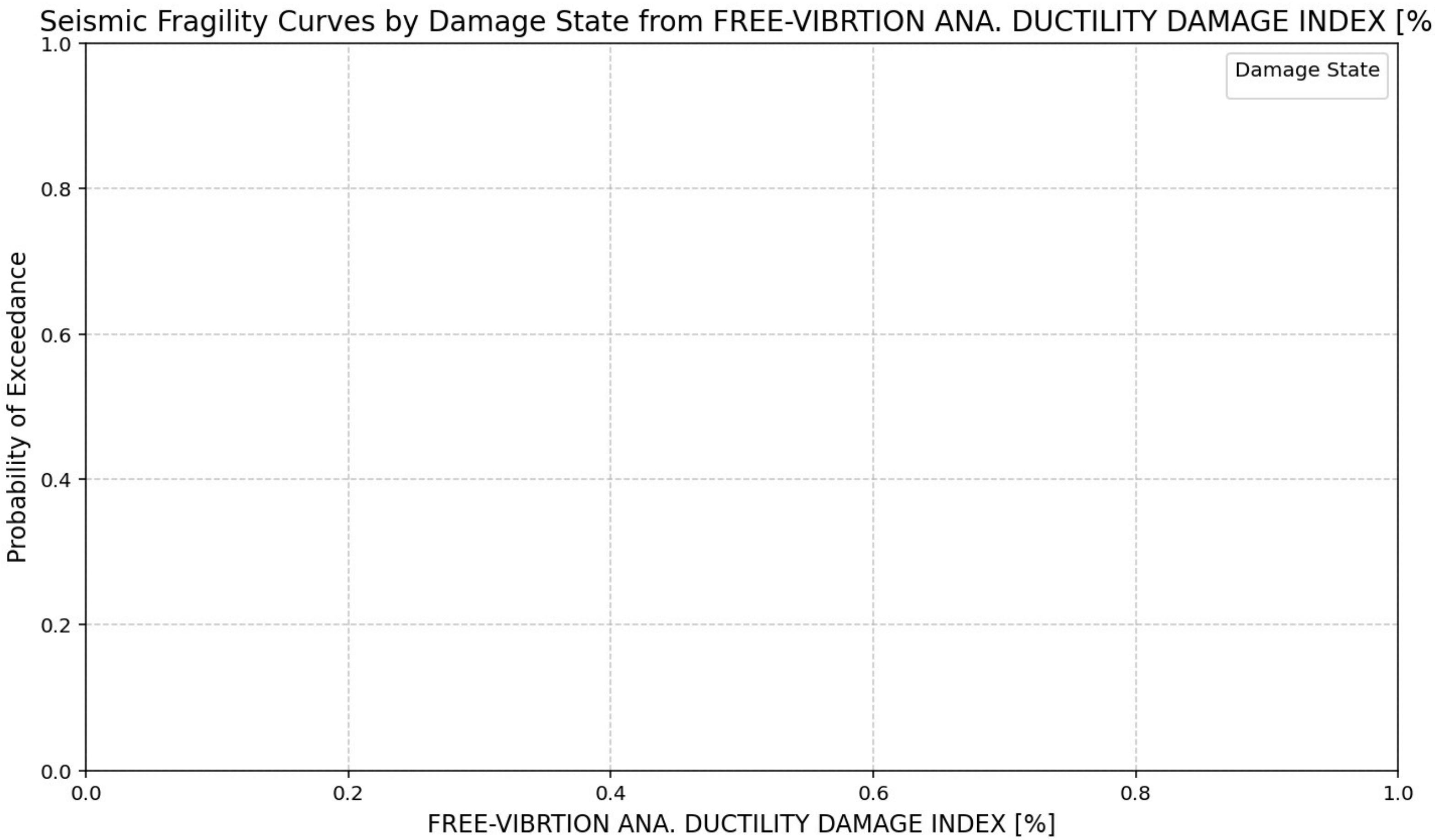
DISSIPATED ENERGY CAPACITY INDEX: 1.087 [%]

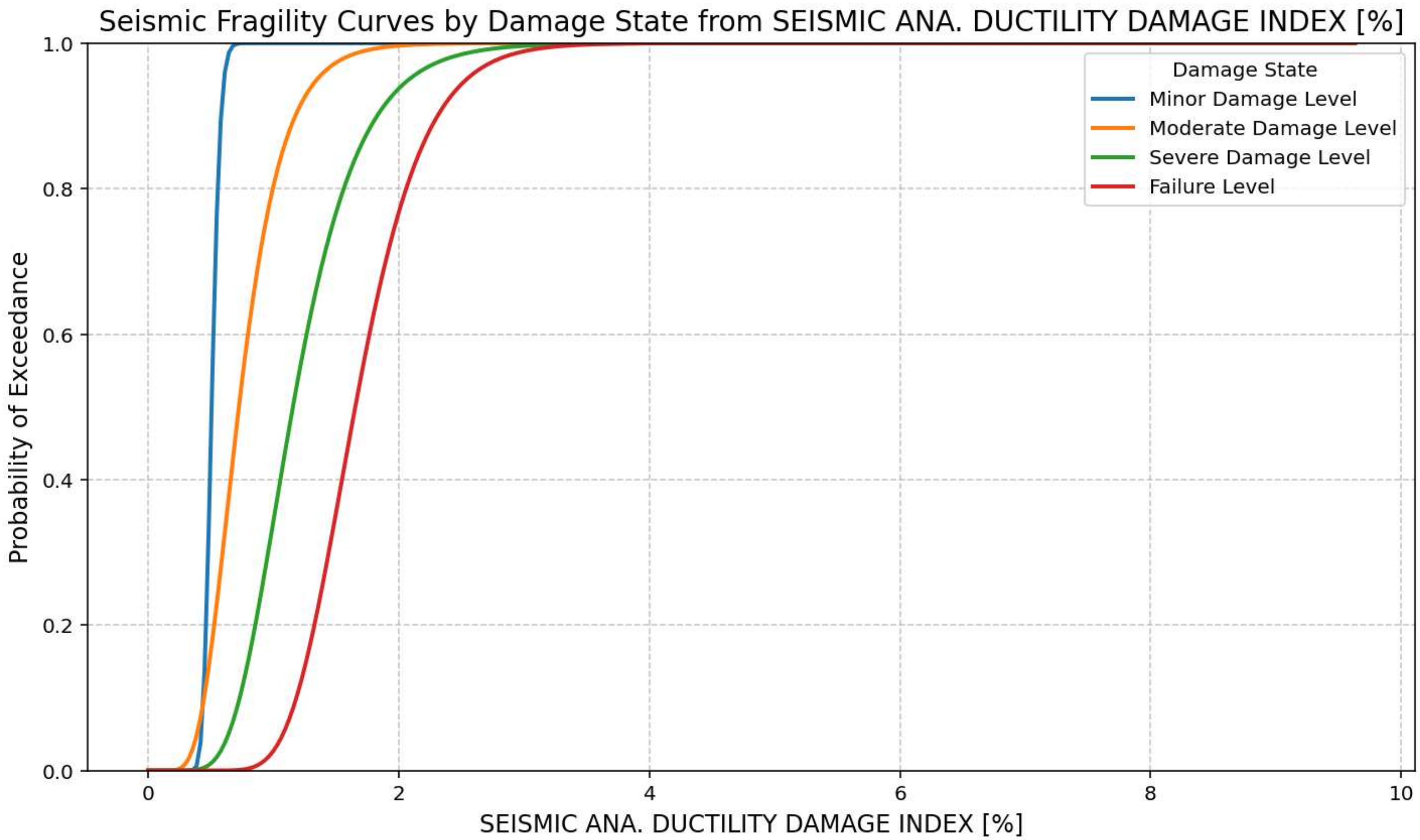
ZONE 1: VERY MINOR DAMAGE


Last Data of BaseShear-Displacement Analysis - Ductility Ratio: 3.5761 - Over Strength Factor: 1.1164
Base Reaction [N]
Displacement [m]
0
50000
100000
150000
200000
250000
300000
350000
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
Curve
Bilinear Fitted

```
+===========================+
=   Analysis curve fitted =
    Disp       Base Shear
---------------------------
[[0.00000000e+00 0.00000000e+00]
 [1.00669128e-01 3.20310863e+05]
 [3.60000000e-01 3.57591396e+05]]
+===========================+
+--------------------------------------------------+
 Structure Elastic Stiffness :     3181818.18
 Structure Plastic Stiffness :     993309.43
 Structure Tangent Stiffness :     143756.63
 Structure Ductility Ratio :      3.58
 Structure Over Strength Factor:   1.12
+--------------------------------------------------+
Over Strength Coefficient (Ω0):      1.1164
Displacement Ductility Ratio (μ):    3.5761
Ductility Coefficient (Rμ):          3.5761
Structural Behavior Coefficient (R): 3.9923
Structural Ductility Damage Index in X Direction: -20.1835 (%)
```


Seismic Fragility Curves by Damage State from DYN. TIME-HISTORY ANA. DUCTILITY DAMAGE INDEX [%]
Damage State
Probability of Exceedance
0.0
0.2
0.4
0.6
0.8
1.0
DYN. TIME-HISTORY ANA. DUCTILITY DAMAGE INDEX [%]


Seismic Fragility Curves by Damage State from FREE-VIBRTION ANA. DUCTILITY DAMAGE INDEX [%]
Damage State
1.0
0.8
0.6
0.4
0.2
0.0
Probability of Exceedance
0.0
0.2
0.4
0.6
0.8
1.0
FREE-VIBRTION ANA. DUCTILITY DAMAGE INDEX [%]



Seismic Fragility Curves by Damage State from SEISMIC ANA. DUCTILITY DAMAGE INDEX [%]
Probability of Exceedance
SEISMIC ANA. DUCTILITY DAMAGE INDEX [%]
0.0
0.2
0.4
0.6
0.8
1.0
0
2
4
6
8
10
Damage State
Minor Damage Level
Moderate Damage Level
Severe Damage Level
Failure Level